EE1101 Signals and Systems : Jan - May 2018

Course Instructor: Dr. Deepa Venkitesh — Room# RJN301

Note Title — 15-01-2018

3 lec + 1 Tut (Tut on Mon @ 1 pm) — Tutorial Tests, Tutorial Note Book

→ Grading : 20 + 20 + 20 + 40

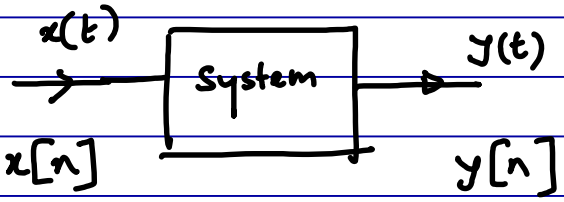
→ TAs

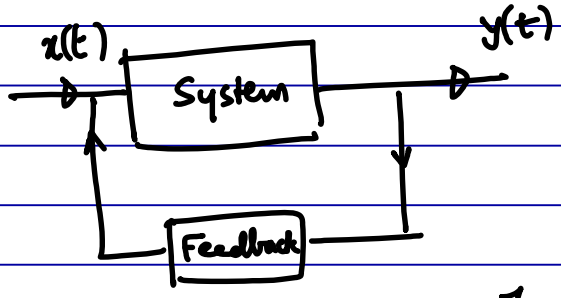
Mobile Phones not allowed during lecture hours

Signals and Systems

$x(t)$ Continuous time signal eg:- Wifi signal

$x[n]$ discrete time signal eg:- Sensex at the end of each day





Analysing systems / signals → Fourier Transforms, Laplace Transforms (Z-transforms)

Principles of Linear Systems & Signals by BP Lathi (2nd Edn)

Signals & Signals : Oppenheim, Willsky & Nawab (2nd Edn)

Signal

Function of one or more variables that conveys information-

Signal function of one variable : 1D signals
One Dimensional Signal

eg:-

>1 Variable :

In this course:-

- One Dimensional Signals, 'time' is the independent variable
- Single valued signal
- Value of the signal — real/complex

Classification of Signals

(CT) (DT)

(1) Continuous time & discrete time signals
(2) analog and digital signals
(3) Even and odd signals

(4) periodic & aperiodic signals

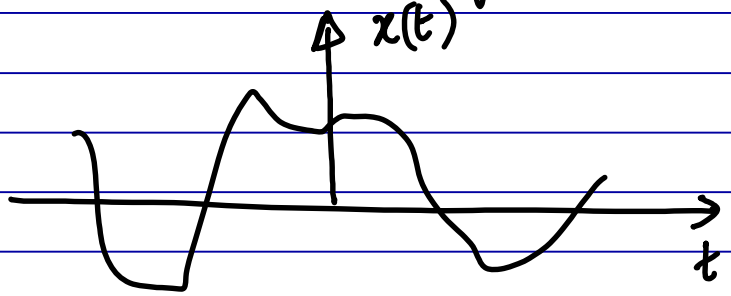
(5) causal & non-causal signals

(6) Deterministic & Random signals

(7) Energy & power Signals

(1) CT

$x(t)$ is a CT signal if it is defined for all t



eg:- audio signal

DT

$x[n]$ defined only at discrete instants of time

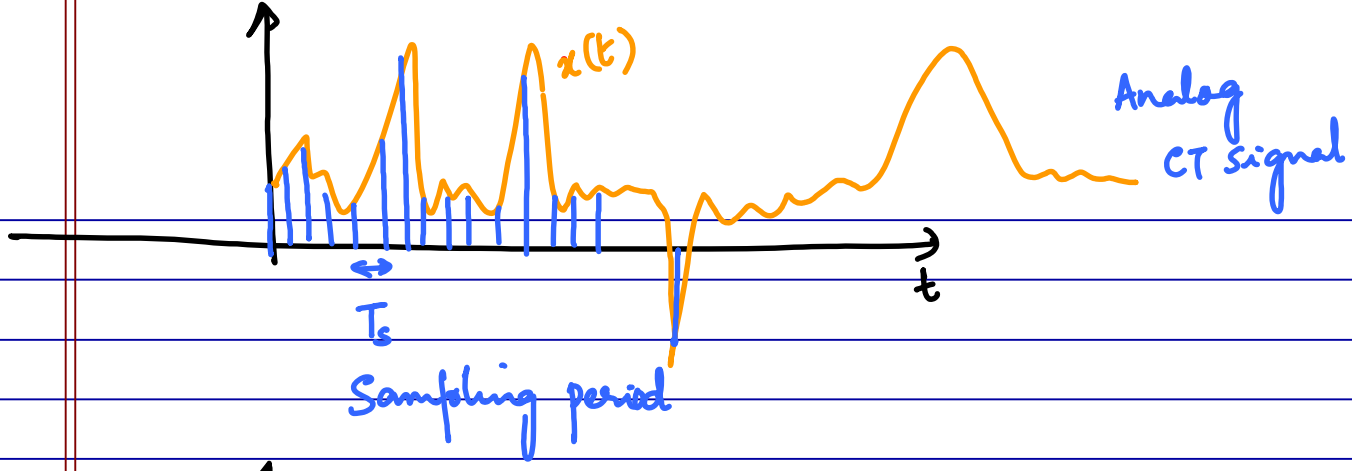
Independent variable takes only discrete values, at which the signal is defined

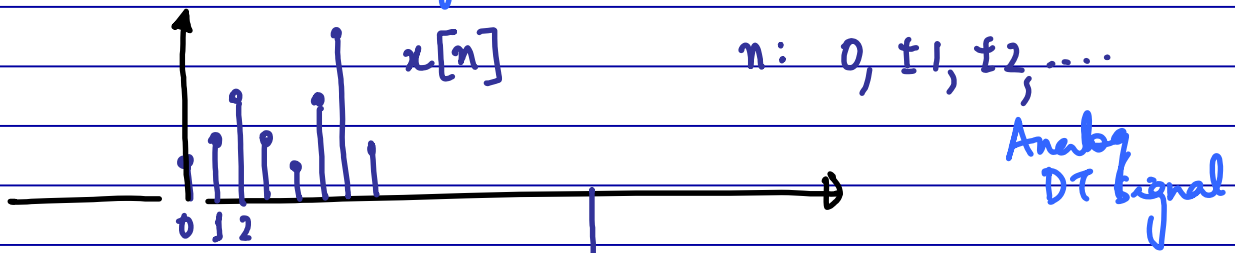
(uniformly spaced)

Time Series, Sequence

$\cdots\ x[-3],\ x[-2],\ x[-1],\ x[0],\ x[1],\ x[2],\ x[3],\ \cdots$

Sampling : Choose values of the signal at specific instances





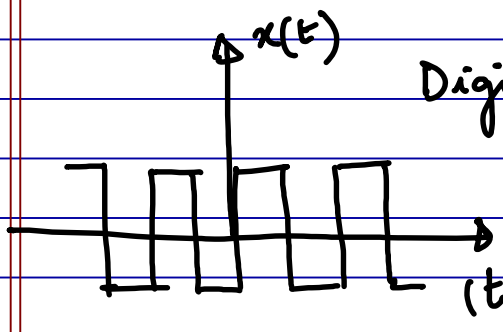
$t \rightarrow$ CT

$n \rightarrow$ DT

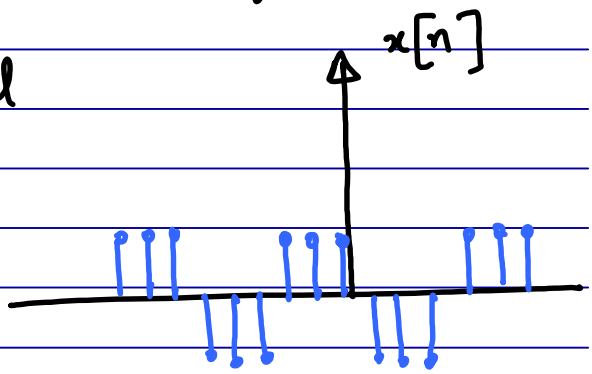
II Analog and Digital —

→ Signal amplitude can take { infinite number of values
                            { finite number of values

" " " "



Digital CT signal



Digital DT signal

III Even & Odd Signals

If $x(-t) = x(t) \quad \forall t$, even

$x(-t) = -x(t) \quad \forall t$, odd

$x[-n] = x[n] \quad \forall \; n$; even

$x[-n] = -x[n] \quad \forall \; n$; odd

$$x(t) = e^{-2t}\cos t$$
$$x(-t) = e^{2t}\cos t$$

$x(t)$ Neither even nor odd

let $x(t) = x_e(t) + x_o(t)$ — (1) $\quad x_e(t)$: even fn, $x_o(t)$: odd fn

Goal: Find $x_e(t)$, $x_o(t)$

$$x(-t) = x_e(-t) + x_o(-t) = x_e(t) - x_o(t) \text{ — (2)}$$

$$x_e(t) = \frac{x(t) + x(-t)}{2}$$

$$x_o(t) = \frac{x(t) - x(-t)}{2}$$

$$x(t) = e^{-2t}\cos t = x_e(t) + x_o(t)$$

$$x_e(t) = \left(\frac{e^{-2t} + e^{2t}}{2}\right)\cos t$$

$$x_o(t) = \left(\frac{e^{-2t} - e^{+2t}}{2}\right)\cos t$$

Conjugate Symmetry (Complex Signals)

$$x(-t) = x^*(t)$$

HW: Prove that for signals with conjugate symmetry,

real part is even, imaginary part is odd

i.e, if $x(t) = a(t) + j\,b(t)$

IV Periodic & Aperiodic Signals

For a positive constant T_0, if

$x(t) = x(t+T_0) \quad \forall t$, then the signal is periodic

Period :- Smallest possible value of T_0 that satisfies periodicity condition

$x[n+N] = x[n] \quad \forall n$

Period Smallest possible value of N .....

Freq $f = \frac{1}{T_0}$ (H_z)

Angular freq : $2\pi f = \frac{2\pi}{T_0}$ rad/s

Freq $= \frac{1}{N}$

Angular freq $\Omega = \frac{2\pi}{N}$ (rad)

Find if periodic ; if Periodic, find period

$x(t) = \cos^2(2\pi t)$

Periodic ; Period : 0.5 s

$x(t) = \sin^3(\pi t)$

Periodic, Period : (π) s

$x[n] = \cos(2n)$

Non periodic

$x(t) = e^{\cos 2\pi t}$

Non periodic

$x[n] = (-1)^n$

Periodic : Period : 2 samples

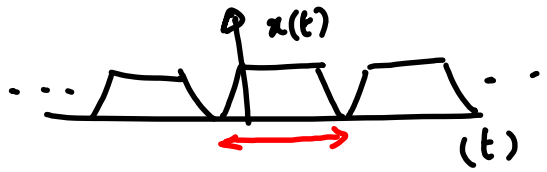
$x[n] = (-1)^{n^2}$

Periodic, Period : 2 samples

Lecture 3

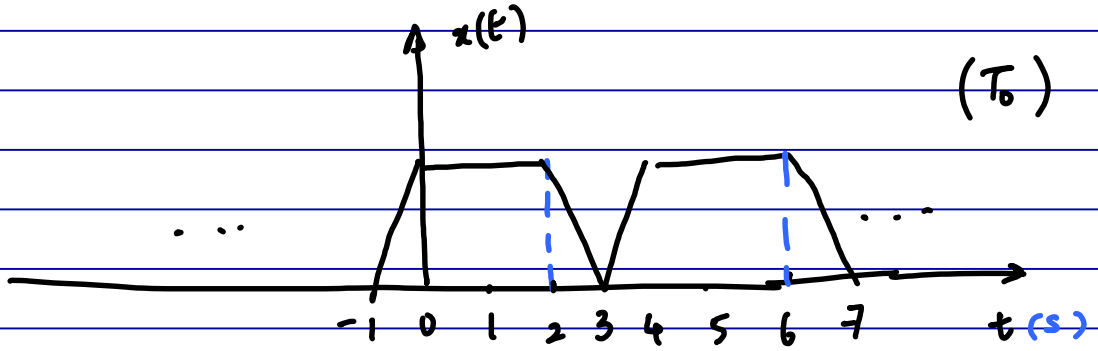
Comments

(a) Periodic Signal must exist from $t = -\infty$ to $t = \infty$



Periodic Signals are 'everlasting'

(b)



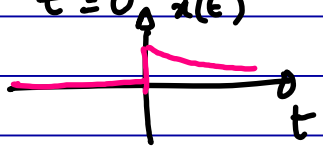
Periodic signal can be reconstructed by knowing its values over one time Period

(c) Area under $x(t)$: $\int_a^{a+T_0} x(t)\,dt = \int_b^{b+T_0} x(t)\,dt$

V Causal & non causal

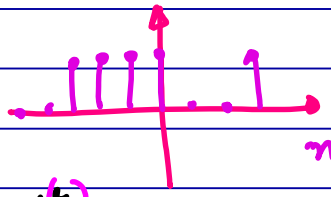
Causal :

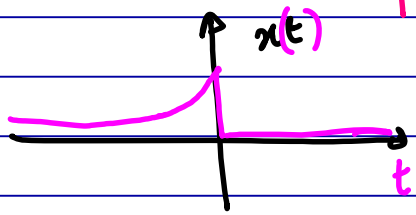
Signal that does not start before $t=0$ $x(t)=0 \quad \forall\, t<0$



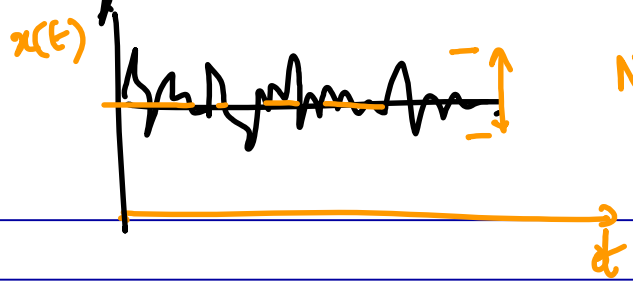
Non causal $x(t) \neq 0 \quad \forall\, t<0$

anti-causal $x(t) = 0 \quad \forall\, t>0$





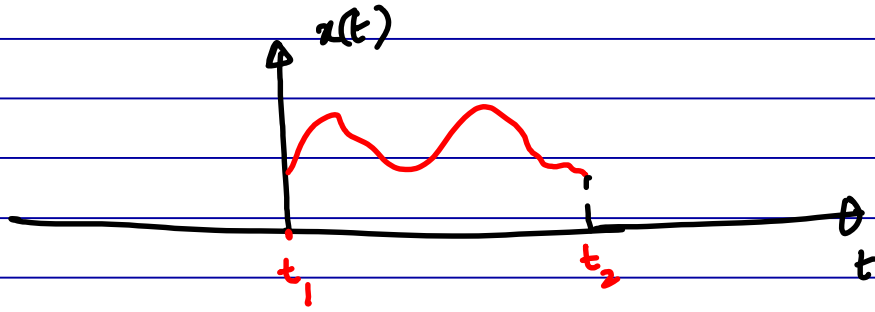
VI Deterministic and Random Signals



Noise signal

Typically the statistics about random signals are known (mean, variance)

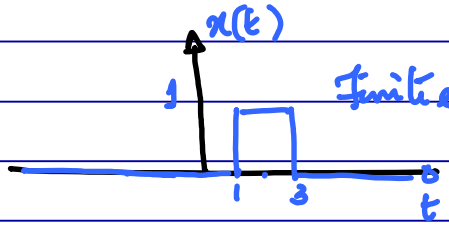
VIII Power & Energy signals



$$\text{Energy} = \int_{t_1}^{t_2} |x(t)|^2 \, dt \qquad \text{Power} = \frac{1}{(t_2 - t_1)} \int_{t_1}^{t_2} |x(t)|^2 \, dt$$

$$E = E_\infty = \int_{-\infty}^{\infty} |x(t)|^2 \, dt \qquad E_\infty = \sum_{n=-\infty}^{\infty} |x[n]|^2$$

$$P = P_\infty = \lim_{T \to \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 \, dt \qquad P_\infty = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x[n]|^2$$

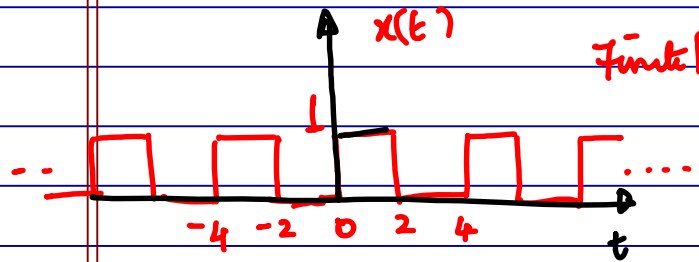


Finite energy

Energy Signals

Signals with $0 < E_\infty < \infty$

Zero Power



Finite Power

Power Signals

Signals with $0 < P_\infty < \infty$

Infinite Energy

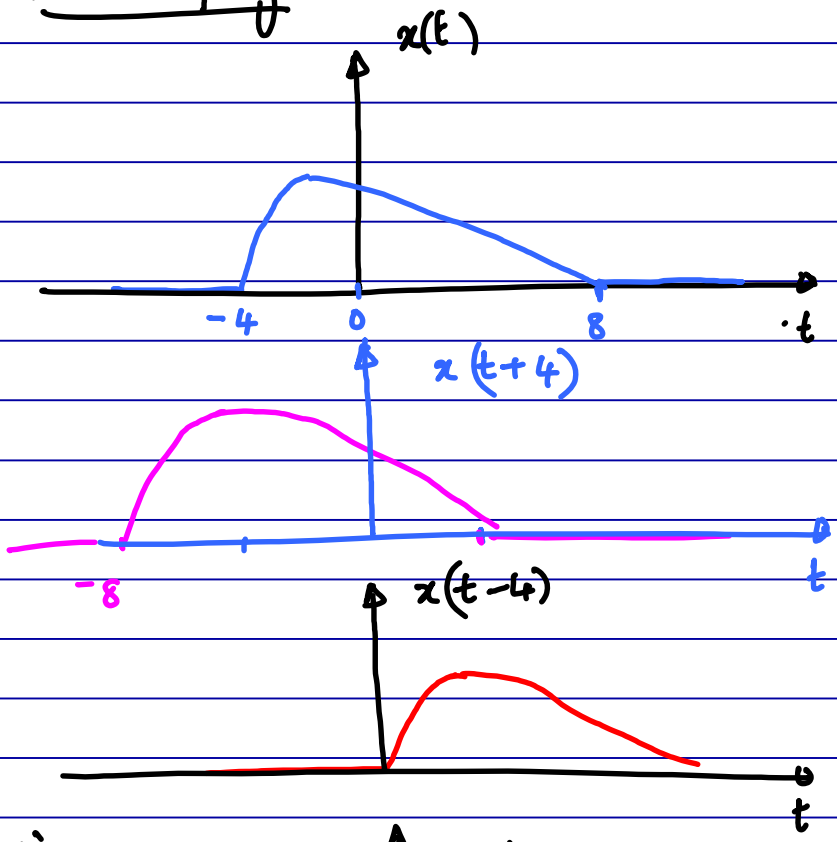
Useful Signal Operations

(operations on x-axis 't')

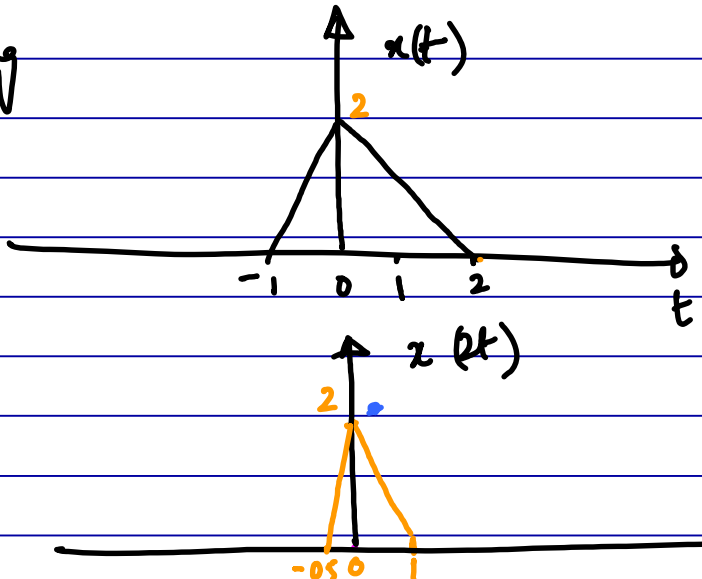
(1) Time Shifting (2) Time Scaling (3) Reflection

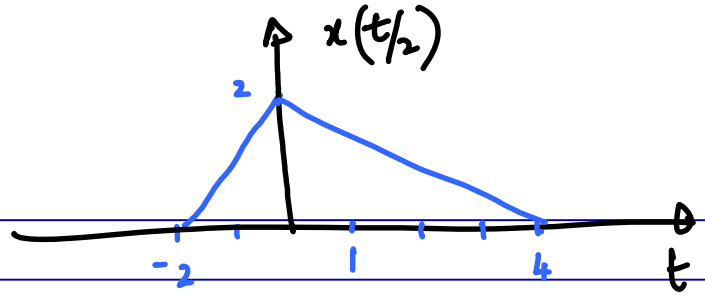
(4) Combination of (1), (2), (3)

(1) Time Shifting

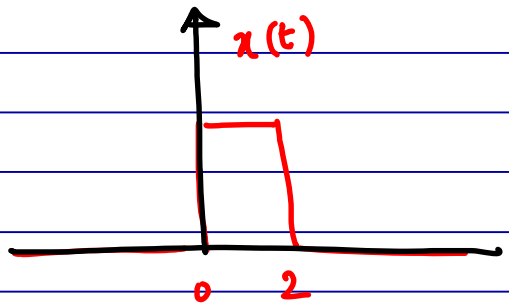


(2) Scaling





(3) Shifting + scaling



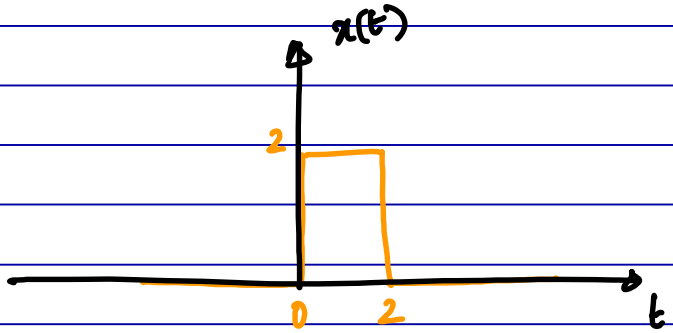
$x(2t+3)$

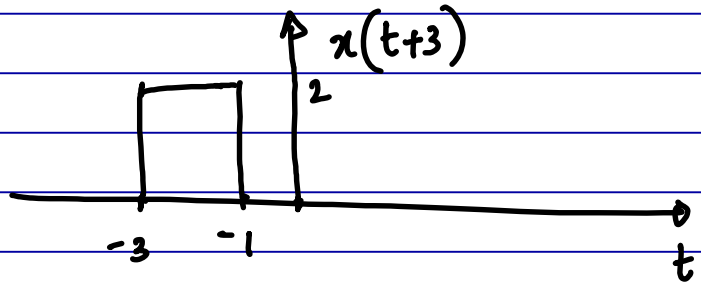
18.01.18

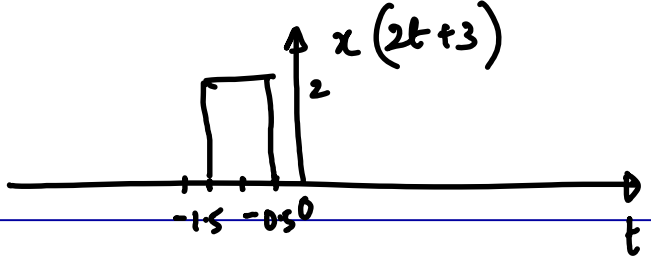
$y(t) = x(at - b)$

(1) Shift $x(t-b)$
(2) scale $x(at-b)$

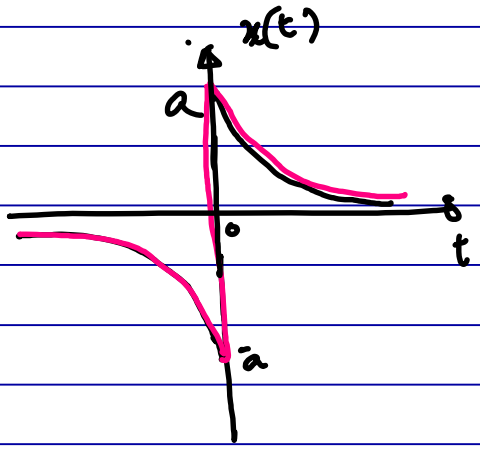
(1) scale $x(at)$
(2) shift by $\frac{b}{a}$

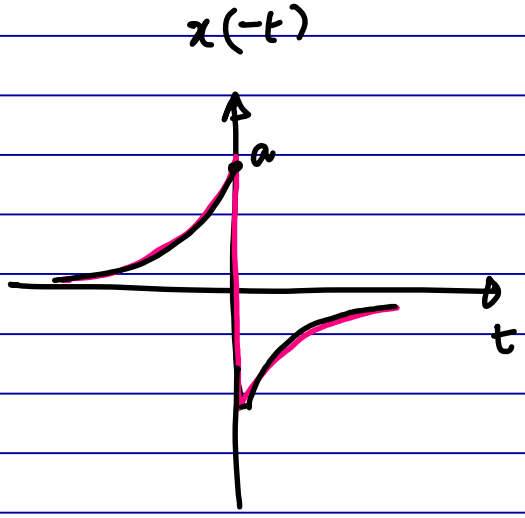






(3) Reflection $x(-t)$

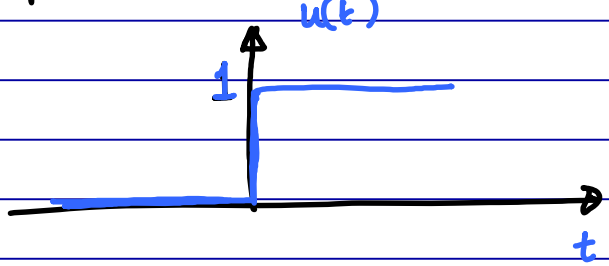




Elementary Signals

(1) Exponential (2) Sinusoidal
(3) unit step function
(4) Dirac Delta function
(5) Ramp function

Unit Step Function



$u(t) = 0 \quad \forall\ t < 0$

$= 1 \quad \forall\ t > 0$

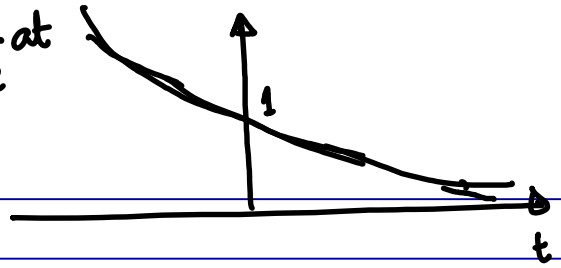
At $t = 0$, discontinuous

Oliver Heaviside

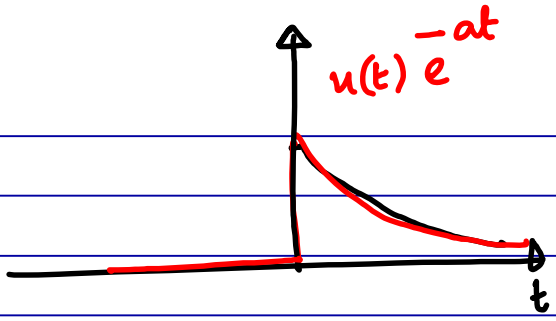
(1) Causal

Any signal can be made causal by multiplying it with u(t)

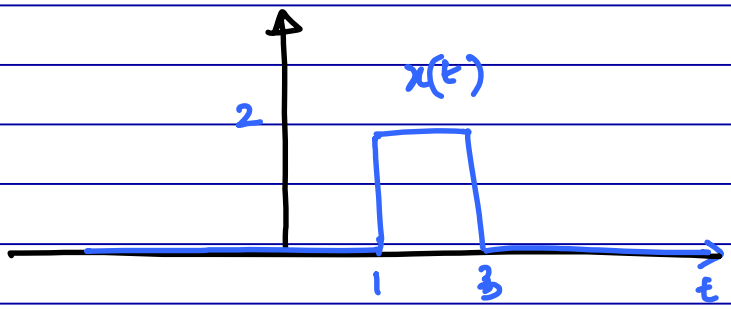
$x(t) = e^{-at}$



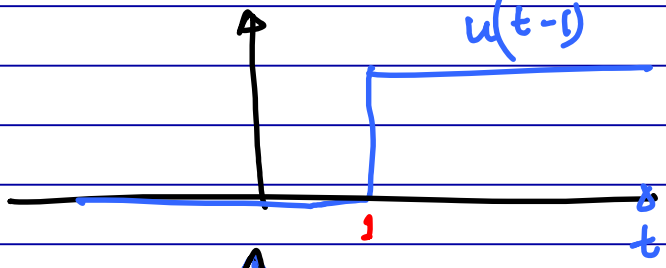
$x(t)\,u(t)$



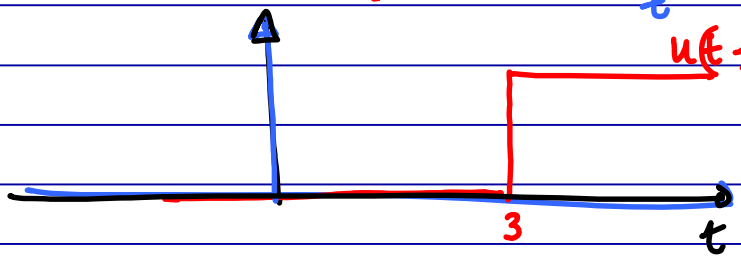
(2) use $u(t)$ for 'windowing'



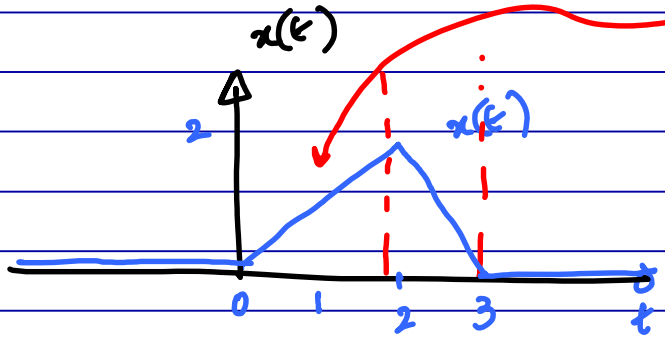
$u(t-1)$



$x(t) = u(t-1) - u(t-3)$



Eg. 2



eqn of the st line

$t\,[u(t) - u(t-2)]$

$+\,2(t-3)\,[u(t-2) - u(t-3)]$

$y = mt + c = -2t + 6$

$y = -2t + c = -2(t-3)$

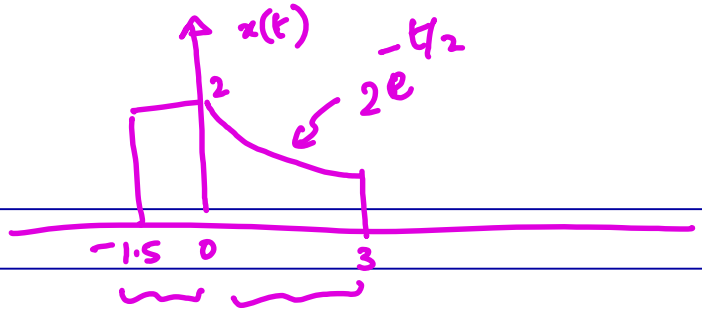
simplify

At $t = 3$, $y = 0$, $\Rightarrow c = 6$

$t\,u(t) + (-3t + 6)\,u(t-2)$

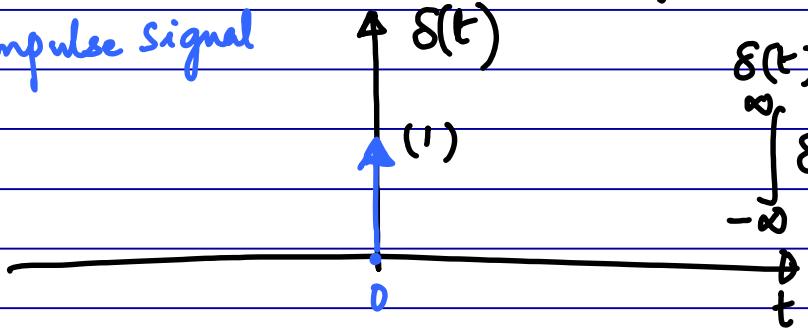
$+\,2(t-3)\,u(t-3)$

HW



II Delta function (Dirac Delta function)

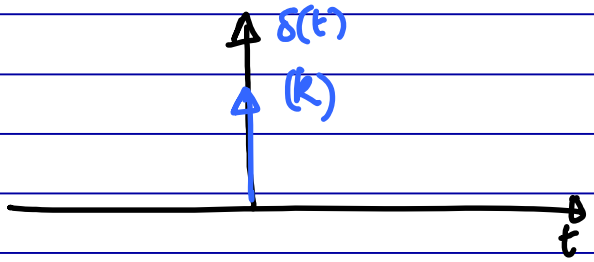
Unit Impulse signal



$$\delta(t) = 0 \quad \forall\, t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t)\,dt = 1$$

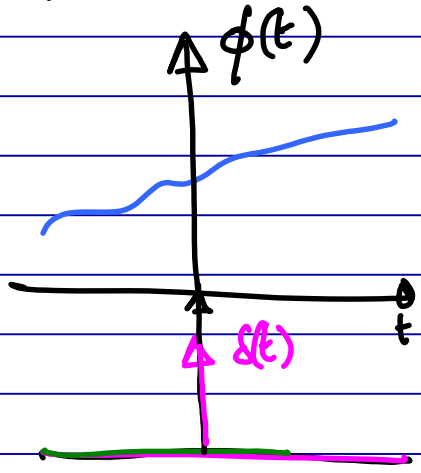
$$K\,\delta(t) \Rightarrow \int_{-\infty}^{\infty} \delta(t)\,dt = K$$



(1) Multiplication of function by impulse

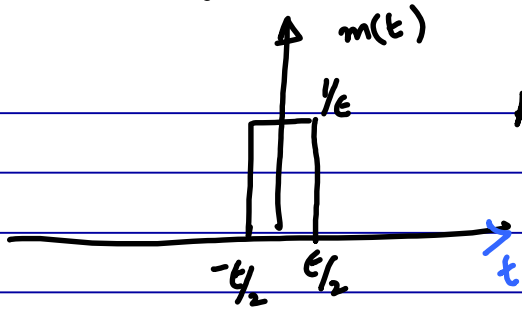
$\phi(t)$: Continuous function

$$\phi(t)\,\delta(t) = \phi(0)\,\delta(t)$$



Visualising impulse fn

22.1.18



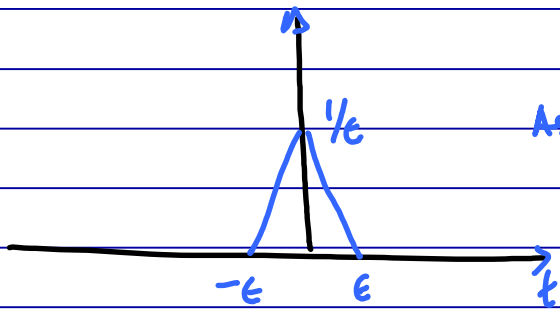
As $\epsilon \to 0$, $m(t) \to \delta(t)$

$\delta(t) = 0$ everywhere except $t = 0$

At $t = 0$, $\delta(t)$ is undefined

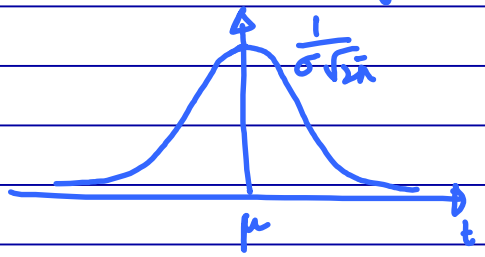
Width $\to \epsilon$ Area 1

Height $\to 1/\epsilon$



As $\epsilon \to 0$ the triangular fn $\to \delta(t)$

Gaussian μ, Mean, Variance: σ^2 $g(t) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(t-\mu)^2}{2\sigma^2}}$

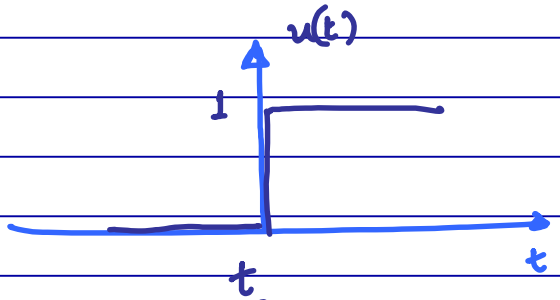


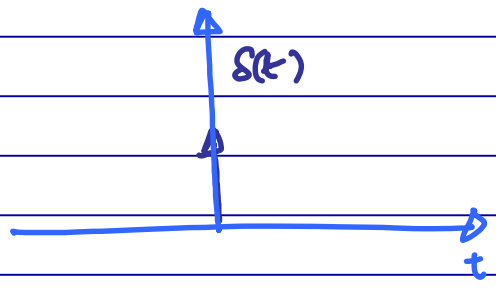
Hint: Find FWHM

$\int_{-\infty}^{\infty} g(t)\,dt = 1$ (Verify)

As $\sigma \to 0$, $g(t) \to \delta(t)$

HW Find an exponential fn representation for $\delta(t)$





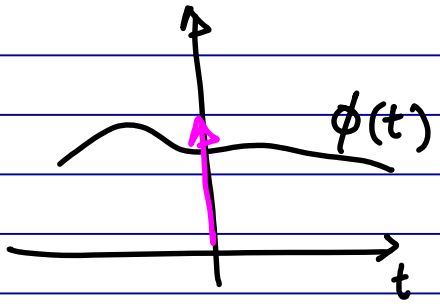
$\int_{t=-\infty}^{t} \delta(\tau)\,d\tau = 0 \quad t < 0$

$= 1$ } Defn of u(t)

$= 1 \quad t > 0$

$$\int_{-\infty}^{t} \delta(\tau)\, d\tau = u(t)$$

$$\frac{d}{dt}\left(u(t)\right) = \delta(t)$$



Sampling property
Sifting property of $\delta(t)$

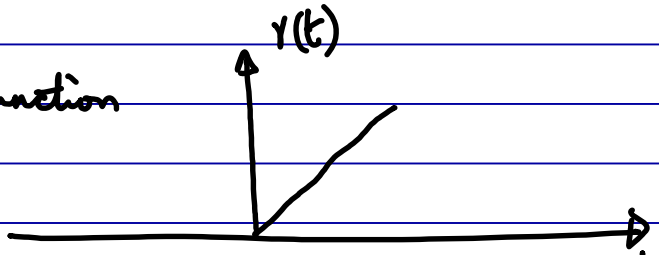
$$\phi(t)\,\delta(t) = \phi(0)\,\delta(t)$$

$$\int_{-\infty}^{\infty} \phi(t)\,\delta(t)\,dt = \int_{-\infty}^{\infty} \phi(0)\,\delta(t)\,dt = \phi(0)\int_{-\infty}^{\infty} \delta(t)\,dt$$

$$= \phi(0)$$

$$\boxed{\phi(0) = \int_{-\infty}^{\infty} \phi(t)\,\delta(t)\,dt}$$

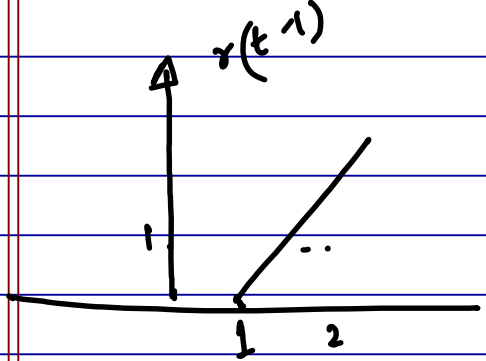
Unit Ramp function

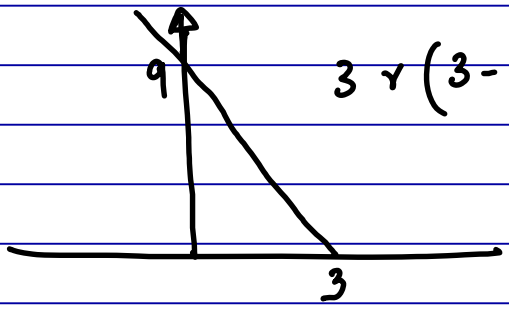


$r(t) = t \quad \forall\ t \geq 0$

$r(t) = 0 \quad \forall\ t < 0$

$r(t) = t\,u(t)$

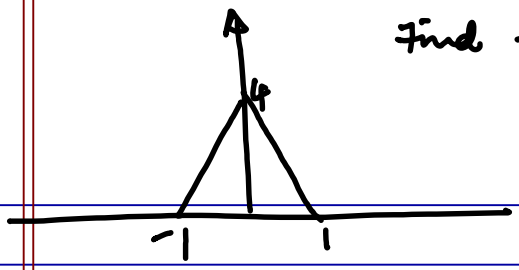


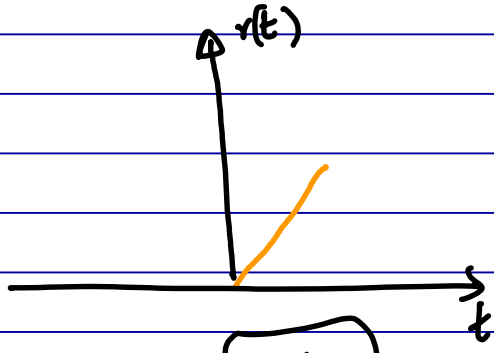


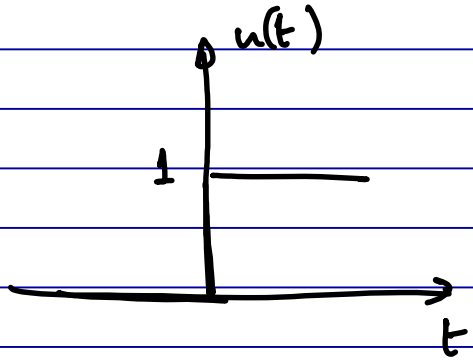
$3\,r(3-t)$

HW

Find the representation in terms of $r(t)$







$$\int_{-\infty}^{t} u(t)\,dt = r(t)$$

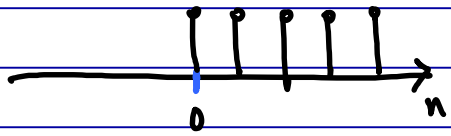
$$\frac{d}{dt} r(t) = u(t)$$

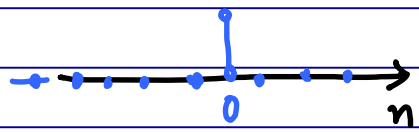
Define step fn, impulse fn for discrete time

$$u[n] = 1 \quad \forall\ n \geq 0$$
$$= 0 \quad \forall\ n < 0$$

$$\delta[n] = 1 \quad n = 0$$
$$= 0 \quad \forall\ n \neq 0$$

$u[n]$ $\qquad$ $\delta[n]$





$\delta[n] = u[n] - u[n-1]$ ✓ (Differentiation)

$u[n] = \sum_{k=0}^{\infty} \delta[n-k]$ (Integration)

$\delta[k]$ $\delta[-k]$, $\delta[n-k]$

Exponential function

$$f(t) = \bar{A} e^{\bar{s}t}$$

$\bar{A}$, $\bar{s}$ } complex

$$\bar{A} = A e^{j\phi}, \quad \bar{s} = \sigma + j\omega$$

$$f(t) = A e^{\sigma t} \cdot e^{j(\omega t + \phi)}$$

$\bar{s}$: complex frequency

$$f^*(t) = A e^{\sigma t} e^{-j(\omega t + \phi)}$$

$$f(t) + f^*(t) = 2A e^{\sigma t} \cos(\omega t + \phi)$$

Cases : $\bar{s} = 0, \ \phi = 0$

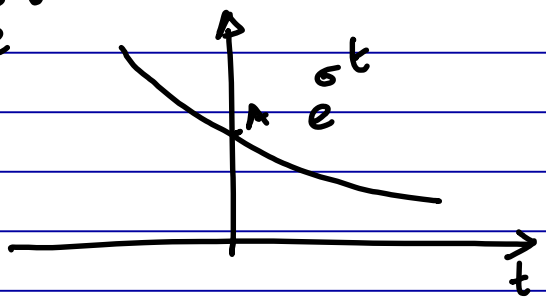
$f(t) = A$ (dc)

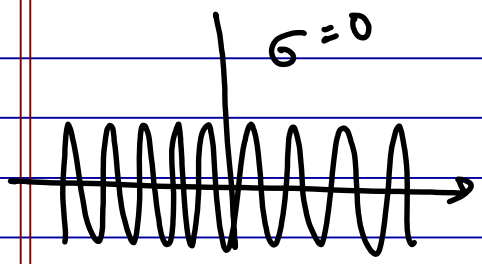
$\sigma = 0 \quad \omega \neq 0, \ \phi \neq 0$

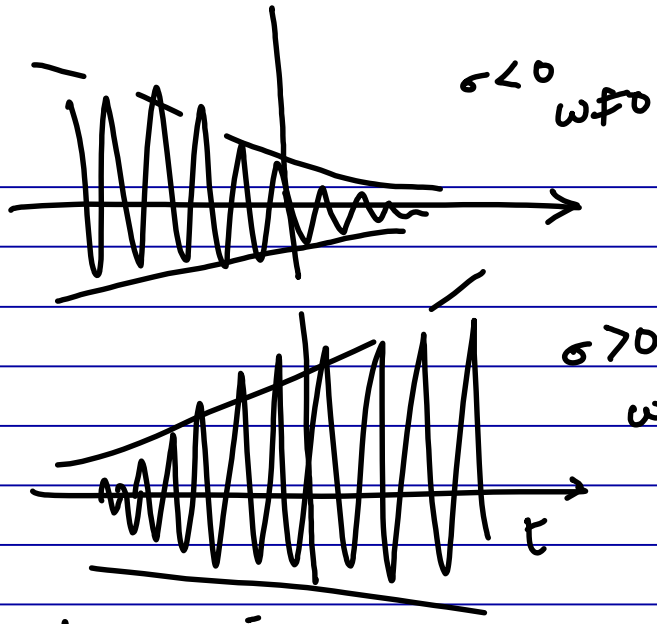
$f(t) = A e^{j(\omega t + \phi)}$ (Sinusoidal)

$\sigma \neq 0, \ \omega = 0 \quad \phi = 0$

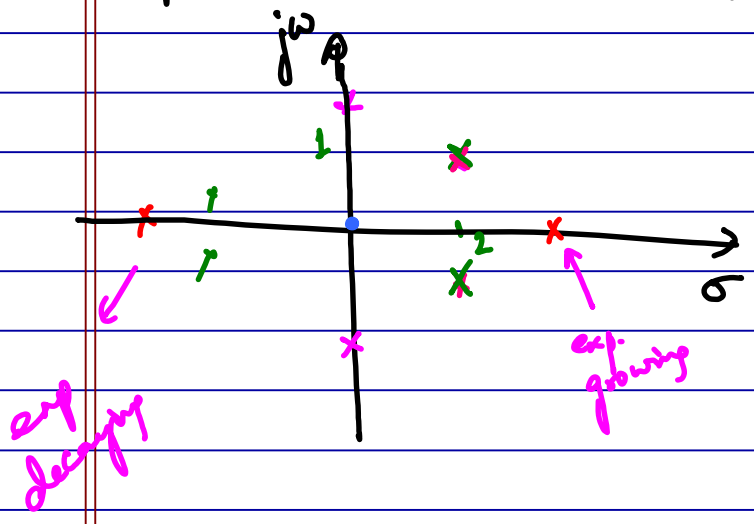
$f(t) = A e^{\sigma t}$







Complex plane representation of s



real sinusoidal/oscillatory appear as pairs

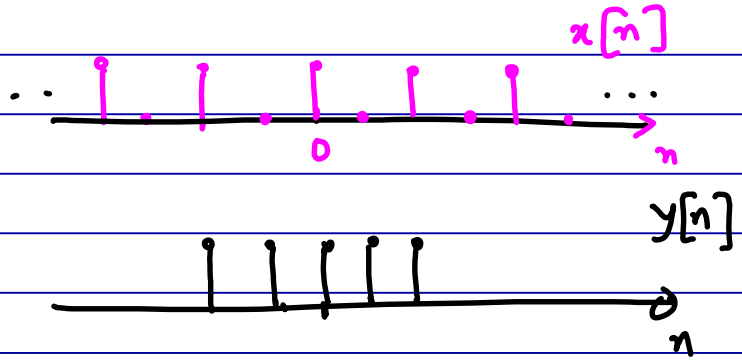
25.1.18

1 Is affine scaling reversible? No

(linear)

$y(t) \rightarrow x(2t)$

$y[n] = x[2n]$



2. $y[n] = x[n]\,\delta[n] = x[0]$ FALSE

$y[n] = x[0]\,\delta[n]$

3. Why do we need a fn like $\delta(t)$

4. Show that $\delta(at) = \frac{1}{|a|}\delta(t)$

$\delta(-t) = \delta(t)$

$\delta(t)\cdot\delta(t)$: Not defined

Discrete time Complex exponentials

$x[n] = C\alpha^n \quad ; \quad \alpha = e^{\beta}$

$x[n] = C e^{\beta n}$ [C is real, β is complex in general]

$x[n] = C e^{j\omega_0 n}$ $\boxed{\beta = j\omega_0}$

$x[n] = C\left[\cos(\omega_0 n) + j\sin(\omega_0 n)\right]$

$C = 1$

Power of this sinusoidal signal : $\frac{1}{|C|^2}$

$$x(t) = \sum_{k=m}^{n} D_k e^{j\omega_k t}$$

$$P = \sum_{k=m}^{n} |D_k|^2$$

$$P = \frac{1}{T}\int_0^T \sum_{k=m}^{n} D_k e^{j\omega_k t} \cdot \sum_{k=m}^{n} D_k^* e^{-j\omega_k t}\, dt$$

$$\sum_{m=}^{n} |D_k|^2$$

$D_0 \quad D_1 \quad D_2 \quad D_3 \quad \dots \quad D_{10}$

$D_0^* \quad D_1^* \quad D_2^* \quad D_3^* \quad \dots \quad D_{10}^*$

$$\sum\sum D_k D_l^* e^{j\omega_k t} e^{-j\omega_l t}$$

$$\frac{1}{T}\int_0^T \sum\sum D_k D_l^* e^{j\omega_k t} e^{-j\omega_l t}\, dt$$

Qn In DT periodic signals, does the rapidity of oscillation increase with ω_0?

Possible range of ω_0: $[-\pi, \pi)$ or $[0, 2\pi]$

Qn Is a DT sinusoid always periodic?

If $x[n]$ is periodic

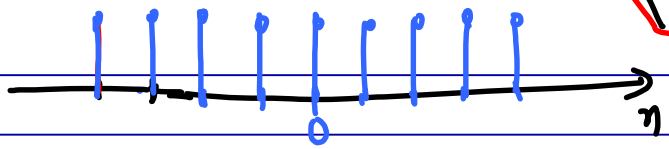
$$x[n+N] = x[n]$$

$$e^{j\omega_0(n+N)} = e^{j\omega_0 n} \Rightarrow e^{j\omega_0 N} = 1 \Rightarrow \omega_0 N = k2\pi$$

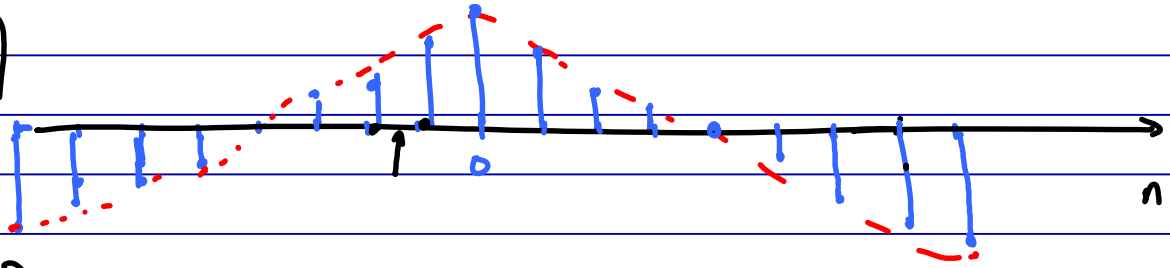
$$\Rightarrow \frac{\omega_0}{2\pi} = \frac{k}{N}$$

DT seq. periodic only when $\frac{\omega_0}{2\pi}$ is rational

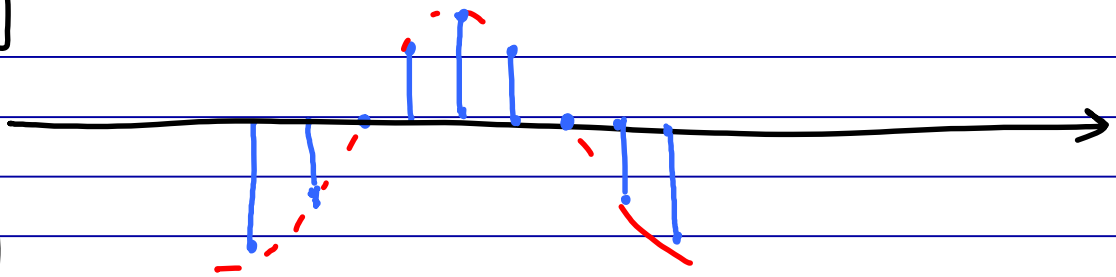
$x[n] = \cos[0\,n]$



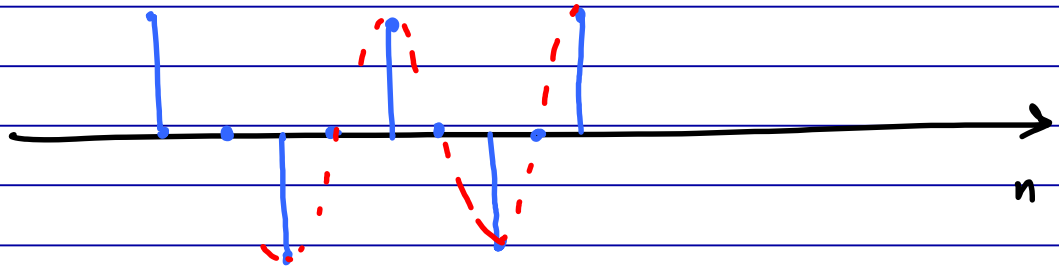
$\cos\left[\frac{\pi}{8}n\right]$



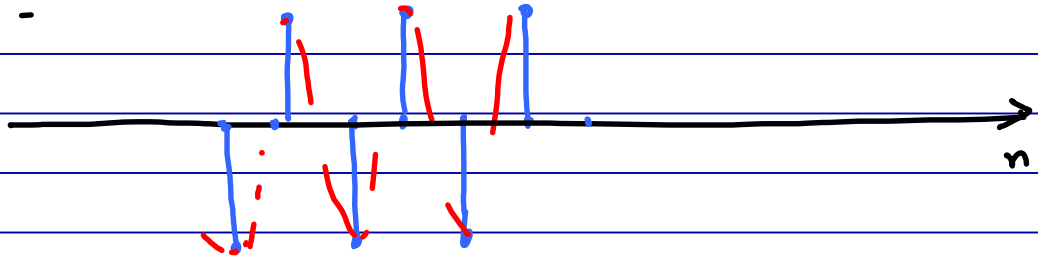
$\cos\left[\frac{\pi}{4}n\right]$



$\cos\left[\frac{\pi}{2}n\right]$



$\cos[\pi n]$

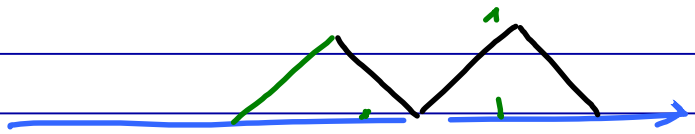


$\cos\left(\frac{3\pi}{2}n\right)$

$\cos\left(\frac{7\pi}{4}n\right)$

rapidly Oscillations increase $\omega_0 = 0 \rightarrow \bar{\pi}$

decreases $\bar{\pi} \rightarrow 2\bar{\pi}$



$[0, 2\bar{\pi}]$ $[-\bar{\pi}, \bar{\pi}]$

$\sin\left(\frac{6\bar{\pi}(n+N)}{35}\right) = \sin\left(\frac{6\bar{\pi}n}{35}\right)$

$x[n] = \cos(3n)$

29/1/18

- Find if the following signals are periodic; find period if they are periodic

$x[n] = 5\sin(2n)$

$\frac{6\bar{\pi}N}{35} = 2\bar{\pi}k$

$N = \frac{k35}{3}$

✓ $x[n] = 5\cos(0)$ → $N = 1$ sample

✓ $x[n] = 5\sin\left(\frac{6\bar{\pi}n}{35}\right)$ $N = 35$ sample

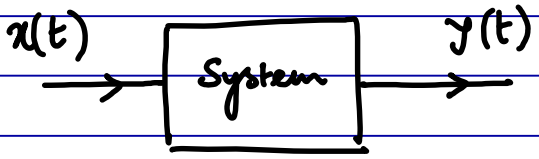
✓ $x[n] = 5\cos(6\bar{\pi}n)$ $N = 1$ sample

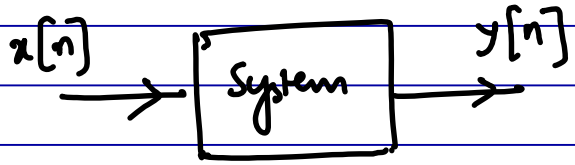
- Find the smallest angular frequency for which the signal with period $N = 8$ is periodic

periodic signal

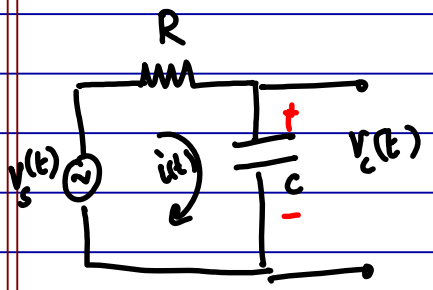
$\cos(\omega_0 n) = \cos[\omega_0(n+8)]$ $\omega_0 = \frac{\bar{\pi}}{4}$

Systems



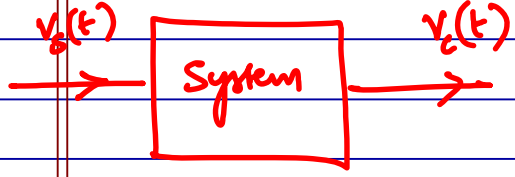




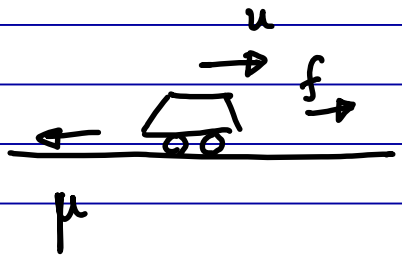


$$i(t) = \frac{V_s(t) - V_c(t)}{R} = C\frac{dV_c}{dt}$$

$$\frac{V_s}{RC} - \frac{V_c}{RC} = \frac{dV_c}{dt}$$



$$\frac{dV_c}{dt} + \frac{V_c}{RC} = \frac{V_s}{RC}$$

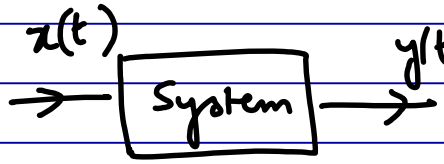


$$f(t) - \mu u = m\frac{du}{dt}$$

$$\frac{du}{dt} + \frac{\mu}{m}u = \frac{f}{m}$$

Classification

(1) CT and DT systems



(2) Memoryless systems

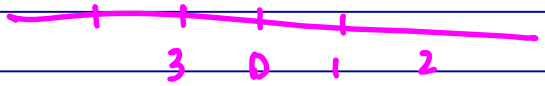
$$y(t) = H\{x(t)\}$$

eg:- Voltage across resistor

H: System operator

System with memory:

$$v_c = \frac{1}{C}\int_{-\infty}^{t} i\,dt$$



Moving average



$$y[n] = \{x[n-1] + x[n] + x[n+1]\}\frac{1}{3}$$

(3) Causal vs non Causal system

✗ Are all causal systems memoryless?

✓ Are all memoryless systems causal?

Output at any time depends only on the input at the present and past times.

Causal system → Non anticipative.

Eg:-1) $y[n] = x[-n]$ Causal? No

for $n = -4$, $y[-4] = x[4]$

Since $y[-4]$ is dependent on the value of x at a later index, $n = 4$, this is not a causal system

(2) $y(t) = x(t)\cos(t+1)$

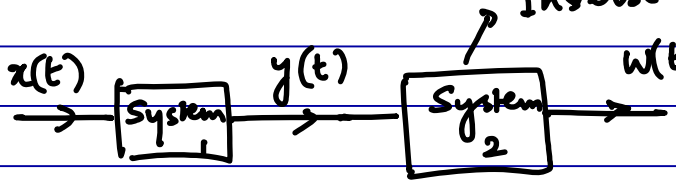
$$g(t) = \cos(t+1)$$

$$y(t) = x(t)\, g(t)$$ Causal

Since the output at any instant depends only on the input at that instant.

(4) **Invertible / Non invertible Systems**

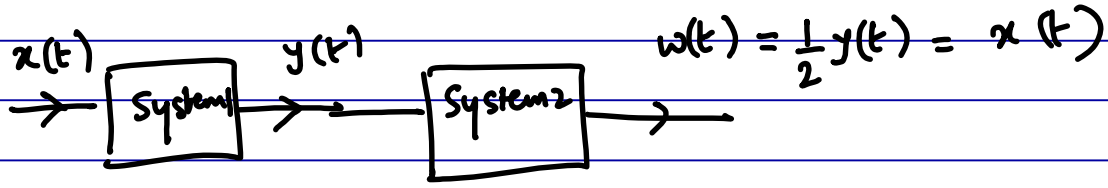
this ~~is~~ ~~that~~ four



System 1 is invertible

Eg:- Error control systems
Modulation

$$y(t) = 2x(t)$$



$$y[n] = \sum_{k=-\infty}^{n} x[k]$$ Accumulator Invertible

$$w[n] = y[n] - y[n-1]$$ }

$$y[n] = 5$$ Non Invertible

(5) **Stability**

For stable systems:- If $|x(t)| \leq B_x < \infty$

$$-B_x \leq x(t) \leq B_x$$

then, $|y(t)| \leq B_y < \infty$

$$-B_y \leq y(t) \leq B_y$$

BIBO stability (Bounded i/p, Bounded o/p)

eg:- $y(t) = t\,x(t)$

Consider bounded i/p eg:- $x(t) = u(t)$

then $y(t) = t\,u(t)$

$y(t)$ is not bounded

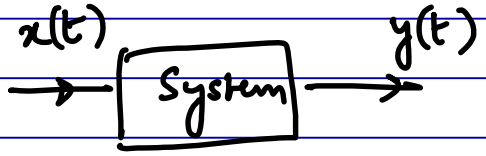
$\Rightarrow$ system not stable

eg:- $y(t) = e^{x(t)}$ BIBO Stable

Let $-B \leq x(t) \leq B$ $(B \neq \infty)$

then $e^{-B} \leq y(t) \leq e^{B}$

VI Time Invariance



① $y_1(t) = H\{x_1(t)\}$

$x_1(t) \rightarrow y_1(t)$

② Define $x_2(t) = x_1(t - t_0)$

③ $y_2(t) = H\{x_2(t)\}$

$= H\{x_1(t-t_0)\}$

④ Check if $y_2(t) = y_1(t - t_0)$

If yes → Time invariant.

$$y(t) = \sin\left(x(t)\right)$$

$$y_1(t) = \sin\left(x_1(t)\right) \quad \text{—}$$

Time shift i/p: $x_2(t) = x_1(t - t_0)$

Apply Transfer fn: $y_2(t) = \sin\left(x_2(t)\right) = \sin\left(x_1(t - t_0)\right)$

Time shift of $y_1(t)$: $y_1(t - t_0) = \sin\left(x_1(t - t_0)\right)$

Identical

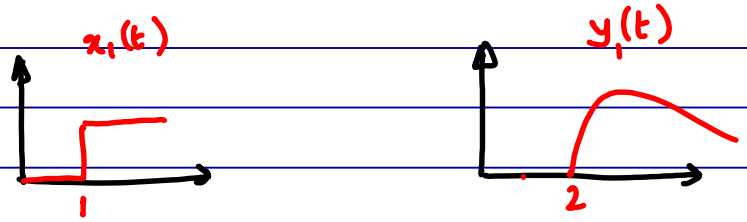
$\Rightarrow$ Time invariant

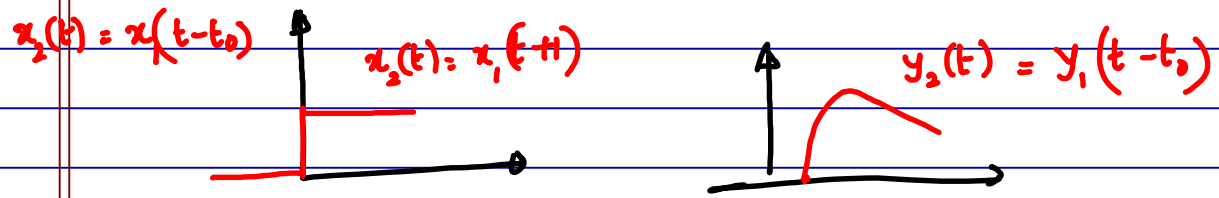
(1) The concept you enjoyed the most

(2) The concept that you did not understand

(3) Any other comments

31-10-18





Eg: $y[n] = n\,x[n]$ Time Invariance?

Let $x_1[n] = \delta[n]$

$y_1[n] = n\,\delta[n] = 0$

(1) Timeshift input: $x_2[n] = x_1[n-1]$

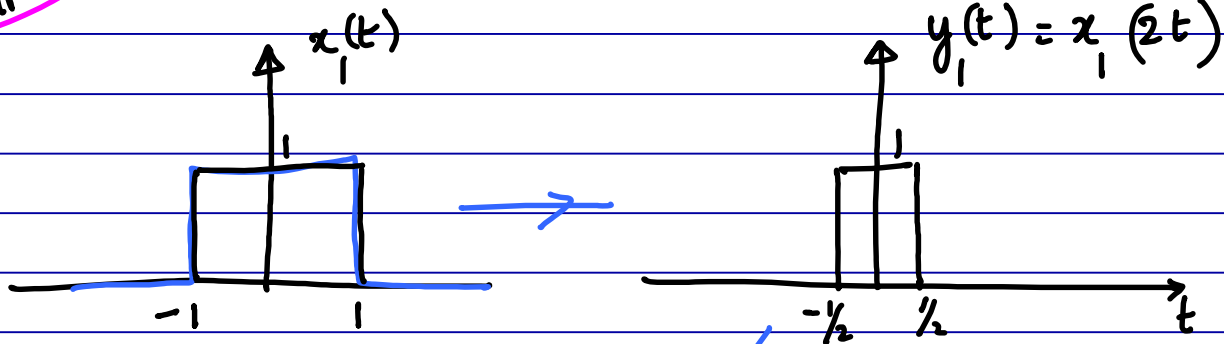
(2) Find the output for time shifted input: $y_2[n] = n\,x_2[n] = n\,x_1[n-1] = n\,\delta[n-1] = 1\cdot\delta[n-1]$

But, $y_1[n-1] = (n-1)\,\delta[n-1] = 0$

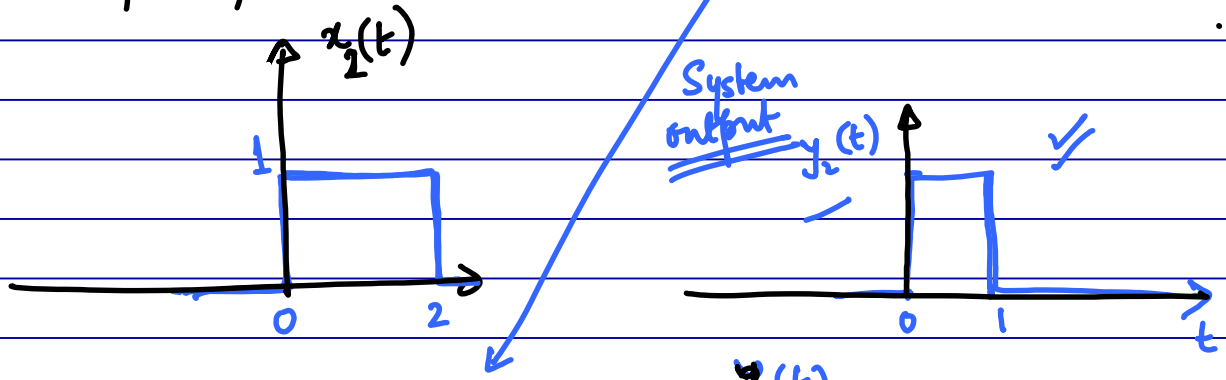
(3) Find time shifted o/p

Not Time invariant

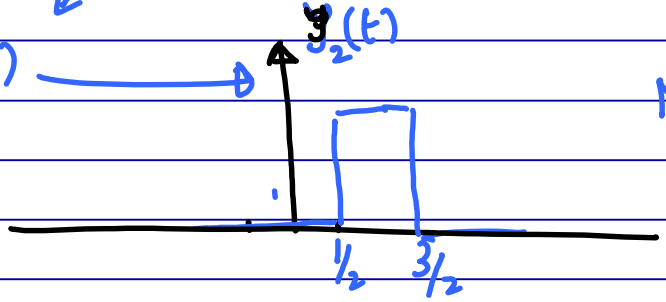
Eg:



(2) $x_2(t) = x_1(t-1)$



Time shifted $y_1(t)$ → $y_2(t)$



Not Time Invari' ant

Eg: (v) Voltage across inductor : $x(t)$

(i) current through inductor : $y(t)$

$V = L\frac{di}{dt}$

$$i = \frac{1}{L}\int_{-\infty}^{t} v\,dt$$

$$y(t) = \frac{1}{L}\int_{-\infty}^{t} x(t)\,dt = \frac{1}{L}\int_{-\infty}^{t} x(\tau)\,d\tau$$

$$\boxed{y_1(t) = \frac{1}{L}\int_{-\infty}^{t} x_1(\tau)\,d\tau}$$

$$x_2(\tau) = x_1(\tau - t_0)$$

Response for time shifted input

$$y_2(t) = \frac{1}{L}\int_{-\infty}^{t} x_2(\tau)\,d\tau = \frac{1}{L}\int_{-\infty}^{t} x_1(\tau - t_0)\,d\tau$$

$\tau - t_0 = u$

$d\tau = du$

$$= \frac{1}{L}\int_{-\infty}^{t-t_0} x_1(u)\,du$$

$$= \frac{1}{L}\int_{-\infty}^{t-t_0} x_1(\tau)\,d\tau$$

Time shifted O/P

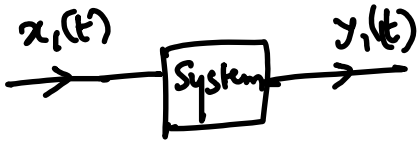
$$y_1(t-t_0) = \frac{1}{L}\int_{-\infty}^{t-t_0} x_1(\tau)\,d\tau$$

O/p of the system for time shifted input = time shifted O/p of the system

$\therefore$ TI

VII Linear Systems

→ additivity
→ homogeneity

If 

(Superposition holds good)

$x_2(t) \rightarrow y_2(t)$

Then, if $a x_1(t) + b x_2(t) \longrightarrow a y_1(t) + b y_2(t)$,

then the system is linear

a, b are complex constants

Eg: $y(t) = t\, x(t)$

$y_1(t) = t\, x_1(t)$, $y_2(t) = t\, x_2(t)$

Let $x_3(t) = a x_1(t) + b x_2(t)$

Output for input $= x_3(t)$ $y_3(t) = t\, x_3(t) = t\left(a x_1(t) + b x_2(t)\right)$

$= a t x_1(t) + b t x_2(t)$

$= a y_1(t) + b y_2(t)$

∴ The system is linear

Time Variant (need to show with the steps)

LTI Systems → linear time invariant

Eg: 1 $y(t) = x^2(t)$

Linearity
Time Invariance
Causality
Memory
Invertible
Stability

2 $y(t) = \frac{dx}{dt}$ → not causal

LTI

Left Diff → Causal

Right Diff → not causal

Has memory
Not invertible

3 $y(t)$: Voltage across capacitor in an RC circuit.

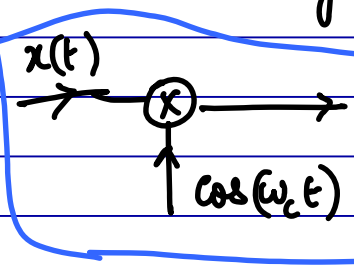
1-2-18

i/p : i(t) o/p v(t) $v(t) = \frac{1}{C}\int_{-\infty}^{t} i(t)dt$

Qn 4 $y[n] = Re\{x[n]\}$ Check for additivity, homogeneity, linearity

Qn 5 $y[n] = 2x[n] + 3$ Check for linearity

Qn 6 Show that moving average is BIBO stable (HW)

Qn 7 

CT systems Modulation

Check for Memory, causality, linearity, time invariance, stability

Not TI, Linear, Stable, No memory, Causal, "

Qn 8 $y[n] = x[-n]$

Scaling in general - not time invariant

Qn 9 Evaluate $\int_{-\infty}^{\infty} e^{x-1} \cos\left(\frac{\pi}{2}(x-5)\right) \delta(x-3)\,dx$ $= e^{2} \int_{-\infty}^{\infty} \delta(x-3)dx = e^{2}$

Qn 10 $\frac{dy}{dt} + 3t\,y(t) = t^2 x(t)$

If $\frac{dy}{dt}$ is small, $y(t) = \frac{t^2}{3}x(t)$

Not stable

✓✓ Check if linear, time invariant, stable ?

Q1 $y(t) = x^2(t)$ $y_1(t) = x_1^2(t)$, $y_2(t) = x_2^2(t)$

Linearity If $x_3(t) = a x_1(t) + b x_2(t)$,

then $y_3(t) = \left(a x_1(t) + b x_2(t)\right)^2 =$

$= a^2 x_1^2(t) + b^2 x_2^2(t) + 2ab\,x_1(t)\,x_2(t)$

$\neq a y_1(t) + b y_2(t)$ × Additivity × Homogeneity

$\therefore$ Not linear

Time Invariance

$$y_1(t) = x_1^2(t)$$

Let $x_2(t) = x_1(t - t_0)$

System o/p for input $= x_2(t)$ | $y_2(t) = x_2^2(t) = x_1^2(t - t_0)$

Time shifted o/p $y_1(t - t_0) = x_1^2(t - t_0)$

$\therefore$ Time Invariant

Memory :- No memory
Since the output at any instant depends only on the value of i/p at that instant

Causality :- Memoryless, hence causal

Invertibility :- Eg, If $x(t) = t$
$x(t) = -t$

gives the same o/p
$\therefore$ not invertible

Stability For bounded inputs ie,

$$-B_x \leq x(t) \leq B_x \quad ; \quad B_x \neq \infty$$

$$0 \leq y(t) \leq B_x^2$$

$y(t)$ is bounded

$\therefore$ Stable

Qn2 $y(t) = \frac{dx}{dt}$ $\quad y_1(t) = \frac{dx_1(t)}{dt} \xrightarrow{\text{ft}\ \Delta t \to 0} \frac{x_1(t+\Delta t) - x_1(t)}{\Delta t}$

Define $x_2(t) = x_1(t-t_0)$

System o/p for input $= x_2(t)$ $\quad y_2(t) = \frac{dx_2(t)}{dt} = \frac{d}{dt}\left(x_1(t-t_0)\right) = \frac{dx_1}{dt}$

Time shifted output : $y_1(t-t_0) = \frac{d\,x_1(t-t_0)}{d(t-t_0)} = \frac{dx_1}{dt}$

$\therefore$ Time invariant

$y(t) = 2x(t)+3$

Not linear

$y(t) = 3t + \frac{5(t)}{2} + \frac{dx}{dt}$

Qn4 $y[n] = Re\{x[n]\}$ $\quad x_1[n] = r_1[n] + j s_1[n]$

$y_1[n] = r_1[n]$ $\quad x_2[n] = r_2[n] + j s_2[n]$

$y_2[n] = r_2[n]$

Additivity : $x_3[n] = x_1[n] + x_2[n]$

$Re\{x_3[n]\}$

$y_3[n] = r_1[n] + r_2[n] = y_1[n] + y_2[n]$

Additivity

Homogeneity

$$x_3[n] = a\,x_1[n]$$

$$y_3[n] = \mathrm{Re}\{x_3[n]\} = \mathrm{Re}\{a\,x_1[n]\}$$

$$= \mathrm{Re}\{a\,(x_1[n] + j\,s_1[n])\}$$

Let $a = j$ $\quad y_3[n] = -s_1[n] \neq a\,y_1[n]$

$$a\,y_1[n] = j\,\mathrm{Re}\{x_1[n]\}$$

$$= j\,r_1[n]$$

$\therefore$ Not homogeneous

$\therefore$ Not linear

$$y_1(t) = x_1(t)\cos(\omega_c t)$$

$$x_2(t) = x_1(t - t_0)$$

Sq o/p for input $x(t)$: $\quad y_2(t) = x_2(t)\cos(\omega_c t) = x_1(t-t_0)\cos\omega_c t$

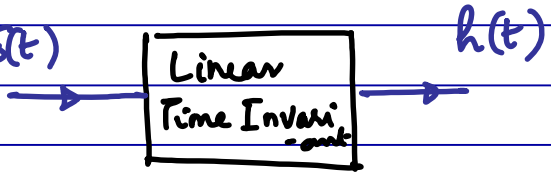
Shifted o/p: $\quad y_1(t-t_0) = x_1(t-t_0)\cdot\cos(\omega_c(t-t_0))$

6-2-18

Time domain Representation Of "LTI Systems"

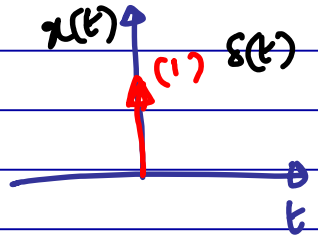
(1) Differential / Difference equations
(2) System function H(s) (4)
(3) Frequency response function H(jω) (3)
(4) Impulse response ✓ (1)
(5) Step response (2)

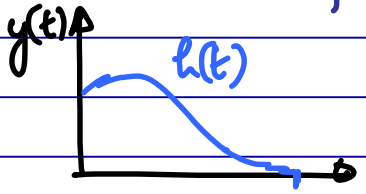
Impulse Response



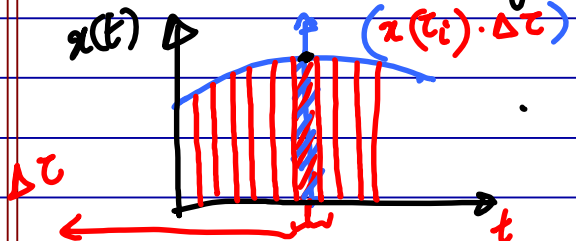
Zero initial Stored energy

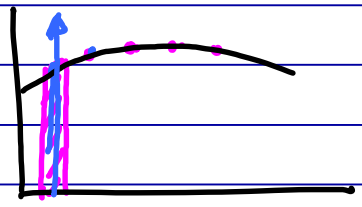
Impulse response : h(t) is the output of the LTI system when the input is δ(t)





Objective :- To find y(t) for a general input x(t)





$$x(t) \simeq \sum_i \underbrace{(x(\tau_i)\,\Delta\tau)\cdot\delta(t-\tau_i)}_{\tau_i}$$

Input → Output

$\delta(t) \longrightarrow h(t)$

$\delta(t-\tau_i) \longrightarrow h(t-\tau_i)$ (Time invariance)

$x(\tau_i)\,\Delta\tau \cdot \delta(t-\tau_i) \longrightarrow x(\tau_i)\,\Delta\tau \cdot h(t-\tau_i)$ (homogeneity)

Let $\Delta\tau \to 0$

$\sum_i x(\tau_i)\,\Delta\tau \cdot \delta(t-\tau_i) \longrightarrow \sum_i x(\tau_i)\,\Delta\tau \cdot h(t-\tau_i)$ (Additivity)

$x(t) \longrightarrow \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau = y(t)$ (Linearity)

$$\therefore\ y(t) = \int_{\tau=-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau$$

Convolution Integral

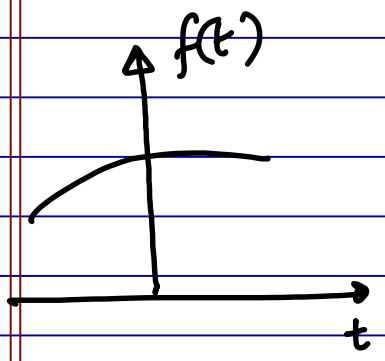
$$y(t) = x(t) * h(t)$$

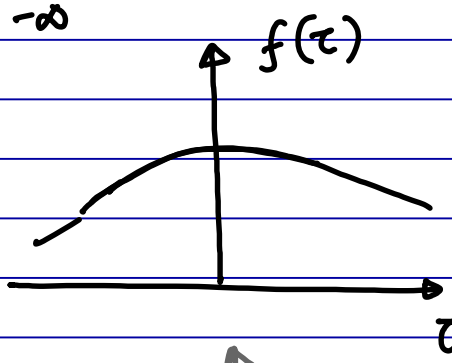
$y(t)$ = $x(t)$ convolved with $h(t)$

HW Check if $x(t) * h(t) = h(t) * x(t)$ Yes

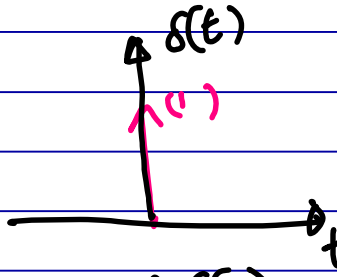
Eg:- Find $y(t) = f(t) * \delta(t) = \int_{-\infty}^{\infty} f(\tau)\,\delta(t-\tau)\,d\tau$

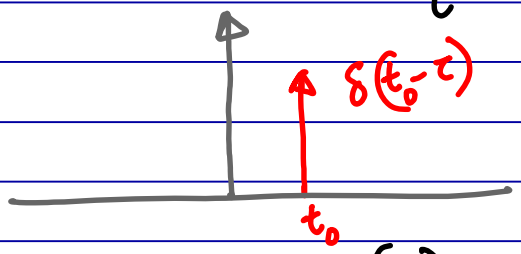
$$y(t) = \int_{-\infty}^{\infty} f(\tau)\,\delta(t-\tau)\,d\tau$$



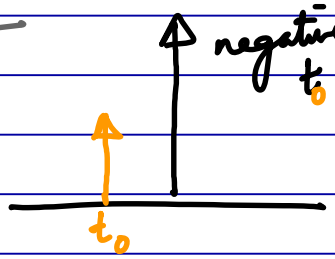


Reflect
Shift
Multiply
add

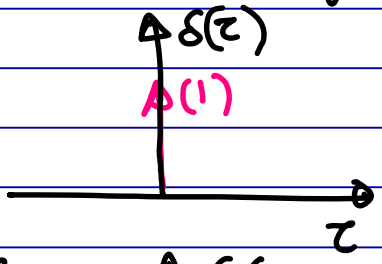




(positive t)



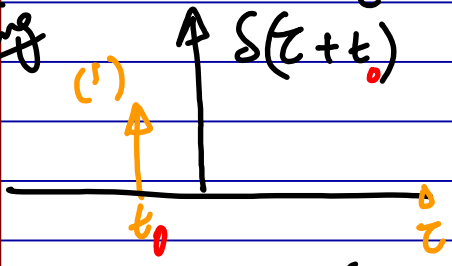
$y(t_0) = f(t_0)$



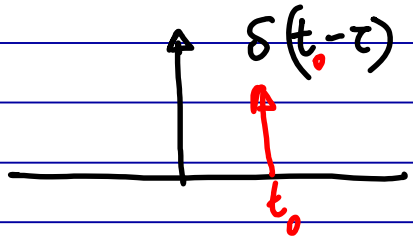
$\delta(t-\tau)$

As t is continuously varied from a negative value to positive value, we get the value of the function at that specific t as the answer

Shifting

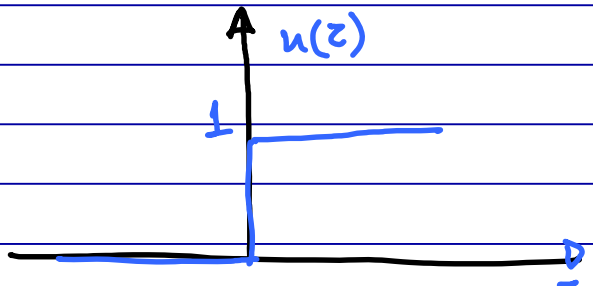


Scaling



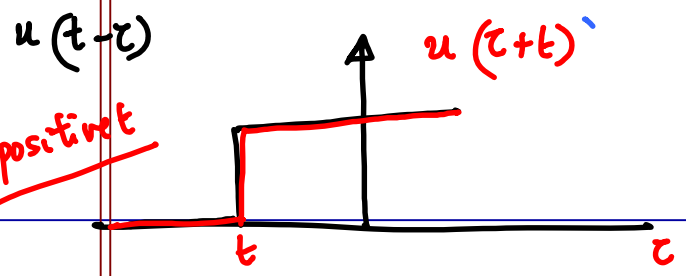
$y(t) = f(t)$

(2)



$u(t) \circledast u(t)$

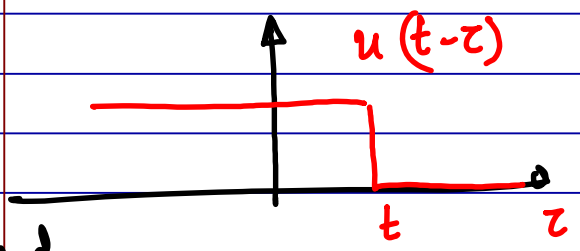
$= \int_{-\infty}^{\infty} u(\tau)\,u(t-\tau)\,d\tau$

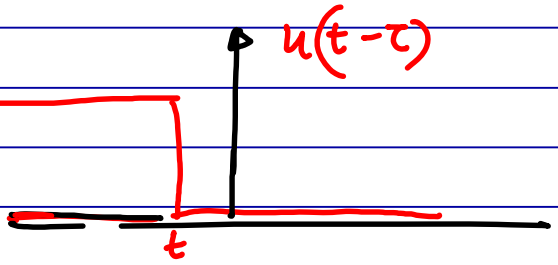


$$\int_{-\infty}^{\infty} u(\tau)\,u(t-\tau)\,d\tau$$

for all t

Negative t



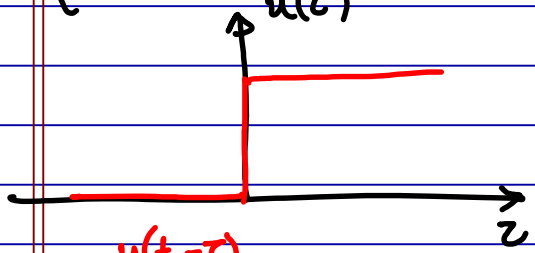


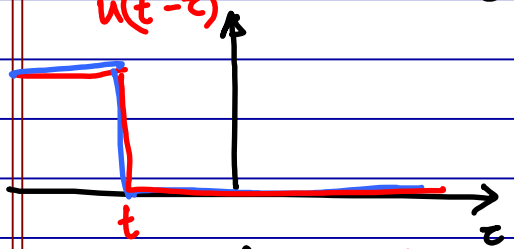
repeated for all t's

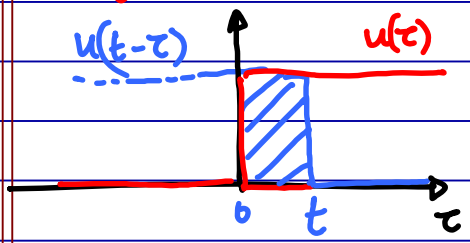
- Step 1: Reflect
- Step 2: Time shift →
- Step 3: Multiply
- Step 4: Add

} Integration

$$g(\tau) = u(\tau)\,u(t-\tau)$$







For $t<0$

$$g(\tau) = 0$$

$$y(t) = \int_{\tau=-\infty}^{\infty} g(\tau)\,d\tau = 0$$

For $t>0$

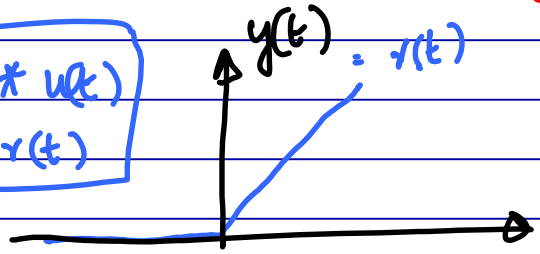
$$g(\tau) = 1 \quad \text{for} \quad 0<\tau<t$$

$$y(t) = \int_{0}^{t} 1\,d\tau = t$$

$$y(t) = 0 \quad \text{for } t<0$$

$$= t \quad \text{for } t>0$$

$u(t) * u(t) = r(t)$



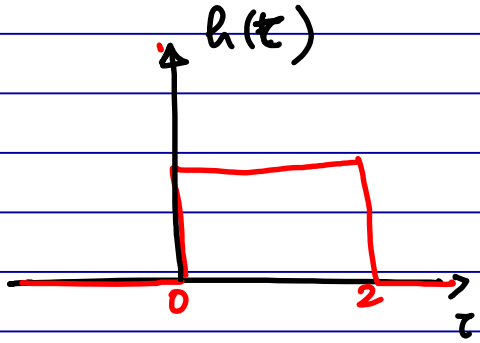
Eg3

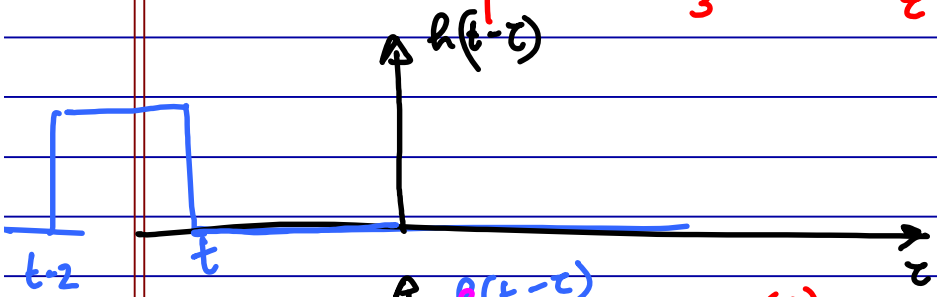
$x(t) = u(t-1) - u(t-3)$

$h(t) = u(t) - u(t-2)$; impulse response of an LTI system

Find $y(t)$



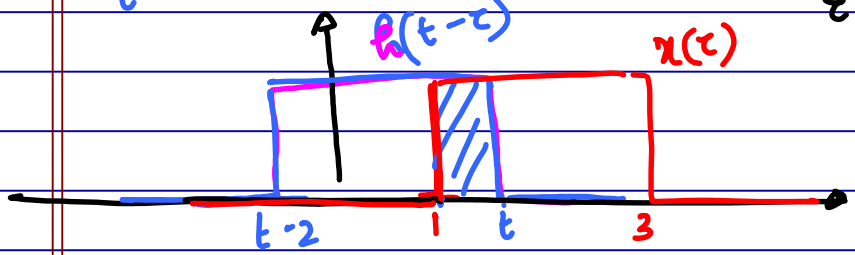




For $t < 1$

$g(\tau) = 0$

$\therefore y(t) = 0$

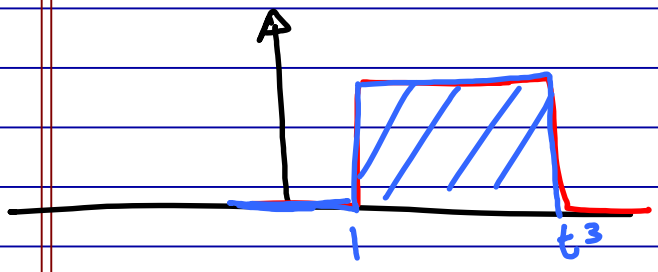


$1 < t < 3$

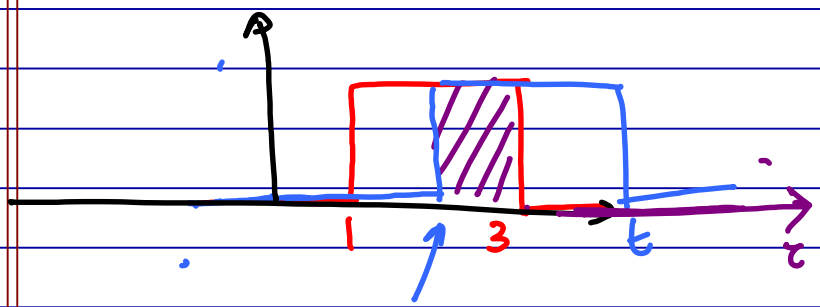
$g(\tau) = 1 \quad 1 < \tau < t$

$y(t) = \int_1^t 1 \, d\tau$

$y(t) = t - 1$



$3 < t < 5$



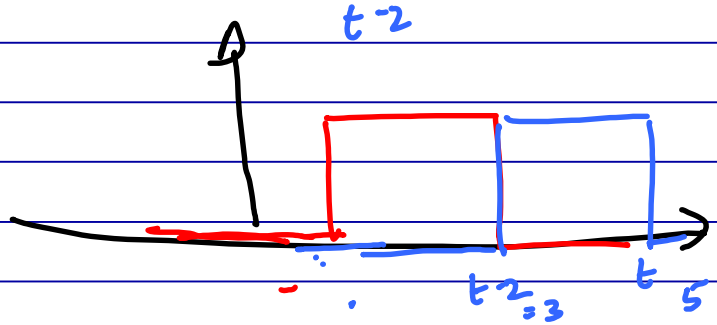
$g(\tau) = 1 \quad t-2 < \tau < 3$

$y(t) = \int_{t-2}^{3} 1 \, d\tau$

$= 3 - (t-2)$

$= 5 - t$

$t > 5$



$g(\tau) = 0$

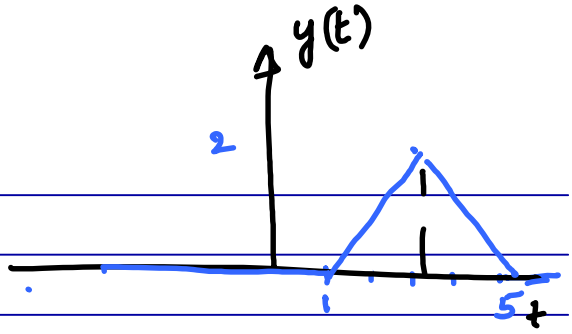
$y(t) = 0$

$y(t) = 0 \quad t < 1$

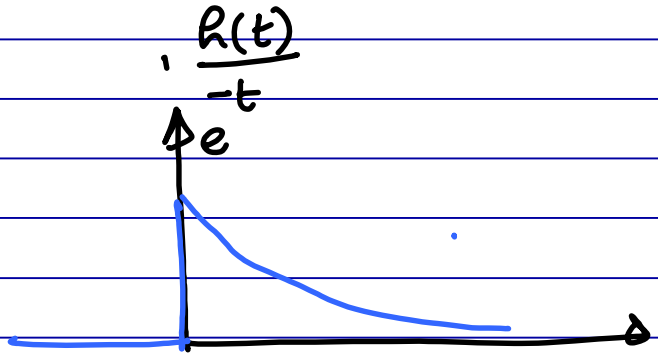
$= t-1 \quad 1 < t < 3$

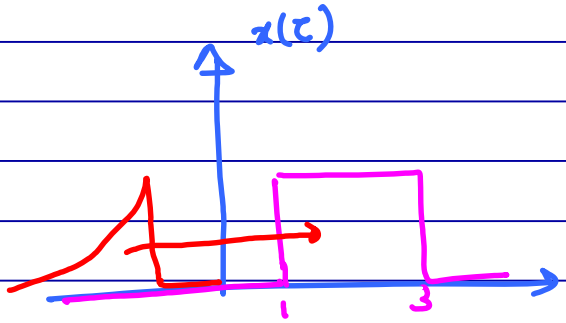
$= 5-t \quad 3 < t < 5$

$= 0 \quad t > 5$

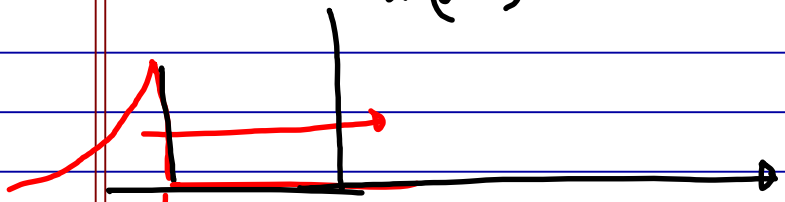


Eg:





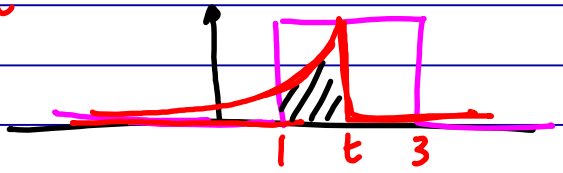
$h(t-\tau)$



$t < 1$

$g(\tau) = 0$

$g(t) \cdot x(\tau) \, h(t-\tau)$



$1 < t < 3$

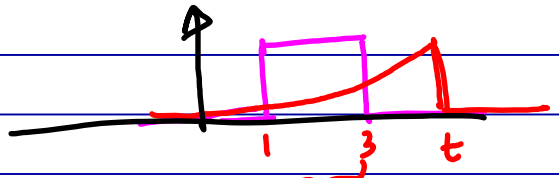
$g(\tau) = e^{-(t-\tau)}$

$1 < \tau < t$

$y(t) = \int_1^t e^{-(t-\tau)} \, d\tau$

$= e^{-t} \left[e^{\tau} \right]_1^t$

$= e^{-t} (e^t - e)$



$t > 3$

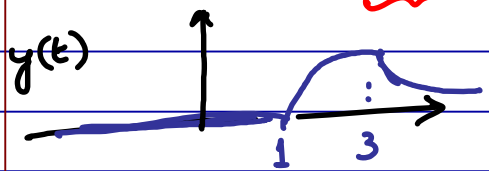
$g(\tau) = 1 \cdot e^{-(t-\tau)}$

$y(t) = \int_1^3 e^{-(t-\tau)} \, d\tau$

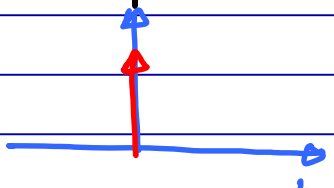
$1 < \tau < 3$

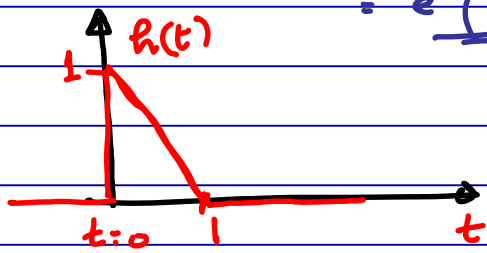
$= e^{-t} \left[e^{\tau} \right]_1^3$

$= e^{-t} (e^3 - e)$

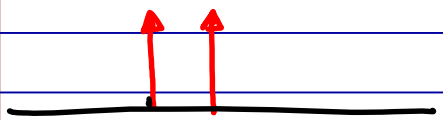


Interpretation

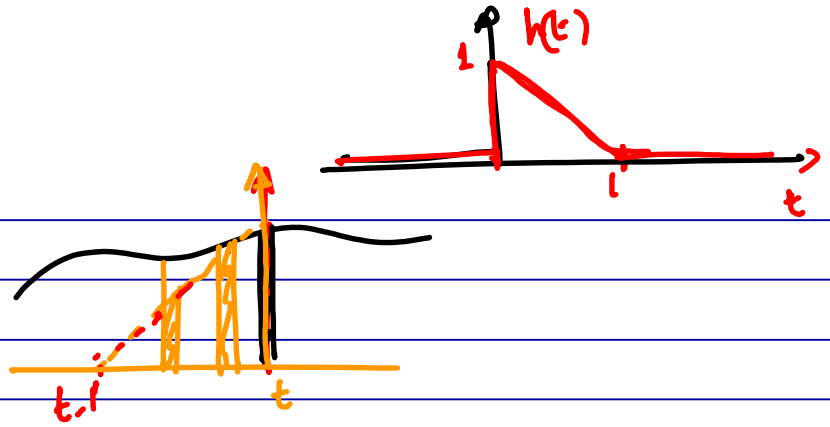


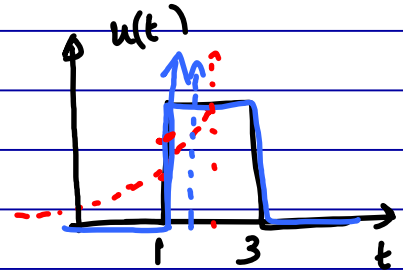


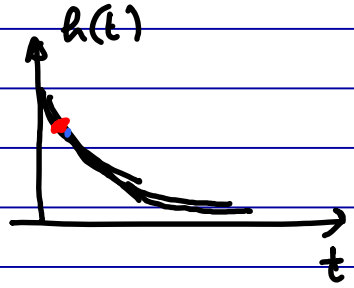




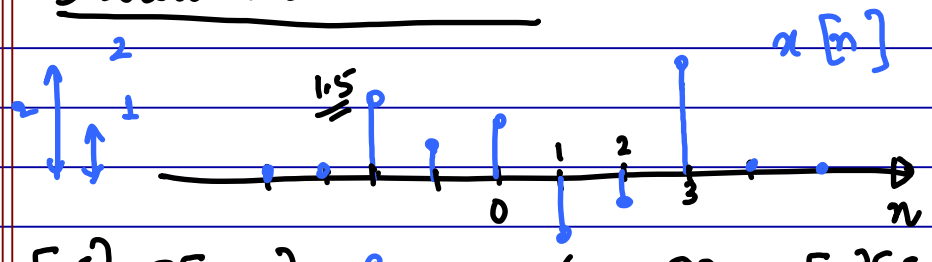
h(t)



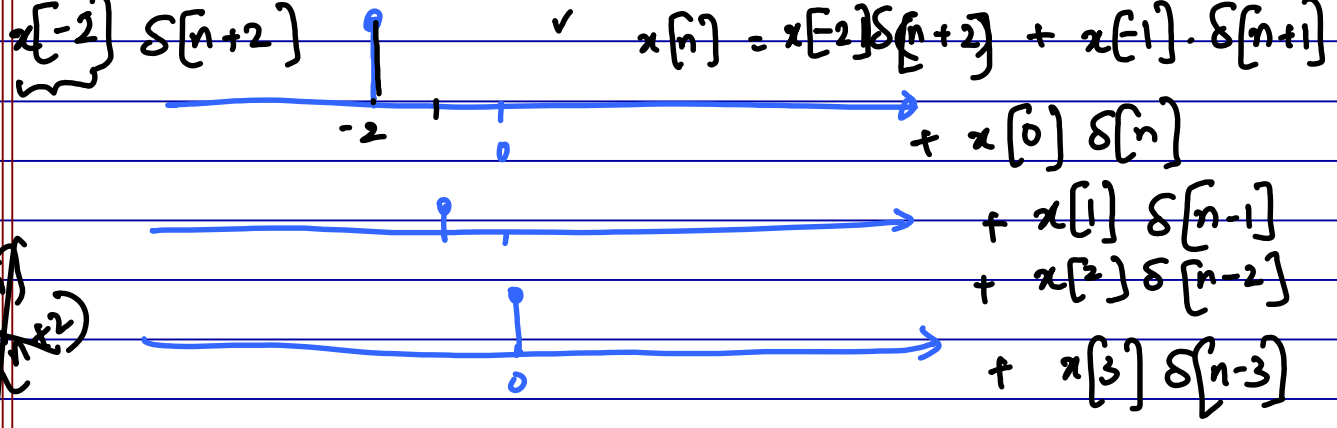


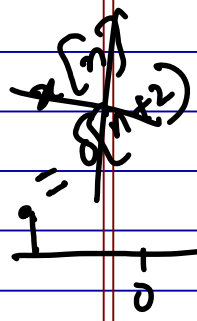


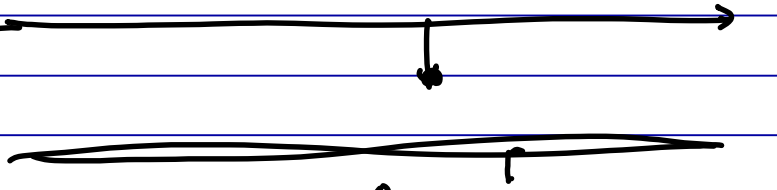
Discrete time convolution



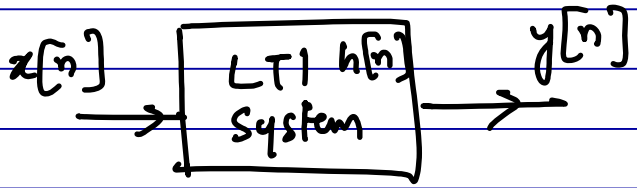
$\delta[n] = 1$







$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$



Impulse Response: $h[n]$ — $y[n] = h[n]$ when $x[n] = \delta[n]$ (i/p / o/p)

$$\delta[n] \longrightarrow h[n]$$

$$\delta[n-k] \longrightarrow h[n-k]$$ Time invariance

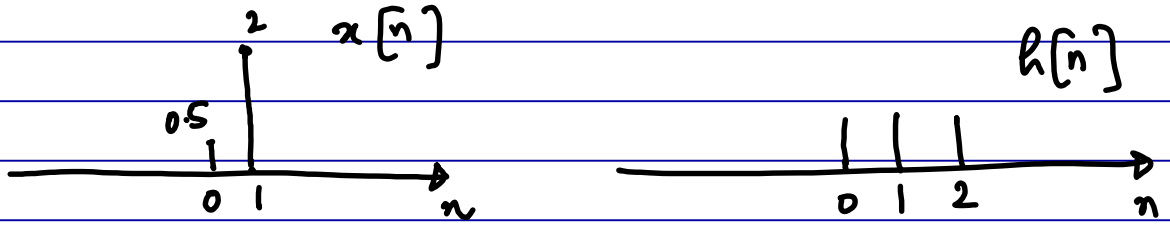
$$x[k]\,\delta[n-k] \longrightarrow x[k]\,h[n-k]$$ Homogeneity

$$\sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k] \longrightarrow \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$

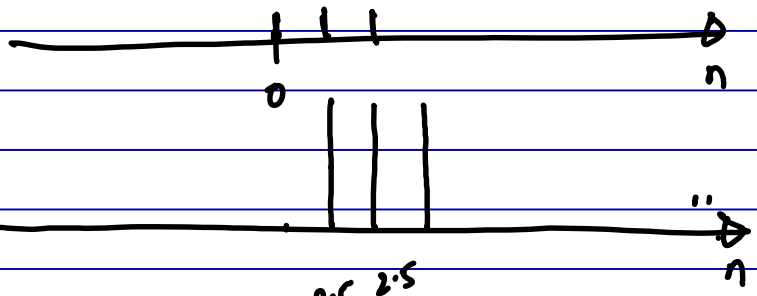
$$x[n] \longrightarrow \sum_{k=-\infty}^{\infty} x[k]\,h[n-k] = y[n]$$ Linearity

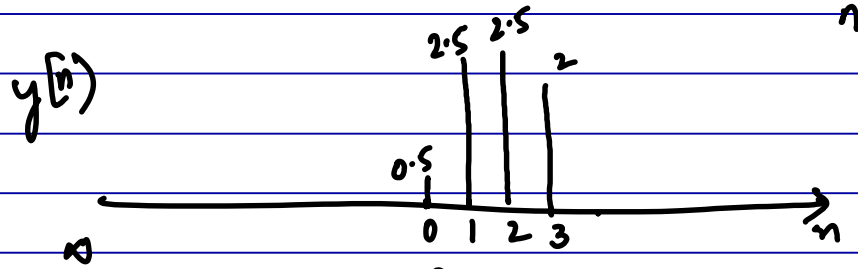
$\sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$ — Convolution sum

Eg:

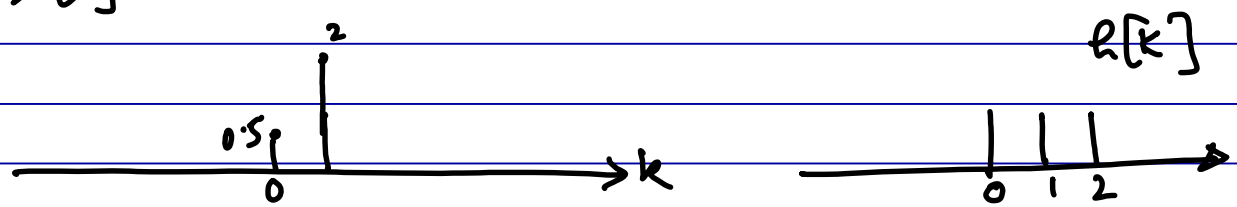


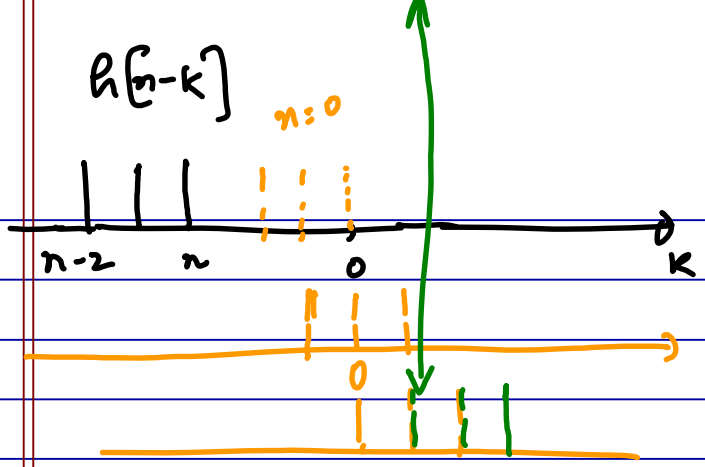
Due to $x[0]$





$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$





$n = 0$

$$g[k] = x[k]\,h[n-k]$$

$$y[n] = \sum_{k=0}^{\infty} g[k]$$

$= 0.5$

$n = 1$

$g[k]$ exists for $k = 0, 1$

$$y[n] = \sum_{k=0,1} g[k]$$

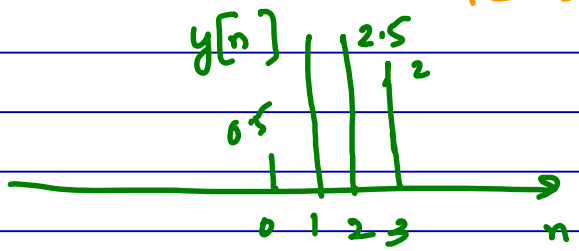
$0.5 + 2 = 2.5$

$n = 2$

$y[n] = 2.5$

$n = 3$

$y[n] = 2$

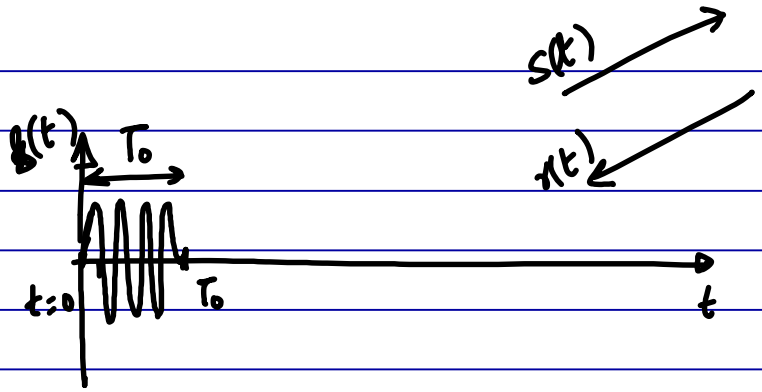


Output $y[n] = \frac{1}{4}\{x[n] + x[n-1] + x[n-2] + x[n-3]\}$

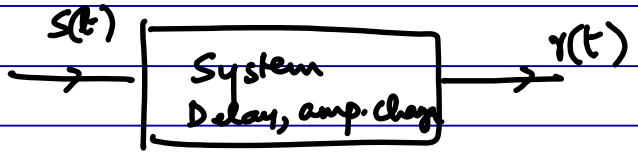
Find $h[n]$

Find $y[n]$ when $x[n] = u[n] - u[n-10]$

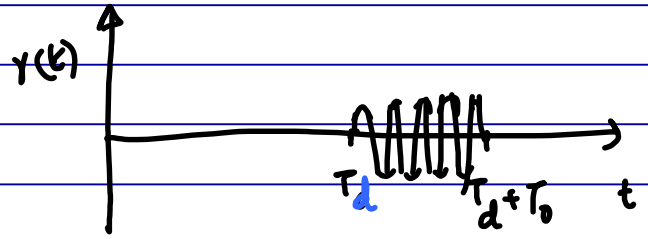
Radars

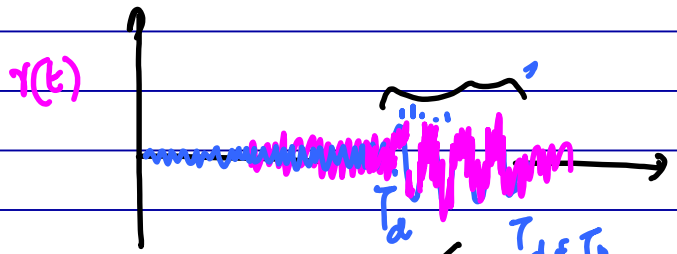


$$s(t) = \sin(\omega_c t) \qquad 0 < t < T_0$$

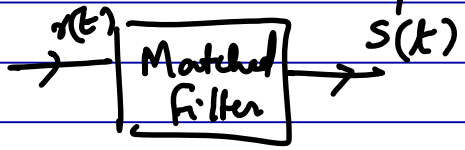


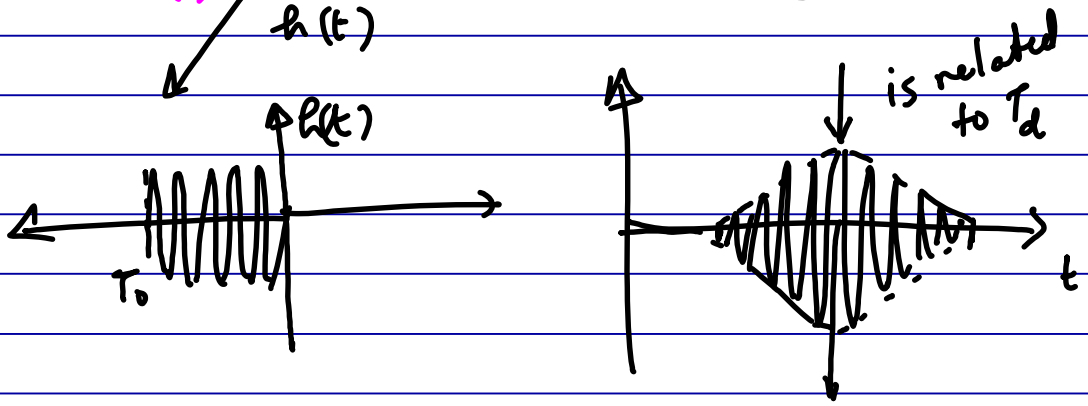
$$h(t) = k\,\delta(t - T_d)$$



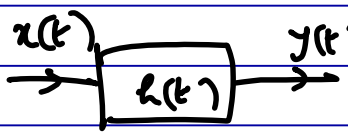


$r(t) * h(t)$





Properties of Systems (based on Impulse Response)



$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau$$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

$x(t) * \delta(t) = x(t)$

$x(t) * \delta(t-t_0) = x(t-t_0)$

$x(t-t_1) * \delta(t-t_2) = x(t-t_1-t_2)$

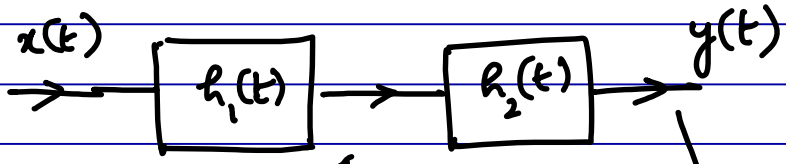
(1) Commutative Property

$$x(t) * h(t) = h(t) * x(t)$$

Prove – change of variables

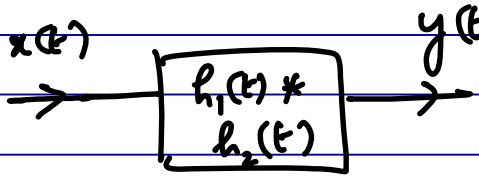
(2) Associati Property

$$\big(x(t) * h_1(t)\big) * h_2(t) = x(t) * \big(h_1(t) * h_2(t)\big)$$



$x(t) * h_1(t)$

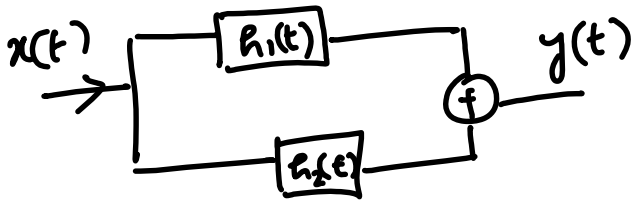
$\big(x(t) * h_1(t)\big) * h_2(t)$

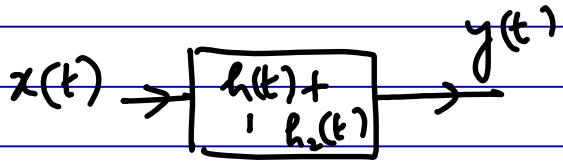


Distributive property

$$x(t) * \big(h_1(t) + h_2(t)\big)$$

$$= x(t) * h_1(t) + x(t) * h_2(t)$$

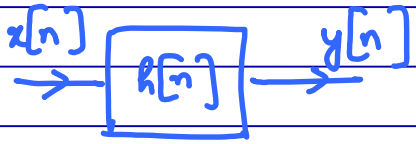




4 What is the impulse response of a memoryless system?

$K\,\delta(t)$

$K\,\delta[n]$



$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$

$$= \cdots\cdots\; x[-2]\,h[n+2] + x[-1]\,h[n+1] + x[0]\,h[n] + x[1]\,h[n-1] + x[2]\,h[n-2] \cdots$$

Output $y[n]$ should depend only on $x[n]$ and not on $x[n-1]$ or $x[n-2]$ or $x[n+1]$ ....

5 What is the impulse response of causal systems?

$$y[n] = \sum_{k=-\infty}^{\infty} x[n-k]\,h[k]$$

$$y[n] = \cdots + h[-2]\,x[n+2] + h[-1]\,x[n+1] + h[0]\,x[n] + h[1]\,x[n-1] + h[2]\,x[n-2] + \cdots$$

For causal system, $y[n]$ must depend only on the past values of $x[n]$

Contributions from terms with $x[n+1]$, $x[n+2]$ 0

$\Rightarrow$ **$h[n]$ is a causal signal**

$\Rightarrow h[-1] = h[-2] = \cdots 0$, $h[k] = 0 \quad \forall\ k < 0$

$h(k)$ is a causal signal for CT causal systems

6. **Stability**

If $|x[n]| \le M_x \quad (M_x < \infty)$

then $|y[n]| \le M_y \quad (M_y < \infty)$

for stable systems

$$y[n] = \sum_{k=-\infty}^{\infty} x[n-k]\,h[k] \qquad |a+b| \le |a| + |b|$$

$$\Rightarrow |y[n]| = \left| \sum_{k=-\infty}^{\infty} x[n-k]\,h[k] \right| \qquad |a||b| = |ab|$$

$$\le \sum_{k=-\infty}^{\infty} |x[n-k]\,h[k]|$$

$$\le \sum_{k=-\infty}^{\infty} |x[n-k]|\,|h[k]|$$

$$|y[k]| \le M_x \sum_{k=-\infty}^{\infty} |h[k]|$$

If $y[k]$ is bounded,

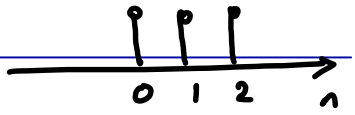
$\Rightarrow \sum_{k=-\infty}^{\infty} |h[k]|$ is finite summable

C.T Systems $\int_{-\infty}^{\infty} |h(t)|\,dt$ must be finite (integrable)

$$\left.\begin{aligned} &\sum_{k=-\infty}^{\infty} |h[k]| < \infty \\ &\int_{-\infty}^{\infty} |h(t)|\,dt < \infty \end{aligned}\right\}$$ Condition for stability

Example $h[n] = u[n]$

$u[n]$



$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\,u[n-k]$$

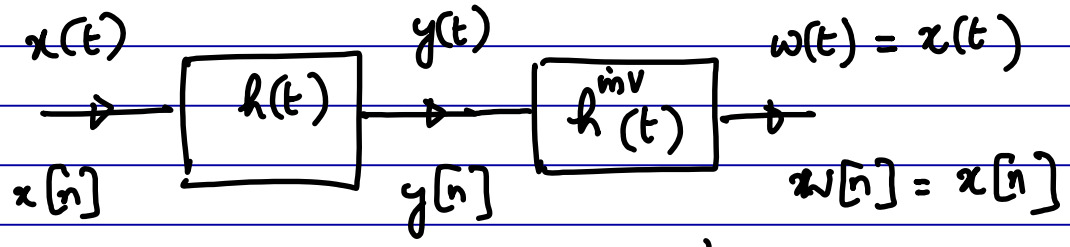
$u[n-k] = 0$ for $k > n$

$= 1$ for $k \le n$

$$y[n] = \sum_{k=-\infty}^{n} x[k]$$ (Accumulator)

$$x[n] = y[n] - y[n-1]$$ (Inverse system)

7 Invertibility



$$x(t) * \left(h(t) * h^{inv}(t)\right) = x(t)$$

$$\Rightarrow \quad h(t) * h^{inv}(t) = \delta(t)$$

$$h[n] * h^{inv}[n] = \delta[n]$$

Systems — **Impulse response**

$$y(t) = x(t-t_0) \longrightarrow h(t) = \delta(t-t_0)$$

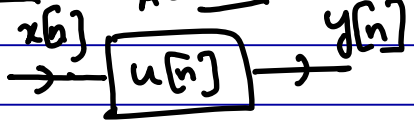
Inverse system — **Impulse response**

$$y(t) = x(t+t_0) \qquad h^{inv}(t) = \delta(t+t_0)$$

$$h(t) * h^{inv}(t) = \delta(t-t_0) * \delta(t+t_0)$$

$$= \delta(t)$$

System — **Accumulator**



$$x[n] = y[n] - y[n-1]$$

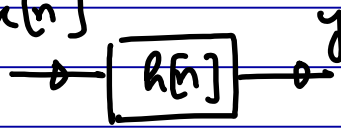
$$h[n] = u[n] \qquad h^{inv}[n] = \delta[n] - \delta[n-1]$$

$$h[n] * h^{inv}[n] = u[n] * (\delta[n] - \delta[n-1])$$

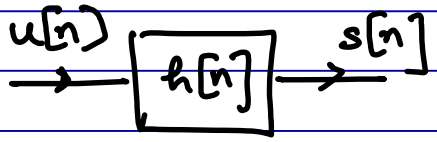
$$= u[n] - u[n-1]$$

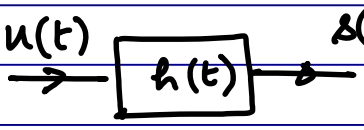
$$= \delta[n]$$

Step Response



Output of the system when the input is unit step function





$$s[n] = u[n] * h[n]$$



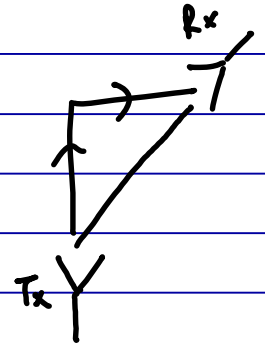
Accumulator

$$s[n] = \sum_{k=-\infty}^{n} h[k] \qquad s(t) = \int_{-\infty}^{t} h(\tau)\, d\tau$$

$$h[n] = s[n] - s[n-1] \qquad h(t) = \frac{ds}{dt}$$

Problems

1. Two path communication model
   echo model

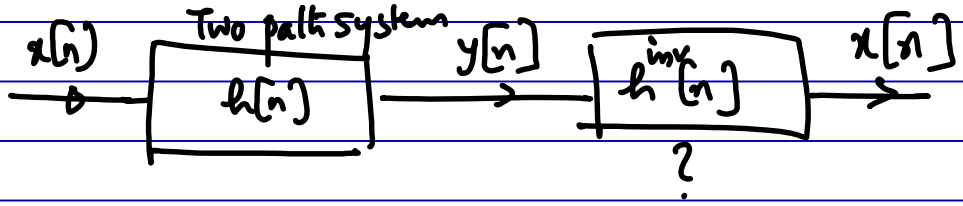


$$y[n] = x[n] + a\,x[n-1]$$

Find the causal inverse system that recovers $x[n]$ from $y[n]$

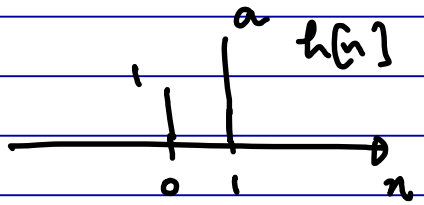
Check if the inverse system is stable, memoryless

Soln



$$h[n] = \delta[n] + a\,\delta[n-1]$$

$$h[0] = 1$$
$$h[1] = a$$
$$h[n] = 0 \qquad n \neq 0,1$$



To find $h^{inv}[n]$

$$h[n] * h^{inv}[n] = \delta[n]$$

$$\sum_{k=-\infty}^{\infty} h[k]\, h^{inv}[n-k] = \delta[n]$$

$$h[0]\, h^{inv}[n] + h[1]\, h^{inv}[n-1] = \delta[n]$$

$n \le 0$ $\quad h^{inv}[n] = 0$

$n = 0$ $\quad h[0]\, h^{inv}[0] + 0 = 1$

$$h^{inv}[0] = 1$$

$n = 1$ $\quad h[0]\, h^{inv}[1] + h[1]\, h^{inv}[0] = 0$

$$h^{inv}[1] + a = 0$$

$$h^{inv}[1] = -a$$

$n = 2$ $\quad h^{inv}[2] = (-a)^2$

$$\vdots \quad h^{inv}[n] = (-a)^n$$

Impulse response of the inverse system: $h^{inv}[n] = (-a)^n u[n]$

$$\neq K\,\delta[n]$$

$\therefore$ Not memoryless

$$\sum_{k=-\infty}^{\infty} h^{inv}[n] = \sum_{k=0}^{\infty} (-a)^n \quad \text{is finite}$$

only for $|a| < 1$

$\Rightarrow$ Stable only for $|a| < 1$

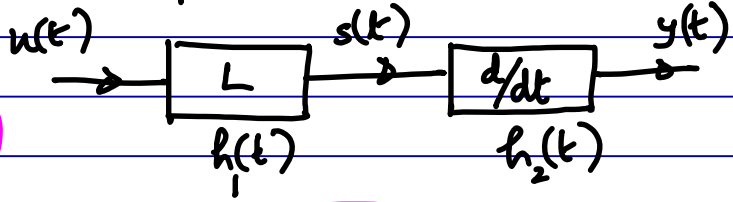
$\Rightarrow$ Stable only when the delayed signal is weaker than original signal

2. Impulse response of an LTI system is $h(t) = e^{-2t} u(t)$

Is the system
Memoryless – Not memoryless
Causal → Yes
Stable — Yes

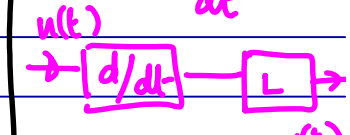
$$\int_0^\infty e^{-2t}\, dt = \frac{1}{2}$$

3. S.T the impulse response of the system L is the derivative of step response

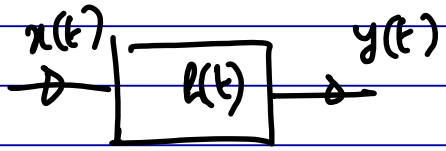


$h_1(t) = \frac{ds}{dt}$

$y(t) = \frac{ds}{dt}$



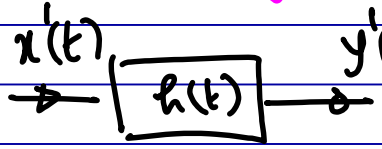
4. Given



$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau)\, d\tau$$

(a) Show that $\frac{dy}{dt} = \frac{dx}{dt} * h(t) = x(t) * \frac{dh}{dt}$

(b)



5. Given $y(t) = \sin \omega_0 t$ for $x(t) = e^{-5t} u(t)$. Find $h(t)$

$x(t) \longrightarrow y(t) \Rightarrow x(t) * h(t) = y(t)$

$x'(t) \longrightarrow y'(t) \Rightarrow x'(t) * h(t) = y'(t)$

$e^{-5t} u(t) * h(t) = y(t)$

$\frac{d}{dt}\left(e^{-5t} u(t)\right) * h(t) = \omega_0 \cos \omega_0 t$

$$\left(-5e^{-5t}u(t) + e^{-5t}\delta(t)\right) * h(t) = \omega_0 \cos\omega_0 t$$

$$-5\underbrace{e^{-5t}u(t) * h(t)} + \underbrace{e^{-5t}\delta(t) * h(t)} = \omega_0 \cos\omega_0 t$$

$$-5y(t) + e^{-5t} h(t) = \omega_0 \cos\omega_0 t$$

$$h(t) = e^{5t}\left(\omega_0 \cos\omega_0 t + 5\sin\omega_0 t\right)$$

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Fourier Series

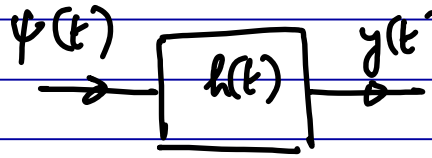
$x(t)$ → sum of impulses (delayed, cont. multiples)

Let $x(t)$ be represented as a linear combination of some basis functions

$$x(t) = a_1\psi_1(t) + a_2\psi_2(t) + a_3\psi_3(t) + \cdots a_n\psi_n(t)$$

(1) { Basis fns should be able to represent the physically realisable signals

(2) Response of LTI system to the basis fn must be 'simple'

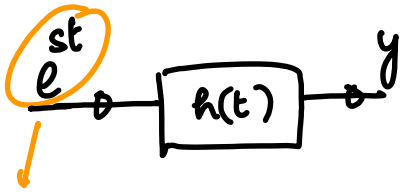


$$y(t) = K\psi(t)$$

$\psi(t)$ → Eigen function of the system

K → complex constant

Eigen value



$H(s)$
System function

$$y(t) = e^{st} * h(t)$$

$$= \int_{-\infty}^{\infty} h(\tau)\, e^{s(t-\tau)}\, d\tau$$

$$= e^{st} \int_{-\infty}^{\infty} h(\tau)\, e^{-s\tau}\, d\tau$$

$$H(s) = \int_{-\infty}^{\infty} h(\tau)\, e^{-s\tau}\, d\tau$$

Eigen fn for Discrete time

$$y(t) = H(s) \cdot e^{st}$$

$\therefore$ Output signal = Constant $\times$ Input signal

$\Rightarrow e^{st}$ is an eigen function of the system with eigen value $H(s)$

Since the response of LTI system is eigen fn in "simple" e^{st},

$$x(t) = a_1 e^{s_1 t} + a_2 e^{s_2 t} + \cdots a_n e^{s_n t}$$

$$x(t) = \sum_{k=1}^{n} a_k e^{s_k t}$$ (Series could be infinite!)

$$y(t) = \sum_{k=1}^{n} a_k H(s_k) e^{s_k t}$$

Issue:-

→ How is arrive at this?

Eg:- $x(t) = \cos 2t$ → $\delta(t-2)$ →

$y(t) = ?$

Express $x(t)$ as a linear combination of eigenfn e^{st}

$$x(t) = \frac{1}{2} e^{j2t} + \frac{1}{2} e^{-j2t}$$

$$y(t) = \frac{1}{2} H(s_1) e^{j2t} + \frac{1}{2} H(s_2) e^{-j2t}$$

$$s_1 = 2j, \quad s_2 = -2j$$

$$H(s_1) = \int_{-\infty}^{\infty} h(\tau) e^{-s_1\tau} d\tau = \int_{-\infty}^{\infty} \delta(\tau-2) e^{-2j\tau} d\tau$$

$$= e^{-4j}$$

$$H(s_2) = e^{4j}$$

$$\therefore y(t) = \frac{1}{2} e^{-4j} e^{j2t} + \frac{1}{2} e^{4j} e^{-j2t}$$

$$= \frac{1}{2} e^{j2(t-2)} + \frac{1}{2} e^{-j2(t-2)}$$

$$= \cos 2(t-2)$$

Periodic Signals

$$x(t+T) = x(t)$$

T is the period

Smallest possible T is the fundamental period

Eg:- $e^{j\omega t}$, $\cos \omega t$

Fundamental period:

$$T = \frac{2\pi}{\omega_0}$$

$e^{j\omega_0 t}$

$1 + e^{j\omega_0 t}$

Is $\cos(2\omega_0 t)$ periodic?

$$a_{-1} e^{-j\omega_0 t} + a_1 e^{j\omega_0 t} + a_2 e^{j2\omega_0 t} + a_3 e^{j3\omega_0 t}$$

In general, $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$ is periodic with period $\frac{2\pi}{\omega_0}$

$\omega_0, -\omega_0 \rightarrow$ fundamental / first harmonic

$2\omega_0, -2\omega_0 \rightarrow$ second harmonic

$n\omega_0, -n\omega_0 \rightarrow n^{th}$ harmonic

Any periodic signal can be represented as

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

Fourier series

$\omega_0 = \frac{2\pi}{T}$

$x(t+T) = x(t)$

a_k = Fourier coefficients

1. Given $x(t) = \sum_{k=-3}^{3} a_k e^{jk \cdot 2\pi t}$

$a_0 = 1, \quad a_{\pm 1} = \frac{1}{4}, \quad a_{\pm 2} = \frac{1}{2} \quad a_{\pm 3} = \frac{1}{3}.$

Find $x(t)$

$$x(t) = \frac{1}{3}\left(e^{-j6\pi t} + e^{j6\pi t}\right) + \frac{1}{2}\left(e^{-j4\pi t} + e^{j4\pi t}\right) + \frac{1}{4}\left(e^{-j2\pi t} + e^{j2\pi t}\right) + 1$$

$$x(t) = \frac{2}{3}\cos(6\pi t) + \cos(4\pi t) + \frac{1}{2}\cos(2\pi t) + 1$$

Fourier coefficients of real signals

$$x(t) = x^*(t)$$

$$\sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} = \sum_{k=-\infty}^{\infty} a_k^* e^{-jk\omega_0 t}$$

identical

RHS : Replace k with $-k$

$$x^*(t) = \sum_{-k=-\infty}^{\infty} a_{-k}^* e^{jk\omega_0 t}$$

$$\therefore a_k = a_{-k}^*$$

$$\boxed{a_k^* = a_{-k}}$$

For real signals

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$= a_0 + \sum_{k=-\infty}^{-1} a_k e^{jk\omega_0 t} + \sum_{k=1}^{\infty} a_k e^{jk\omega_0 t}$$

$$= a_0 + \sum_{k=1}^{\infty} a_k e^{jk\omega_0 t} + a_{-k} e^{-jk\omega_0 t}$$

$$= a_0 + \sum_{k=1}^{\infty} a_k e^{jk\omega_0 t} + a_k^* e^{-jk\omega_0 t}$$

$$x(t) = a_0 + 2\,\mathrm{Re}\sum_{k=1}^{\infty} \left\{ a_k e^{jk\omega_0 t} \right\}$$

Polar form: $a_k = A_k e^{j\theta_k}$

$$\boxed{x(t) = a_0 + 2\sum_{k=1}^{\infty} A_k \cos(k\omega_0 t + \theta_k)}$$

Rectangular form: $a_k = B_k + jC_k$

$$x(t) = a_0 + 2\sum_{k=1}^{\infty} \mathrm{Re}\left\{ (B_k + jC_k) e^{jk\omega_0 t} \right\}$$

Trigonometric form of Fourier series

$$x(t) = a_0 + 2\sum_{k=1}^{\infty} B_k \cos(k\omega_0 t) - C_k \sin(k\omega_0 t)$$

To find a_k

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

n is integer

$$\int_0^T x(t)\, e^{-jn\omega_0 t} = \int_0^T \sum_{k=-\infty}^{\infty} a_k e^{j(k-n)\omega_0 t}$$

$k = n$

$$\int_0^T x(t)\, e^{-jn\omega_0 t}\, dt = \int_0^T \sum_{k=-\infty}^{\infty} a_k \left[\cos((k-n)\omega_0 t) + j\sin((k-n)\omega_0 t)\right] dt$$

$$\int_0^T a_n\, dt = \int_0^T x(t)\, e^{-jn\omega_0 t}\, dt$$

$$a_n = \frac{1}{T}\int_0^T x(t)\, e^{-jn\omega_0 t}\, dt$$

$$\boxed{a_k = \frac{1}{T}\int_0^T x(t)\, e^{-jk\omega_0 t}\, dt}$$

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(1) Example ✓

(2) Convergence Conditions

(3) Properties of Fourier Coeff

limits of integration need not be 0 to T

but, it could be t_1 to $t_1 + T$

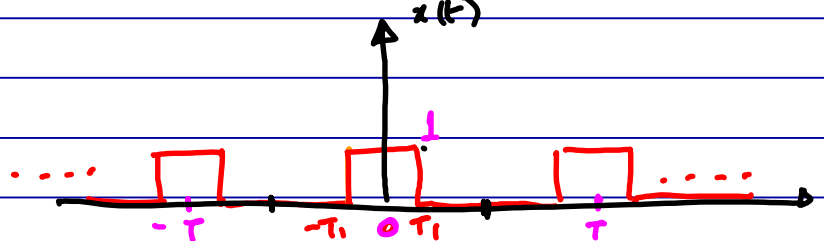
$$a_k = \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt$$

If $x(t+T) = x(t)$,

then $$x(t) = \sum_{k=-\infty}^{\infty} a_k\, e^{jk\omega_0 t}$$ → Synthesis

and $$a_k = \frac{1}{T}\int_T x(t)\, e^{-jk\omega_0 t}\, dt$$ → analysis

Eg



$$x(t+T) = x(t)$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \qquad a_k = \frac{1}{T}\int_T x(t) e^{-jk\omega_0 t}\, dt$$

$$a_0 = \frac{1}{T}\int_{-T/2}^{T/2} x(t)\, dt = \boxed{\frac{1}{T}\int_{-T_1}^{T_1} 1\, dt} = \frac{2T_1}{T}$$

$$a_k = \frac{1}{T}\int_{-T_1}^{T_1} 1\, e^{-jk\omega_0 t}\, dt = \frac{1}{T}\left[\frac{e^{-jk\omega_0 t}}{-jk\omega_0}\right]_{-T_1}^{T_1}$$

$$= \frac{1}{T}\frac{2}{k\omega_0}\left(\frac{e^{jk\omega_0 T_1} - e^{-jk\omega_0 T_1}}{2j}\right)$$

$$= \frac{2}{Tk\omega_0}\cdot \sin(k\omega_0 T_1)$$

$$\omega_0 T = \frac{2\pi}{T} T$$

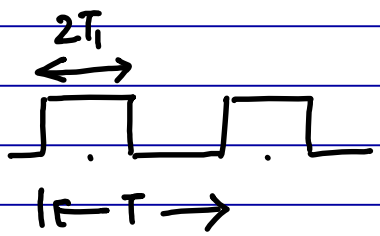
$$a_k = \frac{1}{k\pi}\sin(k\omega_0 T_1)$$

$$a_{+1} = \frac{1}{\pi}\sin(\omega_0 T_1) \qquad a_{-1} = \frac{1}{-\pi}\sin(\omega_0 T_1)$$

$$x(t) = \frac{2T_1}{T} + \underbrace{\frac{1}{\pi}\sin\omega_0 T_1}_{a_{+1}}\left(e^{j\omega_0 t} + \underbrace{e^{-j\omega_0 t}}_{a_{-1}}\right) + \cdots$$

$$x(t) = \frac{2T_1}{T} + \left(\frac{2}{\pi}\sin(\omega_0 T_1)\right)\cos(\omega_0 t) + \left(\frac{2}{2\pi}\sin(2\omega_0 T_1)\right)\cdot\cos(2\omega_0 t) + \cdots$$

$T = 4T_1$



50% Duty cycle

$$\omega_0 T_1 = \frac{2\pi}{T}\cdot T_1 = \frac{\pi}{2}$$

Fourier Coefficients

$$a_k = \frac{1}{k\pi}\sin(k\omega_0 T_1) = \frac{1}{k\pi}\sin\left(\frac{k\pi}{2}\right)$$

$a_0 = \frac{1}{2}$

$a_{\pm 1} = \frac{1}{\pi}$ $\quad$ $a_{\pm 2} = 0$

$a_{\pm 3} = \frac{-1}{3\pi}$ $\quad$ $a_{\pm 4} = 0$

$a_{\pm 5} = \frac{1}{5\pi}$ $\quad$ $a_{\pm 6} = 0$

$a + jb$

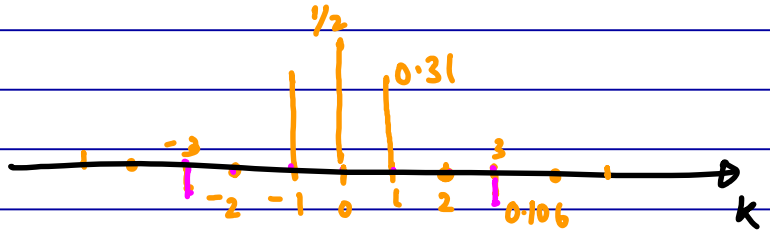
$|a+jb| = \sqrt{a^2+b^2}$

$\angle = \tan^{-1}(b/a)$

Plot (Fourier 'spectrum')

$|a_k|$ vs k $\quad$ $\angle a_k$ vs k

Freq domain

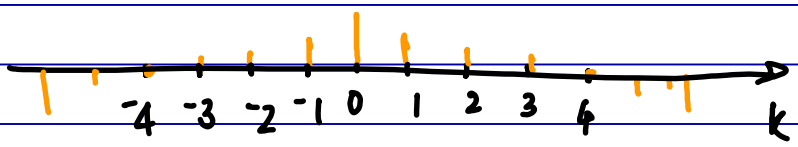


Show the a_k

But show phase spectra with π for (-1)

$T = 8T_1$

- Michelsons harmonic synthesiser



- Gibbs Phenomena

Value at discontinuity $= 1/2$

Convergence Conditions

Dirichlet's conditions

(1) Over any period, $x(t)$ must be absolutely integrable

$$\int_T |x(t)|\,dt < \infty$$

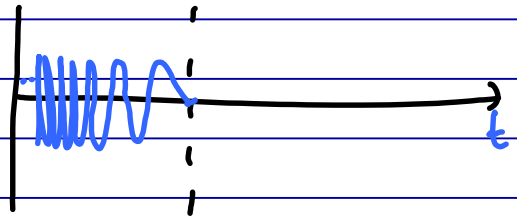
(Coeff must be finite)

$x(t) = \frac{1}{t}$ $\quad 0 < t \leq 1$

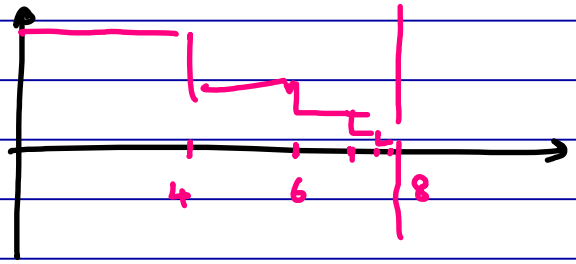


(2) Function must have finite maxima and minima within a period.

$x(t) = \sin\left(\frac{2\pi}{t}\right)$ $\quad 0 < t \leq 1$



(3) In a finite interval of time, there can be only finite number of discontinuities.



Properties of CT Fourier Series

(1) **Linearity**

Given $x(t) \overset{FS}{\longleftrightarrow} a_k$ $\quad x(t+T) = x(t)$

and $y(t) \overset{FS}{\longleftrightarrow} b_k$, with $y(t+T) = y(t)$

$$A\,x(t) + B\,y(t) \overset{FS}{\longleftrightarrow} A\,a_k + B\,b_k = c_k$$

$$c_k = \frac{1}{T}\int_T \left(A\,x(t) + B\,y(t)\right) e^{-jk\omega_0 t}\,dt$$

$$= A\frac{1}{T}\int_T x(t)\,e^{-jk\omega_0 t}\,dt + B\int_T y(t)\,e^{-jk\omega_0 t}\,dt$$

$$= A a_k + B b_k$$

2. **Time Shifting**

$$x(t) \xleftrightarrow{FS} a_k$$

$$x(t-t_0) \xleftrightarrow{FS} b_k = a_k e^{-jk\omega_0 t_0}$$

$$b_k = \frac{1}{T}\int_T x(t-t_0)\, e^{-jk\omega_0 t}\, dt \qquad t - t_0 = u$$

$$= \frac{1}{T}\int_T x(u)\, e^{-jk\omega_0 (u+t_0)}\, du$$

$$= e^{-jk\omega_0 t_0} \underbrace{\int_T x(u)\, e^{-jk\omega_0 u}\, du}_{} = a_k e^{-jk\omega_0 t_0}$$

3. **Time Scaling**

$$x(t) \xleftrightarrow{FS} a_k$$

$$x(\alpha t) \xleftrightarrow{FS} b_k = ?$$

- $x(t+T) = x(t) \Rightarrow$ Period: T

$x(\alpha t) \rightarrow$ Period: $\frac{T}{\alpha}$ freq: $\alpha\omega_0$

$$\Rightarrow \underline{x(\alpha t)} = \sum_{k=-\infty}^{\infty} b_k e^{jk(\alpha\omega_0)t}$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

FS coeff does not change, fundamental freq changes

(4) Time reversal

$$x(t) \xleftrightarrow{FS} a_k$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \qquad x(-t) \xleftrightarrow{FS} b_k$$

$$\rightarrow x(-t) = \sum_{k=-\infty}^{\infty} a_k e^{-jk\frac{2\pi}{T}t}$$

↓ Replace k with $-k$

$$\rightarrow x(-t) = \sum_{k=-\infty}^{\infty} \underbrace{a_{-k}}\, e^{jk\frac{2\pi}{T}t}$$

$$\Rightarrow \boxed{b_k = a_{-k}}$$

If $x(t)$ is even signal $\quad x(-t) = x(t)$

$$a_{-k} = a_k$$

$\Rightarrow$ Fourier coeff are even

For a real signal : $\quad a_k^* = a_{-k} = a_k$ (even)

HW If $x(t)$ is odd — Fourier coeff ?

Prove Fourier coeff are odd, imaginary.

(5) $\quad x(t) \xleftrightarrow{} a_k \; ; \; y(t) \xleftrightarrow{} b_k$

$$x(t) \times y(t) = \underbrace{\sum_{k=-\infty}^{\infty} c_k e^{jk\omega_0 t}}_{} = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \cdot \sum_{l=-\infty}^{\infty} b_l e^{jl\omega_0 t}$$

Coeff of the kth harmonic ?

$$c_k = \sum_{l=-\infty}^{\infty} a_l \, b_{k-l}$$

Fourier coeff of product of two functions
= convolution of the Fourier coeff-

(1) Point / Topic that was clear
(2) Topic that needs to be discussed again
(3) Any other feedback

Periodic Convolution

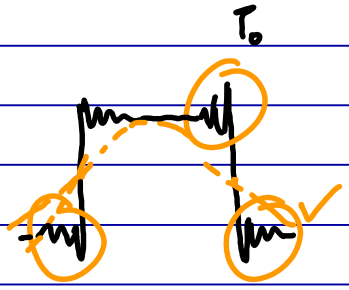
$$x(t) \xleftrightarrow{FS} a_k$$
$$y(t) \xleftrightarrow{} b_k \quad \Big\} \; T_0$$
$$x(t) * y(t) \xleftrightarrow{} T_0 \, a_k \, b_k$$

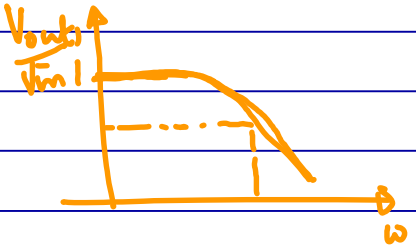
Gibbs Phenomena

$$x(t) = \sum_{k=-\infty}^{\infty} a_k \, e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T_0} \int_{T_0} x(t) \, e^{-jk\omega_0 t} \, dt$$

$f_0 = 500\,Hz$



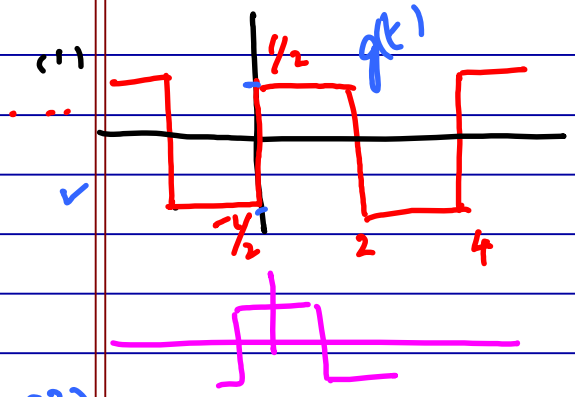


Parsevals Relation

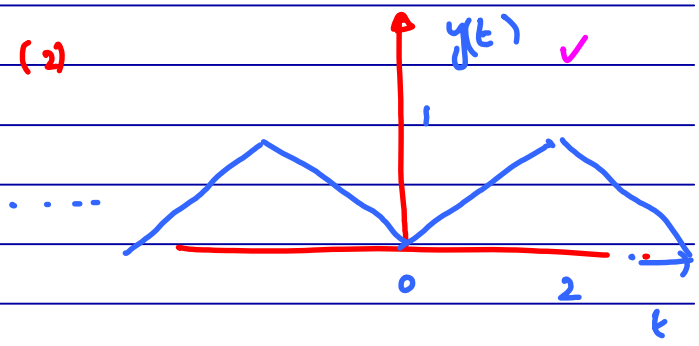
$$\frac{1}{T_0}\int_{T_0} |x(t)|^2\,dt = \sum_{k=-\infty}^{\infty} |a_k|^2$$

Q: Find magnitude & phase spectra

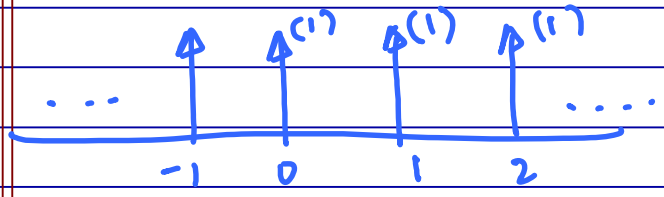
(1)

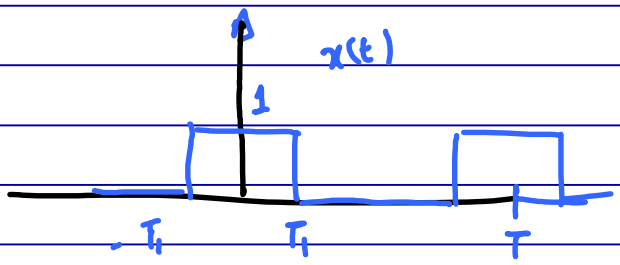


(2)



(3)

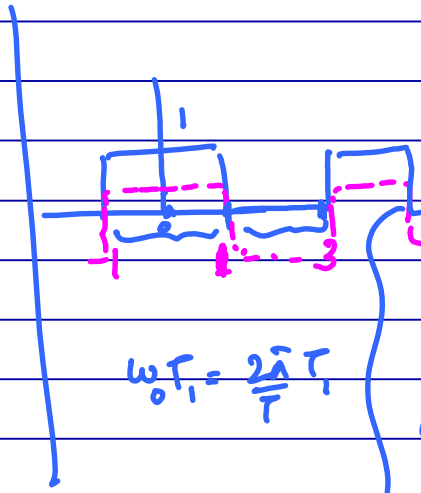




~~g(t) = x~~

$x(t) \longleftrightarrow a_k$

$g(t)$ $x(t-1) - \frac{1}{2} \longleftrightarrow$



$\omega_0 T_1 = \frac{2\pi}{T} T_1$

$$a_k = \frac{\sin(k\omega_0 T_1)}{k\pi}$$

$$a_0 = \frac{2T_1}{T} \qquad \omega_0 = \frac{2\pi}{T}$$

50% duty cycle

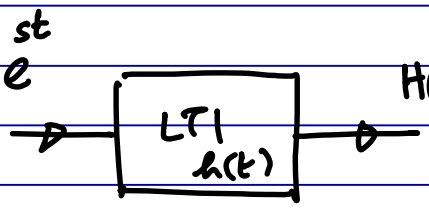
$4T_1 = T$;

$$a_k = \frac{\sin(k\pi/2)}{k\pi} \checkmark$$

$$a_0 = 1/2$$

and implications in

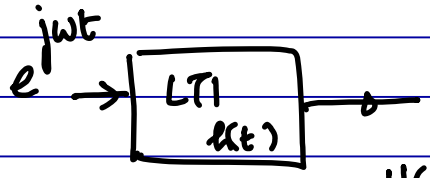
Fourier Series & LTI systems



$H(s)$: System function

$$H(s) = \int_{-\infty}^{\infty} h(\tau) e^{-s\tau} d\tau$$

$s = j\omega$



$H(j\omega)$: Freq response

$$H(j\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau$$

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$= \cdots a_{-2} e^{-j2\omega_0 t} + a_{-1} e^{-j\omega_0 t} + a_0 + a_{+1} e^{j\omega_0 t} + a_{+2} e^{j2\omega_0 t} + \cdots$$

$$y(t) = \cdots a_{-2} H(-j2\omega_0) e^{-j2\omega_0 t} + a_{-1} H(-j\omega_0) e^{-j\omega_0 t} + a_0 H(0) + a_{+1} H(j\omega_0) e^{j\omega_0 t} + \cdots$$

$$y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$$

$H(0)$

$$\int_{-\infty}^{\infty} h(\tau) d\tau$$

Eg: Let $x(t)$ be the input to an LTI system with $h(t) = e^{-t} u(t)$

Non-zero a_k's are

Given $x(t) \xleftrightarrow{FS} a_k$; $a_0 = 1$, $a_{\pm 1} = \frac{1}{4}$

Given $\omega_0 = 2\pi$ $\quad a_{\pm 2} = \frac{1}{2}$, $a_{\pm 3} = \frac{1}{3}$

Find $y(t)$

Soln Find freq response: $H(j\omega) = \frac{1}{1+j\omega}$

$$= \int_0^\infty e^{-t} e^{-j\omega t} dt = \left[\frac{e^{-t(1+j\omega)}}{-(1+j\omega)} \right]_0^\infty$$

$$H(j\omega_0) = \frac{1}{1+j\omega_0}$$

$$H(j2\omega_0) = \frac{1}{1+j2\omega_0}$$

$$H(j3\omega_0) = \frac{1}{1+j3\omega_0}$$

$$y(t) = a_{-3} H(-j3\omega_0) e^{-j3\omega_0 t} + a_{-2} H(-j2\omega_0) e^{-j2\omega_0 t} + a_{-1} H(-j\omega_0) e^{-j\omega_0 t}$$
$$+ a_0 H(j0) + a_1 H(j\omega_0) e^{j\omega_0 t} + a_2 H(j2\omega_0) e^{j2\omega_0 t}$$
$$+ a_3 H(j3\omega_0) e^{j3\omega_0 t}$$

$$= \frac{1}{3}\left(\frac{1}{1-j3\omega_0}\right) e^{-j3\omega_0 t} + \cdots$$

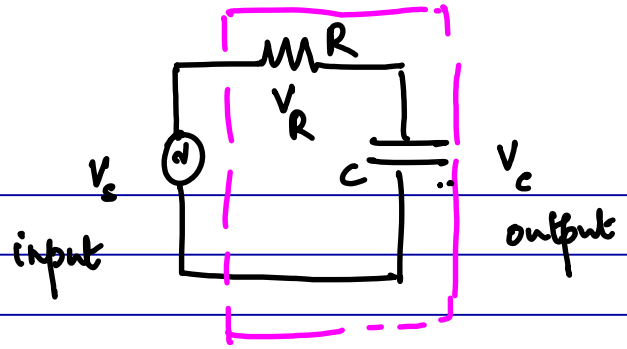
$$b_3 = \frac{1}{3}\frac{1}{1+3j\omega_0} \qquad b_{-3} = \frac{1}{3}\frac{1}{1-j3\omega_0}$$

$$b_2 = \frac{1}{2}\frac{1}{1+2j\omega_0} \qquad b_{-2} = \frac{1}{2}\frac{1}{1-j2\omega_0}$$

$$b_1 = \frac{1}{4}\frac{1}{1+j\omega_0} \qquad b_1 = \frac{1}{4}\frac{1}{1+j\omega_0}$$

$$b_0 = 1$$

Eg:-



$$h(t) = \frac{1}{RC} e^{-t/RC} u(t)$$

$H(j\omega)$ ↑

IFT of $H(j\omega)$

Find freq response

$$V_s = V_R + V_c$$

$$V_R = V_s - V_c \Rightarrow i = \frac{V_s - V_c}{R}$$

$$C \frac{dV_c}{dt} = \frac{V_s - V_c}{R}$$

$$RC \frac{dV_c}{dt} + V_c = V_s$$

If $V_s = e^{j\omega t}$ then $V_c = H(j\omega) e^{j\omega t}$

$$RC\, H(j\omega) \cdot j\omega \cdot e^{j\omega t} + H(j\omega) e^{j\omega t} = e^{j\omega t}$$

$$H(j\omega)\left[1 + j\omega RC\right] e^{j\omega t} = e^{j\omega t}$$

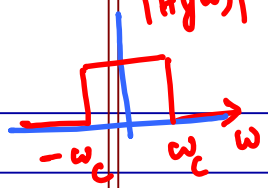
$$H(j\omega) = \frac{1}{1 + j\omega RC} = \frac{1 - j\omega RC}{1 + \omega^2 R^2 C^2}$$

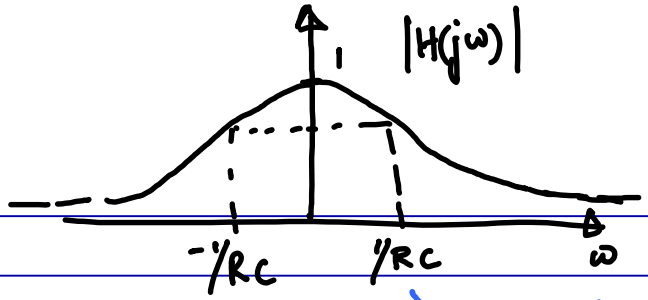
$$|H(j\omega)| = \frac{1}{\sqrt{1 + \omega^2 R^2 C^2}}$$

$$\angle H(j\omega) = \tan^{-1}(-\omega RC)$$

for ω freq

Ideal LPass filter





$\omega = \frac{1}{RC}$

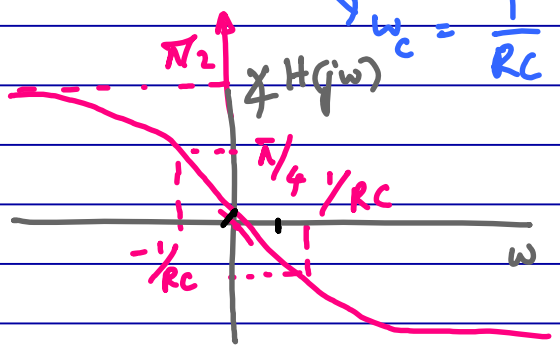
$x(t) \rightarrow \boxed{H(j\omega)} \rightarrow$

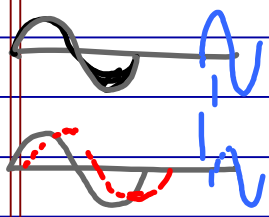
$\sum_k a_k e^{jk\omega_0 t}$

$y(t)$

$\sum_k a_k H(jk\omega_0) e^{jk\omega_0 t}$

$\omega_c = \frac{1}{RC}$

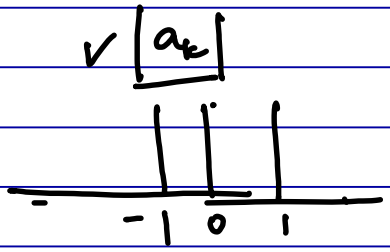


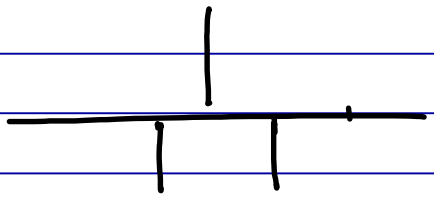


$\angle H(j\omega) =$

a_k's

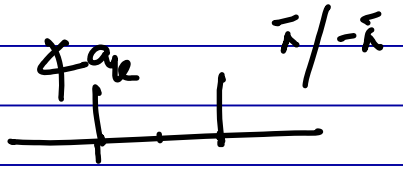
a_k



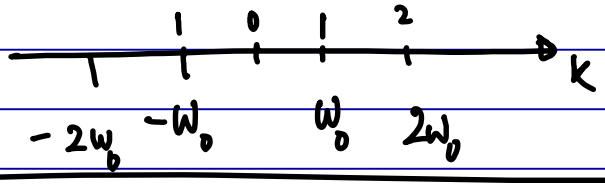


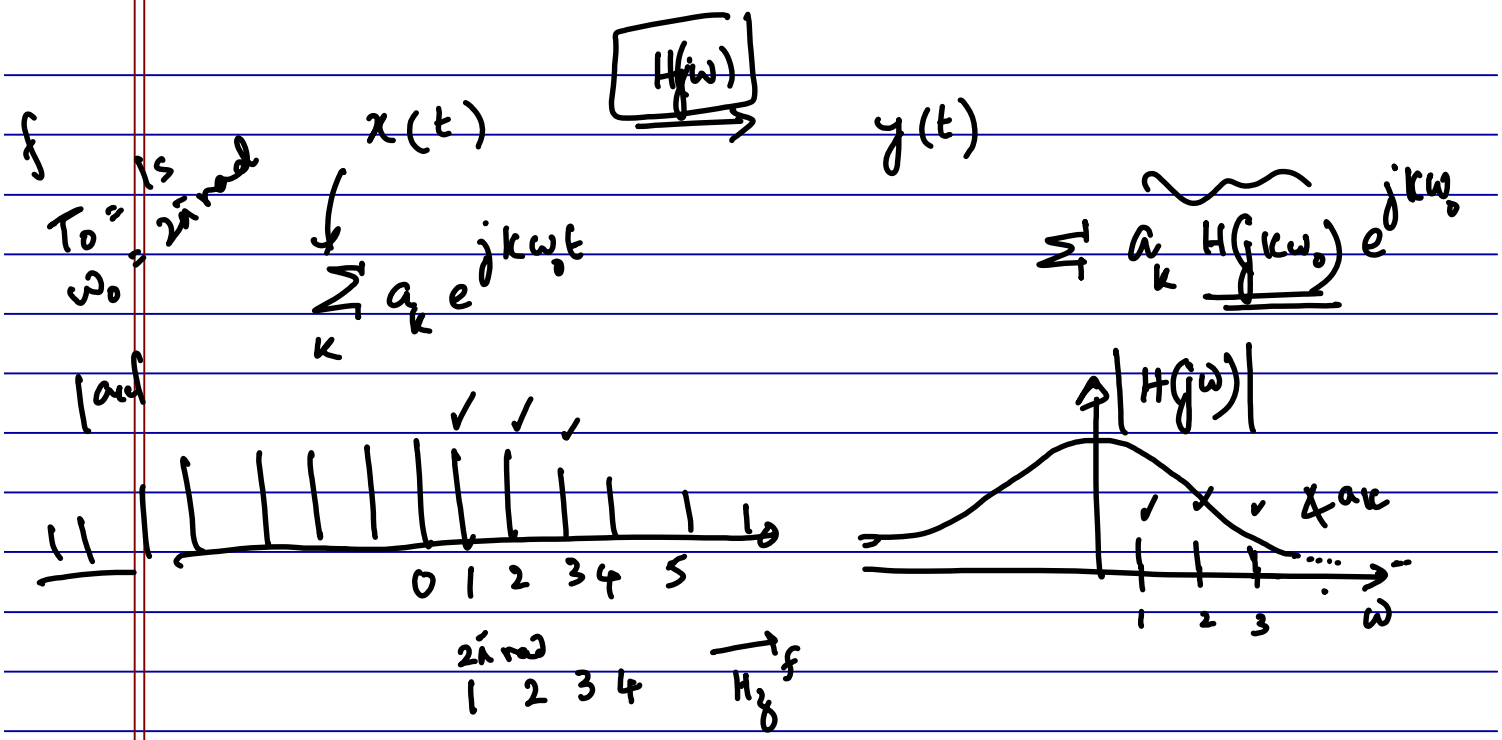
$a_0 = 1$

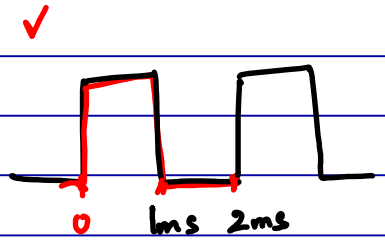
$a_{\pm 1} = \frac{1}{2} e^{j\pi}$



'Spectrum'

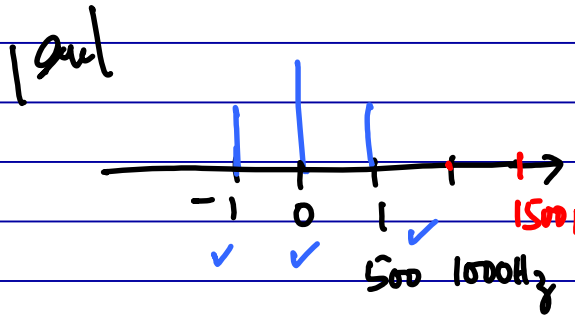






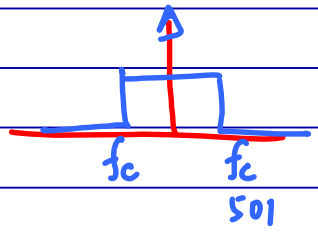
$T = 2ms$

$f = \frac{1}{2ms} = 500\ Hz$



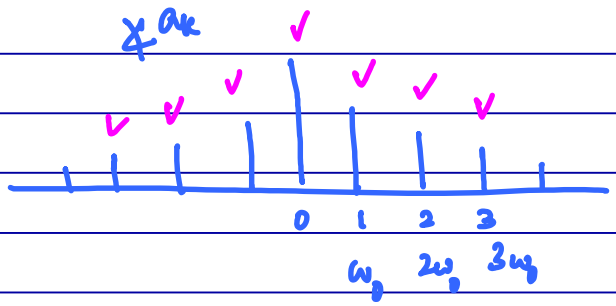
Ideal Low Pass filter

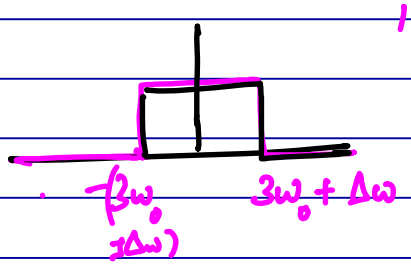
$f_c = 501\ Hz$

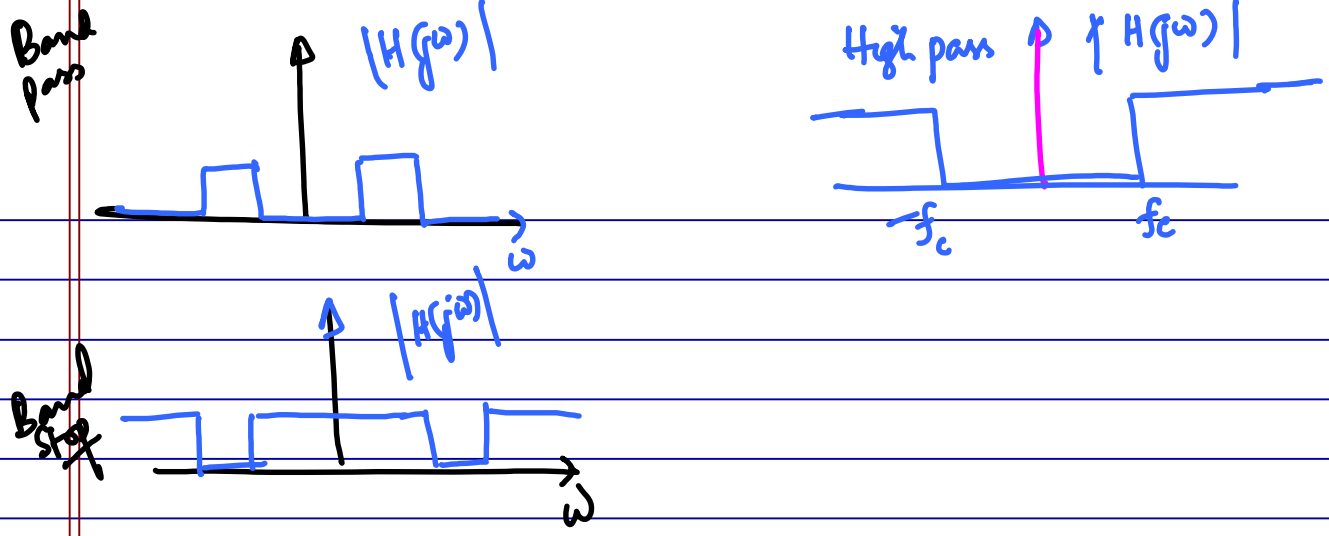


$\sin \omega_0 t = \frac{1}{2j}\left(e^{j\omega_0 t} - e^{-j\omega_0 t}\right)$

eg :-



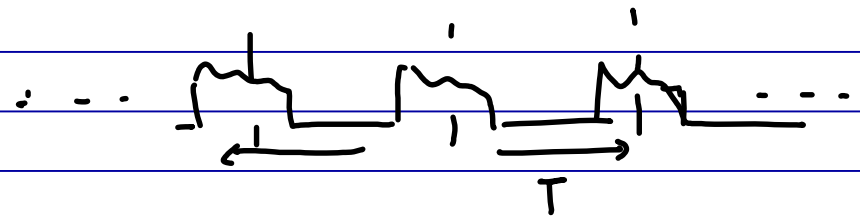




Fourier Tranforms

Periodic signals → Fourier Series of complex exponentials

Aperiodic signals ?



$T \to \infty \Rightarrow \omega_0 = \frac{2\pi}{T} \to 0$

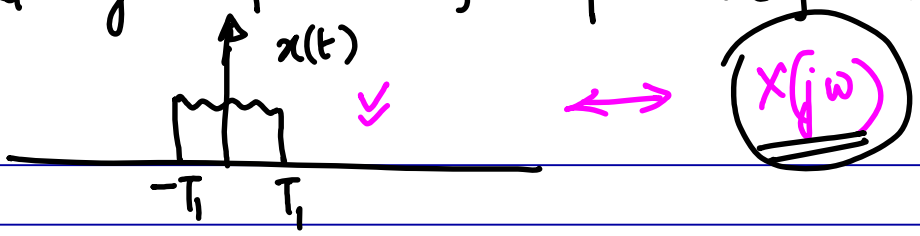
Freq spectrum → discrete description

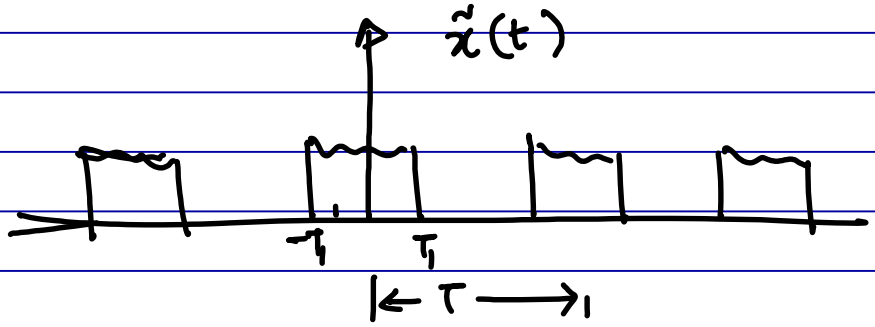
↓

Continuous domain description

Periodic signals → complex exponentials → harmonic relation

Objective : To find eigen expansion for aperiodic fn $x(t)$





$$a_k = \frac{1}{T} X(j\omega)\Big|_{\omega = k\omega_0}$$

$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$a_k = \frac{1}{T}\int_{-T/2}^{T/2} \tilde{x}(t) e^{-jk\omega_0 t}\, dt = \frac{1}{T}\int_{-T/2}^{T/2} x(t) e^{-jk\omega_0 t}\, dt$$

$$a_k = \frac{1}{T}\int_{-\infty}^{\infty} x(t) e^{-jk\omega_0 t}\, dt$$

Define $\quad X(j\omega) = \int_{-\infty}^{\infty} x(t) e^{-j\omega t}\, dt \quad \leftarrow \omega = k\omega_0$

$$a_k = \frac{1}{T} X(jk\omega_0)$$

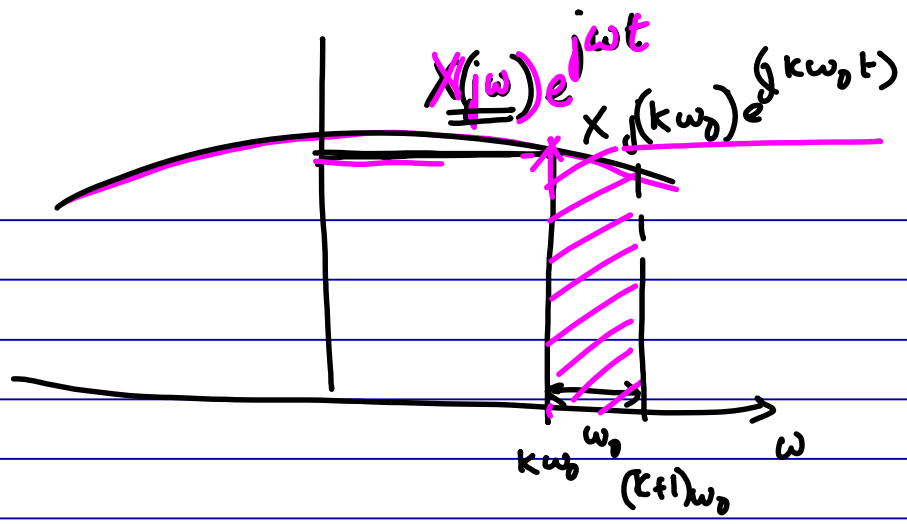
$$\tilde{x}(t) = \sum_{k=-\infty}^{\infty} \frac{1}{T} X(jk\omega_0) e^{jk\omega_0 t}$$

$$\tilde{x}(t) = \frac{1}{2\pi} \sum_{k=-\infty}^{\infty} X(jk\omega_0) e^{jk\omega_0 t}\, \omega_0$$

$T \to \infty$

$\omega_0 \to 0$ $\quad \tilde{x}(t) \to x(t)$ $\quad \omega = k\omega_0$

$$x(t) \quad \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega) e^{j\omega t}\, d\omega$$



$x(t)$ is aperiodic

$x(t)$ → time domain ← $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$

F.T pair: $x(t) \longleftrightarrow X(j\omega)$

$X(j\omega)$ → F.D representation ← $X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$

a_k's are now derivable from $X(j\omega)$

$$a_k = \frac{1}{T} X(j\omega)\Big|_{\omega = k\omega_0}$$

For aperiodic function

$$x(t) \overset{FT}{\longleftrightarrow} X(j\omega)$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

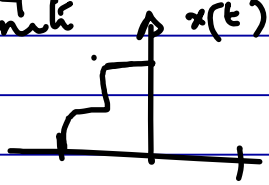
Convergence conditions / Dirichlet Conditions

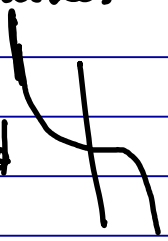
(1) $\int_{-\infty}^{\infty} |x(t)|\, dt < \infty$

$x(t)$ must be absolutely integrable

(2) $x(t)$ must have only finite no: of maxima and minima within a finite interval

(3) $x(t)$ must have only finite number of discontinuities – and each of these discontinuities must be finite





Gibbs phenomena holds good

$\delta(t)$ allowed



eg:-1

$$x(t) = e^{-at} u(t) \quad ; \quad a > 0$$

to satisfy Dirichlet condition

Find Fourier Transform

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$= \int_{0}^{\infty} e^{-at}\, e^{-j\omega t}\, dt$$

$$= \left[\frac{e^{-(a+j\omega)t}}{-(a+j\omega)} \right]_0^{\infty}$$

$\int_{-\infty}^{\infty} |x(t)|\, dt$ $\angle^{\infty}$

$\angle^{\infty}$ $\quad X(j\omega) = \frac{1}{a+j\omega}$

$$\int_0^\infty |e^{-at}|\,dt \quad \int_0^\infty e^{-at} e^{-j\lambda t}\,dt < \infty$$

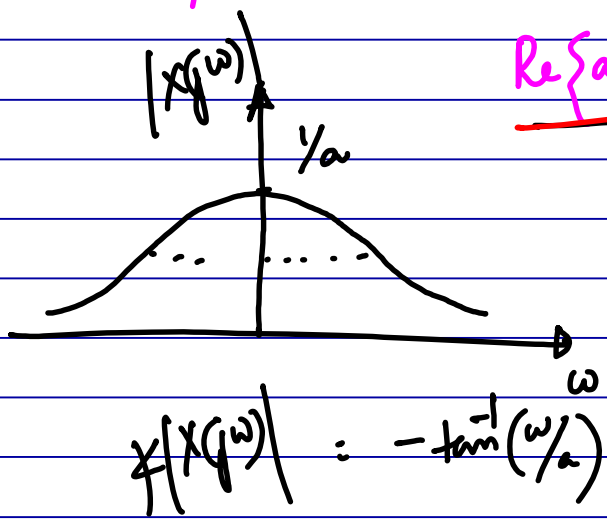
$a = x + j\lambda$

$a = |a| e^{j\theta}$

$x(t) = e^{-at} u(t)$, Given a is complex -

$\mathrm{Re}\{a\} > 0$

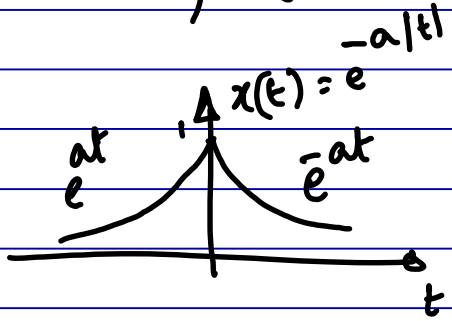
$\frac{1}{\sqrt{a^2+\omega^2}}$



$|X(j\omega)|$

$\not\propto X(j\omega)$

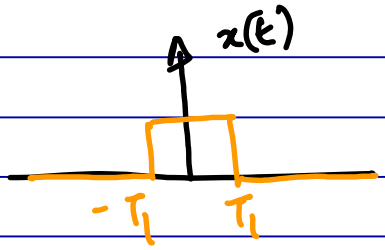
$\not\propto |X(j\omega)| = -\tan^{-1}(\omega/a)$



(2) $x(t) = e^{-a|t|}$

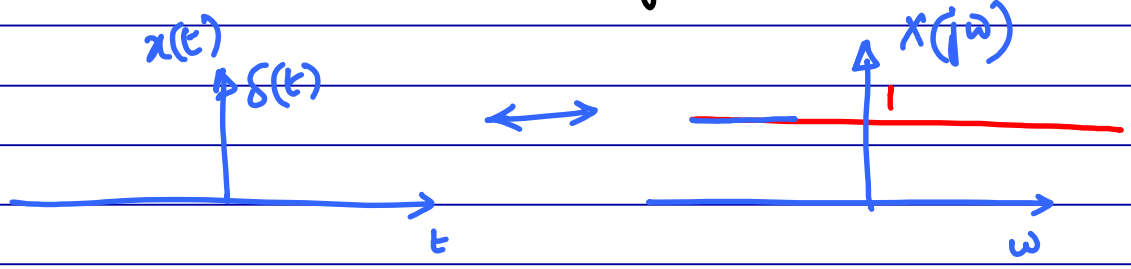
(3) $x(t) = \delta(t)$

(4)

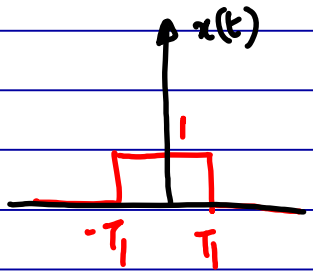


$$X(j\omega) = \frac{2a}{a^2+\omega^2}$$

(3) $x(t) = \delta(t)$ $\quad X(j\omega) = 1$



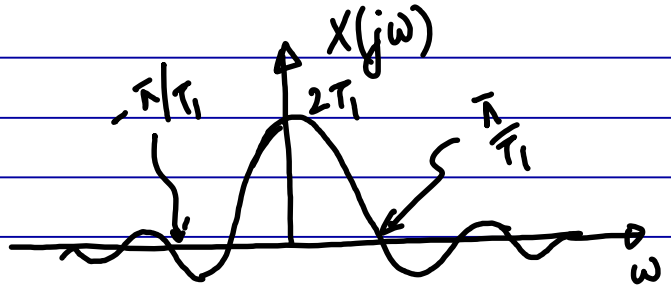
(4)



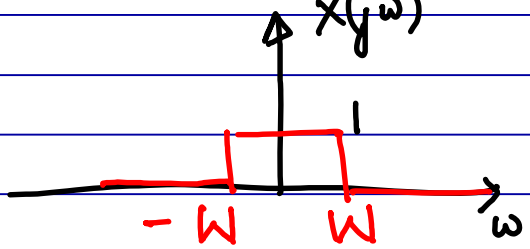
$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\,dt$$

$$= \int_{-T_1}^{T_1} 1 \cdot e^{-j\omega t}\,dt = \left[\frac{e^{-j\omega t}}{-j\omega}\right]_{-T_1}^{T_1}$$

$$X(j\omega) = 2T_1 \left[\frac{\sin(\omega T_1)}{(\omega T_1)}\right] = \frac{e^{j\omega T_1} - e^{-j\omega T_1}}{j\omega} = \frac{2\sin(\omega T_1)}{\omega}$$



(5) Find the inverse Fourier transform of X(jω) given below and plot the same



$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

$$= \frac{1}{2\pi}\int_{-W}^{W} 1 \cdot e^{j\omega t}\, d\omega = \frac{1}{2\pi}\left[\frac{e^{j\omega t}}{jt}\right]_{-W}^{W}$$

$$= \frac{1}{2\pi}\left[\frac{e^{jWt} - e^{-jWt}}{jt}\right]$$

$$= \frac{1}{\pi}\cdot\frac{\sin(Wt)}{t} = \frac{W}{\pi}\left\{\frac{\sin(Wt)}{(Wt)}\right\}$$

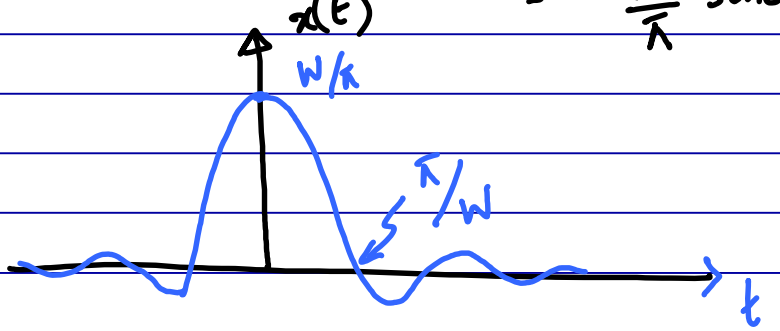
$$= \frac{W}{\pi}\,\mathrm{sinc}\left(\frac{Wt}{\pi}\right)$$

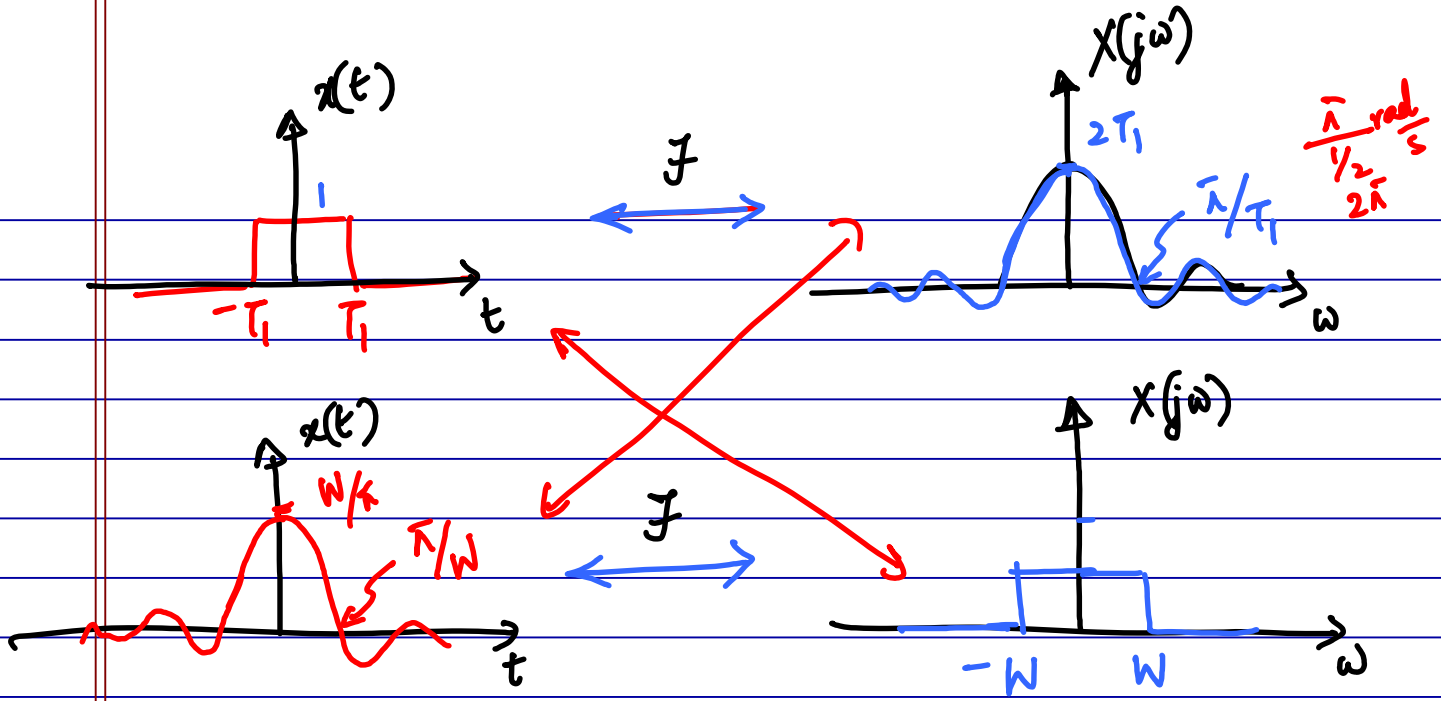
$\mathrm{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$

$\mathrm{sinc}(Wt) = \frac{\sin(\pi Wt)}{\pi Wt}$

$\mathrm{sinc}\left(\frac{Wt}{\pi}\right) = \frac{\sin(\pi \frac{Wt}{\pi})}{\pi \frac{Wt}{\pi}}$

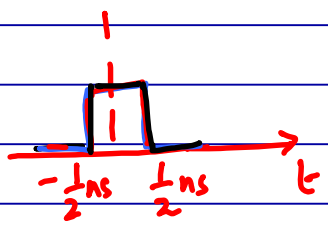
$Wt = \pi$

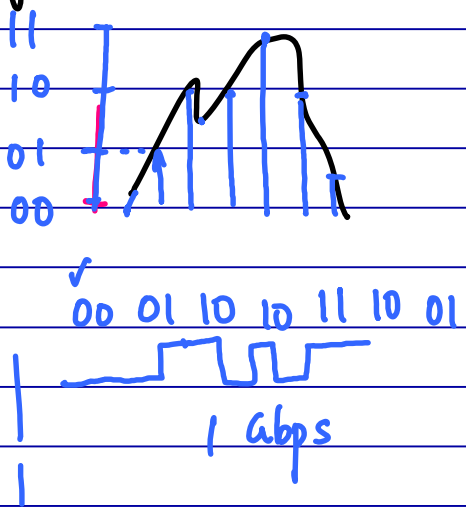




pulse shaping

On-Off key Transmission (Digital)





Properties of Fourier Transform

(1) Linearity

If $x_1(t) \overset{\mathcal{F}}{\longleftrightarrow} X_1(j\omega)$

and $x_2(t) \overset{\mathcal{F}}{\longleftrightarrow} X_2(j\omega)$

then $a\,x_1(t) + b\,x_2(t) \overset{\mathcal{F}}{\longleftrightarrow} a\,X_1(j\omega) + b\,X_2(j\omega)$

(2) Time shifting

If $x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$

then $x(t-t_0) \xleftrightarrow{\mathcal{F}} Y(j\omega) = e^{-j\omega t_0} X(j\omega)$

$$Y(j\omega) = \int_{-\infty}^{\infty} x(t-t_0)\, e^{-j\omega t}\, dt$$

$u = t - t_0$

$$= \int_{-\infty}^{\infty} x(u)\, e^{-j\omega(u+t_0)}\, du$$

$$= e^{-j\omega t_0} \int_{-\infty}^{\infty} x(u)\, e^{-j\omega u}\, du$$

(3) Conjugation and Conjugate symmetry

If $x(t) \xleftrightarrow{\mathcal{F}} X(j\omega)$

then $x^*(t) \longleftrightarrow ? \; X^*(-j\omega)$

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$$

$$X^*(j\omega) = \int_{-\infty}^{\infty} x^*(t)\, e^{j\omega t}\, dt$$

Replace ω with $-\omega$

$$X^*(-j\omega) = \int_{-\infty}^{\infty} x^*(t)\, e^{-j\omega t}\, dt = \mathcal{FT}\{x^*(t)\}$$

If $x(t)$ is real $\quad x^*(t) = x(t)$

$\mathcal{FT}\{x^*(t)\} = X^*(-j\omega) \qquad \mathcal{FT}\{x(t)\} = X(j\omega)$

How are F.T. related?

$$X^*(j\omega) = X(-j\omega)$$

$$X^*(-j\omega) = X(j\omega)$$

Replace ω with $-\omega$, $\quad X^*(j\omega) = X(-j\omega)$

If $x(t)$ is real and even

Let $\quad X(j\omega) = \mathrm{Re}\{X(j\omega)\} + j\,\mathrm{Im}\{X(j\omega)\}$

$$X^*(j\omega) = X(-j\omega)$$

$$\mathrm{Re}\{X(j\omega)\} - j\,\mathrm{Im}\{X(j\omega)\} = \mathrm{Re}\{X(-j\omega)\} + j\,\mathrm{Im}\{X(-j\omega)\}$$

$x(t)$ is real

$$\mathrm{Re}\{X(j\omega)\} = \mathrm{Re}\{X(-j\omega)\}$$ (Real part is even fn of ω)

$$\mathrm{Im}\{X(j\omega)\} = -\mathrm{Im}\{X(-j\omega)\}$$ Imag part is odd fn of ω

If $x(t)$ is real and even

$X(j\omega)$ is real and even

} Proof – Blackboard

If $x(t)$ is real and odd?

$$X(-j\omega) = \int_{-\infty}^{\infty} x(t)\, e^{+j\omega t}\, dt$$

$t = -\tau$

$$= \int_{-\infty}^{\infty} x(-\tau)\, e^{-j\omega\tau}\, d\tau \qquad x(-\tau) = -x(\tau)$$

$$= -\int_{-\infty}^{\infty} x(\tau)\, e^{-j\omega\tau}\, d\tau = -X(j\omega)$$

$\therefore$ If $x(t)$ is odd, $X(-j\omega) = -X(j\omega)$ (odd)

If $x(t)$ is real, $X^*(j\omega) = X(-j\omega)$

$$\therefore \underline{X^*(j\omega)} = X(-j\omega) = \underline{-X(j\omega)}$$

$\Rightarrow$ $X(j\omega)$ must be purely imaginary

$\therefore$ If $x(t)$ is real and odd

$X(j\omega)$ is imaginary and odd

Eg: $x(t) = e^{-a|t|}$

$$= e^{-at}u(t) + e^{at}u(-t)$$

$$x(t) = 2\ \text{Even}\left\{\underbrace{e^{-at}u(t)}\right\}$$

$$y(t) = e^{-at}u(t)$$

$$y(t) \longleftrightarrow Y(j\omega) = \frac{1}{a+j\omega}$$

$$x(t) \longleftrightarrow 2 \cdot \text{Re}\left\{\frac{1}{a+j\omega}\right\} = \underline{\frac{2a}{a^2+\omega^2}}$$

for real $x(t)$

$$x(t) = \underbrace{x_e(t)}_{\text{Re}\{X(j\omega)\}} + \underbrace{x_o(t)}_{\text{Im}\{X(j\omega)\}}$$

$$X(j\omega) = \text{Re}\{X(j\omega)\} + \text{Im}\{X(j\omega)\}$$

(4) Differentiation

$$x(t) \overset{\mathcal{F}.\mathcal{T}}{\longleftrightarrow} X(j\omega)$$

$$y(t) = \frac{d}{dt}(x(t)) \longleftrightarrow j\omega X(j\omega)$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)\, e^{j\omega t}\, d\omega$$

$$y(t) = \frac{dx(t)}{dt} = \frac{1}{2\pi}\int_{-\infty}^{\infty} j\omega X(j\omega)\, e^{j\omega t}\, d\omega$$

Eg:- $\dfrac{dx(t)}{dt} + x(t) = 1$ $\qquad x(t) \longleftrightarrow X(j\omega)$

In Fourier Domain

$$j\omega X(j\omega) + X(j\omega) = \mathcal{F}.\mathcal{T}\{1\} = \mathcal{F}.\mathcal{T}\{1\}$$

$$X(j\omega)(1 + j\omega) = \mathcal{F}.\mathcal{T}\{1\}$$

$$X(j\omega) = \frac{\mathcal{F}.\mathcal{T}\{1\}}{1 + j\omega}$$

Qn Prove Diff property:-

$$x(t) = e^{-at}u(t)$$

$$X(j\omega) = \frac{1}{a + j\omega}$$ } already derived

Show that $\mathcal{F.T}\left\{\frac{d}{dt}\{x(t)\}\right\} = j\omega X(j\omega)$

$$\frac{d}{dt}\{x(t)\} = e^{-at}\delta(t) - a e^{-at}u(t)$$

$$\mathcal{F.T}\left\{\frac{d}{dt}(x(t))\right\} = \mathcal{F.T}\left\{e^{-at}\delta(t)\right\} - a\,\mathcal{F.T}\left\{e^{-at}u(t)\right\}$$

$$= \underbrace{\int_{-\infty}^{\infty} e^{-at}\delta(t)\,e^{-j\omega t}\,dt}_{} - a\int_{-\infty}^{\infty} e^{-at}u(t)\,e^{-j\omega t}\,dt$$

$$= 1 - \frac{a}{a+j\omega} = \frac{j\omega}{a+j\omega} \checkmark$$

$$x(t) \longleftrightarrow X(j\omega)$$

$$\frac{d}{dx}(x(t)) \longleftrightarrow j\omega X(j\omega) = \frac{j\omega}{a+j\omega}$$

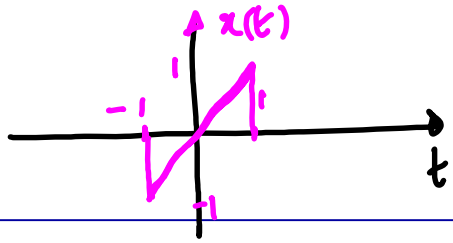
5 Integration

During differentiation, the DC component is 'lost'

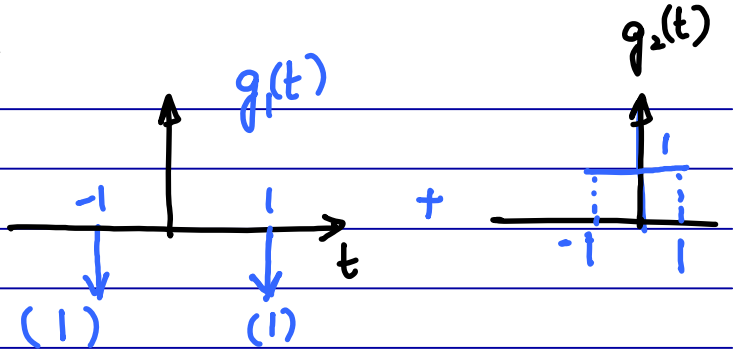
Let $x(t) \overset{\mathcal{F.T}}{\longleftrightarrow} X(j\omega)$

$$y(t) = \int_{-\infty}^{t} x(\tau)\,d\tau \longleftrightarrow \underbrace{\frac{1}{j\omega}X(j\omega)}_{\omega \neq 0} + \underbrace{\pi X(j0)\delta(\omega)}_{\text{to retrieve the dc component}}$$

eg:-



Let's find $g(t) = \frac{dx}{dt}$



$$F.T\{g(t)\} = FT\{g_1(t)\} + F.T\{g_2(t)\}$$

$$= -\left(e^{j\omega} + e^{-j\omega}\right) + \frac{2\sin\omega}{\omega}$$

$$G_1(j\omega) = \frac{2\sin\omega}{\omega} - 2\cos\omega \quad (= 0)$$

$$X(j\omega) = \frac{G_1(j\omega)}{j\omega} + \pi G_1(j0)\,\delta(\omega)$$

$$X(j\omega) = \frac{1}{j\omega}\left[\frac{2\sin\omega}{\omega} - 2\cos\omega\right]$$

Qn Find $F.T\{u(t)\}$

$x(t) \leftrightarrow X(j\omega)$

$$\int_{-\infty}^{t} x(\tau)\,d\tau \leftrightarrow \frac{X(j\omega)}{j\omega} + \pi X(j0)\,\delta(\omega)$$

$u(t) \leftrightarrow U(j\omega)$

$\delta(t) \leftrightarrow 1$; $x(t) = \delta(t)$, $X(j\omega) = 1$

$$\int_{-\infty}^{t} \delta(\tau)\,d\tau \leftrightarrow$$

$$\int_{-\infty}^{t} x(\tau)\,d\tau \leftrightarrow \frac{X(j\omega)}{j\omega} + \pi X(j0)\,\delta(\omega)$$

$$u(t) = \int_{-\infty}^{t} \delta(\tau)\,d\tau \leftrightarrow \frac{1}{j\omega} + \pi \cdot 1 \cdot \delta(\omega)$$

$$\therefore \quad \boxed{U(j\omega) = \frac{1}{j\omega} + \pi\,\delta(\omega)}$$

6. Multiplication

$$x(t) \longleftrightarrow X(j\omega)$$
$$y(t) \longleftrightarrow Y(j\omega)$$

$$z(t) = x(t) \cdot y(t) \longleftrightarrow Z(j\omega) \; ?$$

$$Z(j\omega) = \int_{-\infty}^{\infty} x(t)\, y(t)\, e^{-j\omega t}\, dt$$

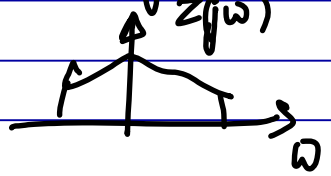
$$= \int_{-\infty}^{\infty} \left[\frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega')\, e^{j\omega' t}\, d\omega'\right] y(t)\, e^{-j\omega t}\, dt$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} X(j\omega')\, e^{-j(\omega-\omega')t}\, y(t)\, dt\, d\omega'$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega')\, Y(j(\omega-\omega'))\, d\omega'$$

$$= \frac{1}{2\pi}\, X(j\omega) * Y(j\omega)$$

$$\mathcal{F.T.}\{x(t)\cdot y(t)\} = \frac{1}{2\pi}\, X(j\omega) * Y(j\omega)$$



7.

$$x(t) \longleftrightarrow X(j\omega)$$
$$y(t) \longleftrightarrow Y(j\omega)$$

$$z(t) = x(t) * y(t) \longleftrightarrow X(j\omega)\cdot Y(j\omega)$$

Prove it yourself

$$Z(j\omega) = \int_{-\infty}^{\infty} \left(x(t) * y(t)\right) e^{-j\omega t}\, dt$$

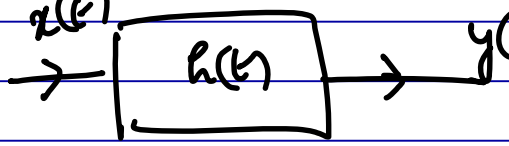
$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(\tau)\, y(t-\tau)\, d\tau\; e^{-j\omega t}\, dt$$

$$= \int_{-\infty}^{\infty} x(\tau)\, e^{-j\omega\tau} \cdot Y(j\omega)\, d\tau$$

$$= Y(j\omega) \int_{-\infty}^{\infty} x(\tau)\, e^{-j\omega\tau}\, d\tau$$

$$F.T.\left\{x(t) * y(t)\right\} = X(j\omega) \cdot Y(j\omega)$$

Implication for LTI systems



$$Y(j\omega) = X(j\omega) \cdot H(j\omega)$$

8

$$x(t) \longleftrightarrow X(j\omega)$$

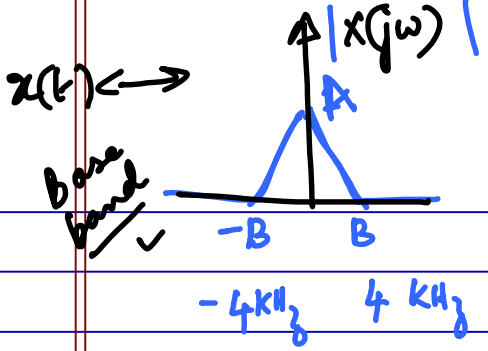
$$e^{j\omega_0 t}\, x(t) \longleftrightarrow X\{j(\omega-\omega_0)\}$$

$$\int_{-\infty}^{\infty} e^{j\omega_0 t}\, x(t)\, e^{-j\omega t}\, dt = \int_{-\infty}^{\infty} x(t)\, e^{-j(\omega-\omega_0)t}$$

$$= X\{j(\omega-\omega_0)\}$$

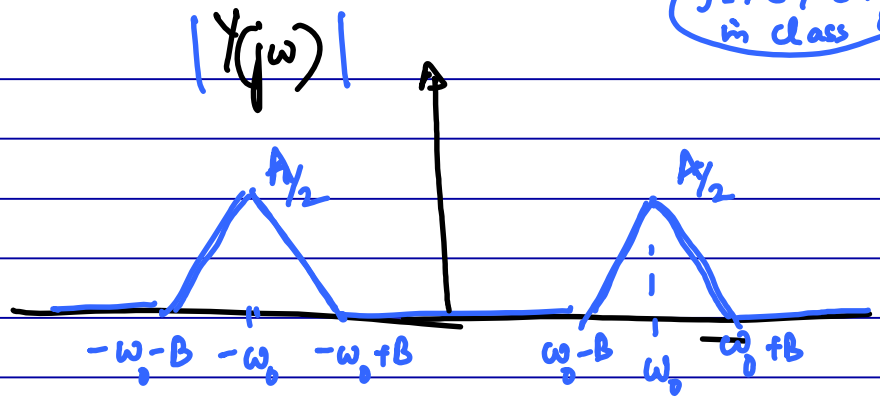
$$\cos\omega_0 t \cdot x(t) \longleftrightarrow \frac{1}{2}\left[X\,j(\omega-\omega_0) + X\,j(\omega+\omega_0)\right]$$

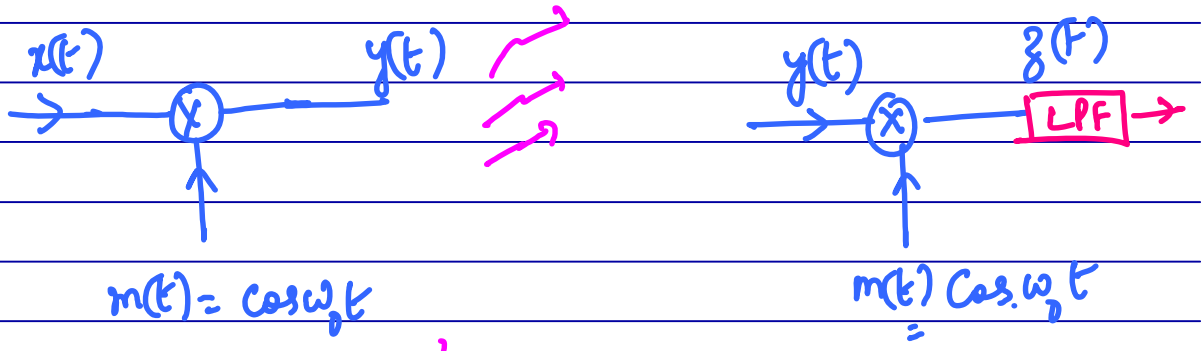
Modulation



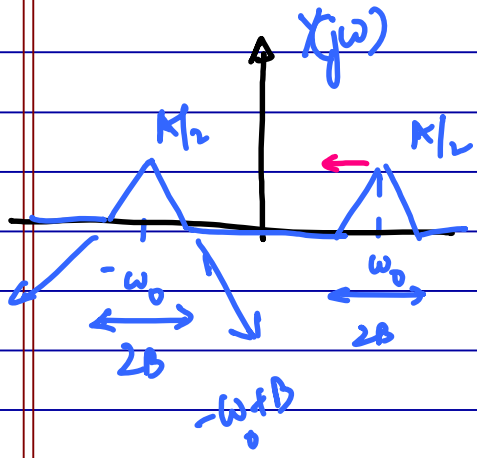
After Modulation $y(t) = x(t)\cos(\omega_0 t)$

$f = 789.8$ MHz in class



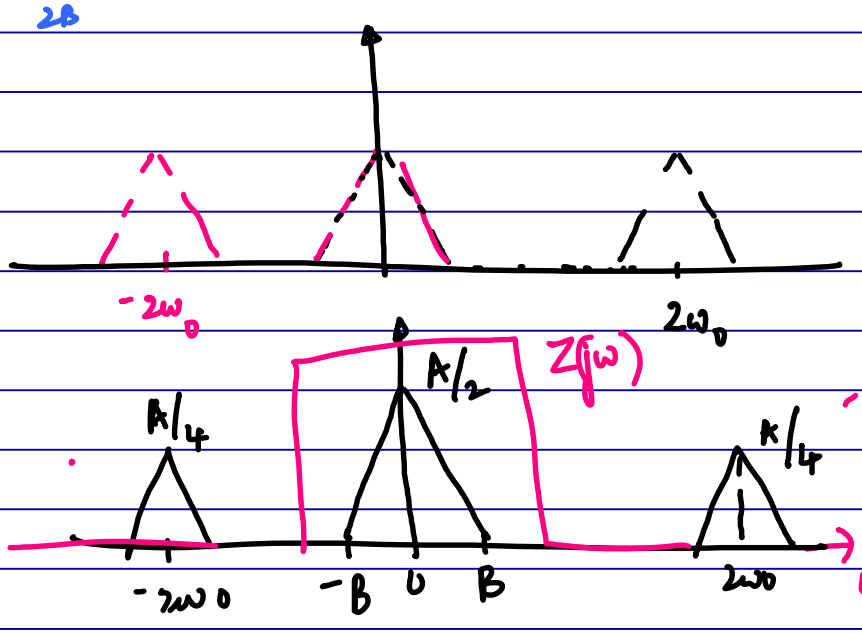


Modulation



$z(t) = y(t)\cos\omega_0 t$

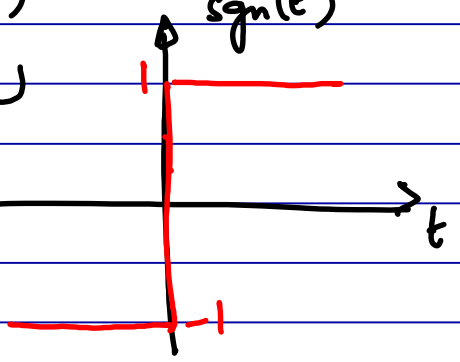
$$\frac{1}{2}\left[Y\{j(\omega-\omega_0)\} + Y\{j(\omega+\omega_0)\}\right]$$



F.T $\{u(t)\}$ revisited (without using integration property)

$$u(t) = \frac{1}{2} + \frac{1}{2}\,\text{sgn}(t)$$

$\frac{1}{2} \rightarrow \pi\,\delta(\omega)$



$1 \longleftrightarrow ?$

$\delta(t) \longleftrightarrow 1$ [Inverse F.T of 1]

$$\delta(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} 1\, e^{+j\omega t}\, d\omega$$

$$2\pi\,\delta(t) = \int_{-\infty}^{\infty} 1\, e^{j\omega t}\, d\omega$$

Change of variable from t to $-t$

$$2\pi\,\delta(t) = \int_{-\infty}^{\infty} 1\, e^{-j\omega t}\, d\omega$$

Interchange ω and t

$$2\pi\,\delta(\omega) = \int_{-\infty}^{\infty} 1\, e^{-jt\omega}\, dt$$

$$2\pi\,\delta(\omega) = \text{F.T}\{1\}$$

$$1 \xleftrightarrow{\text{F.T}} 2\pi\,\delta(\omega)$$

$$\frac{1}{2} \longleftrightarrow \pi\,\delta(\omega)$$

$$\text{sgn}(t) = s(t) \longleftrightarrow S(j\omega)$$

$$\frac{d}{dt}(s(t)) \longleftrightarrow j\omega\, S(j\omega)$$

2 $\delta(t) \longleftrightarrow j\omega\, S(j\omega)$

$$j\omega\, S(j\omega) = 2$$

$$S(j\omega) = \frac{2}{j\omega}$$

$$\therefore \quad U(j\omega) = \mathcal{F.T}\{u(t)\} = \mathcal{F.T}\left\{\frac{1}{2} + \frac{1}{2}\operatorname{sgn}(t)\right\}$$

$$U(j\omega) = \pi\,\delta(\omega) + \frac{1}{j\omega}$$

Proof for integration property

$$x(t) \longleftrightarrow X(j\omega)$$

$$y(t) = \int_{-\infty}^{t} x(\tau)\,d\tau \longleftrightarrow ?$$

$$y(t) = \int_{-\infty}^{t} x(\tau)\,d\tau = \int_{-\infty}^{\infty} x(\tau)\cdot u(t-\tau)\,d\tau$$

But $y(t) = \int_{-\infty}^{\infty} x(\tau)\,u(t-\tau)\,d\tau = x(t) * u(t)$

$$\therefore \quad \mathcal{F.T.}\{y(t)\} = \mathcal{F.T}\{x(t) * u(t)\}$$

$$= X(j\omega)\cdot U(j\omega)$$

$$= X(j\omega)\cdot\left[\pi\,\delta(\omega) + \frac{1}{j\omega}\right]$$

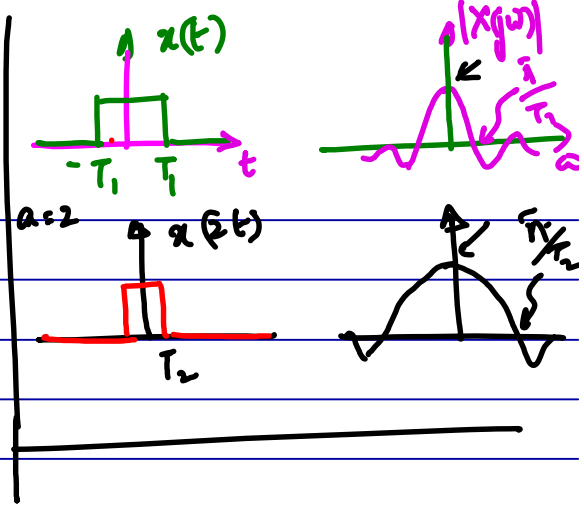
$$\therefore \mathcal{F.T}\left\{\int_{-\infty}^{t} x(\tau)\,d\tau\right\} = \frac{X(j\omega)}{j\omega} + \pi\,X(j0)\,\delta(\omega)$$

Integration property.

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$$x(t) \longleftrightarrow X(j\omega)$$

$$x(at) \longleftrightarrow \frac{1}{|a|} X\left(j\frac{\omega}{a}\right)$$



$$\int_{-\infty}^{\infty} x(at)\, e^{-j\omega t}\, dt$$

$u = at$

$du = a\,dt$

$$\int_{-\infty}^{\infty} x(u)\, e^{-j\frac{\omega}{a}u}\, \frac{du}{a}$$

$$\frac{1}{|a|} X\left(j\frac{\omega}{a}\right)$$

Quiz2. Topics - Fourier Series
Fourier Transforms
Sampling

- Parsevals theorem
- F.T of periodic Signals
- FT and LTI System

(1) Parsevals theorem (Plancheral's theorem for aperiodic signals)

$$\int_{-\infty}^{\infty} |x(t)|^2\, dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2\, d\omega$$

Energy in time domain

Proof

$$\int_{-\infty}^{\infty} |x(t)|^2 \, dt = \int_{-\infty}^{\infty} x(t)\, x^*(t)\, dt$$

$$= \int_{-\infty}^{\infty} x(t) \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(j\omega)\, e^{-j\omega t}\, d\omega \, dt$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(j\omega) \int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt \, d\omega$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} X^*(j\omega)\, X(j\omega)\, d\omega$$

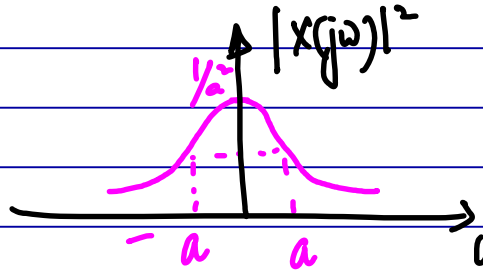
$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} |X(j\omega)|^2 \, d\omega$$

Integration of energy / unit freq

Energy density spectrum $|X(j\omega)|^2$

$e^{-at} u(t) \longleftrightarrow \frac{1}{a+j\omega}$

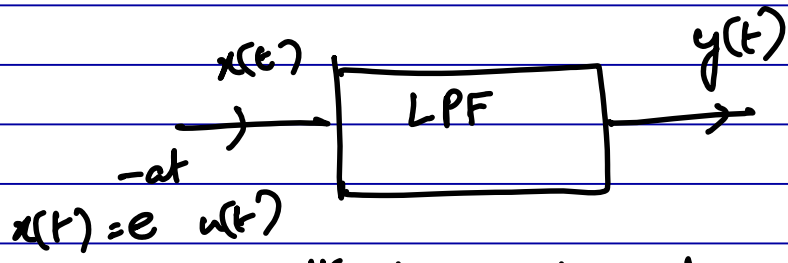
$$|X(j\omega)|^2 = \frac{1}{a^2+\omega^2}$$



Verify Parseval's theorem for

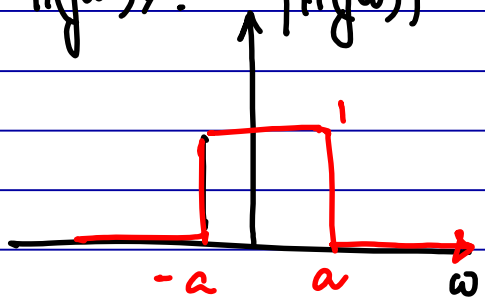
$$LHS = \int_{-\infty}^{\infty} |x(t)|^2 \, dt = \int_{0}^{\infty} e^{-2at} \, dt = \frac{1}{2a}$$

$$RHS \quad \frac{1}{2\pi} \int_{-\infty}^{\infty} \left| \frac{1}{a+j\omega} \right|^2 = \frac{1}{2a}$$



$x(t) = e^{-at} u(t)$

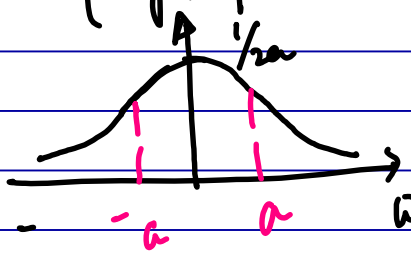
$H(j\omega)$ =: $|H(j\omega)|$



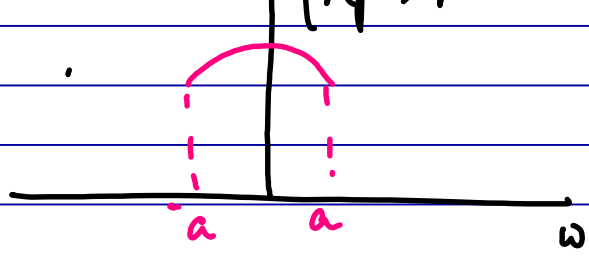
Find energy at input, energy at output

$$Y(j\omega) = X(j\omega) \cdot H(j\omega)$$

$|X(j\omega)|^2$



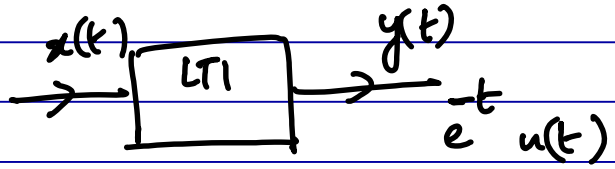
$|Y(j\omega)|^2$



$$\text{Energy at the output} = \frac{1}{2\pi} \int_{-a}^{a} \frac{1}{a^2+\omega^2} \, d\omega$$

$$= \frac{1}{4a}$$

Eg:



$e^{-2t} u(t)$

$e^{-t} u(t)$

Find the impulse response & freq response of the System

$$y(t) = x(t) * h(t)$$

$$X(j\omega) = \frac{1}{2+j\omega} \qquad Y(j\omega) = X(j\omega) \cdot H(j\omega)$$

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{1+1+j\omega}{1+j\omega}$$

$$Y(j\omega) = \frac{1}{1+j\omega}$$

$$= \underbrace{\frac{1}{1+j\omega}} + 1$$

$$h(t) = e^{-t} u(t) + \delta(t)$$

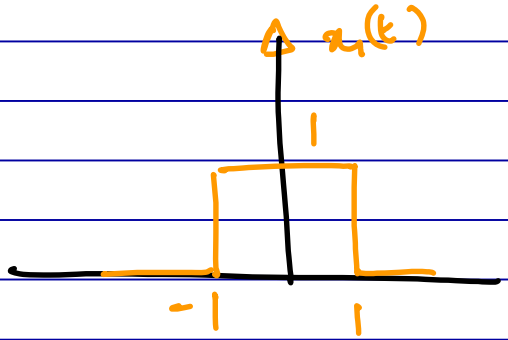
Eg:

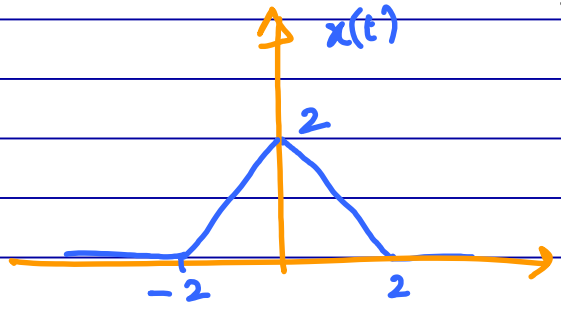
$$X(j\omega) = \frac{4}{\omega^2}\sin^2\omega = \left(\frac{2\sin\omega}{\omega}\right)\left(\frac{2\sin\omega}{\omega}\right)$$

$$= X_1(j\omega)\, X_2(j\omega)$$

Find $x(t) = x_1(t) * x_2(t)$

$$x_1(t) = \text{IFT}\{X_1(j\omega)\}$$





Eg:

$$X(j\omega) = 2\pi\,\delta(\omega - \omega_0)$$

$$x(t) = ?$$

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} 2\pi\,\delta(\omega - \omega_0)\, e^{j\omega t}\, d\omega$$

$= e^{j\omega_0 t}$

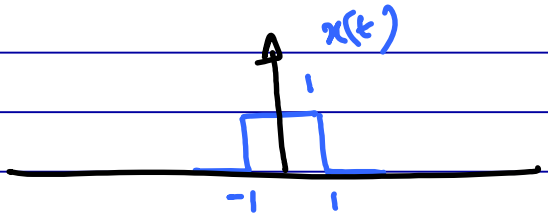
$$e^{j\omega_0 t} \longleftrightarrow 2\pi\,\delta(\omega-\omega_0)$$

$$\sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t} \longleftrightarrow \sum_{k=-\infty}^{\infty} a_k 2\pi\,\delta(\omega-k\omega_0)$$

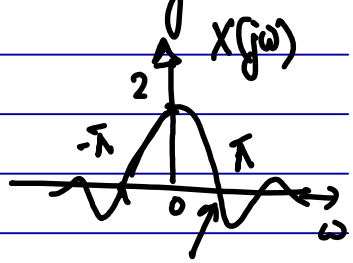
$x(t)$ when $x(t)$ is periodic

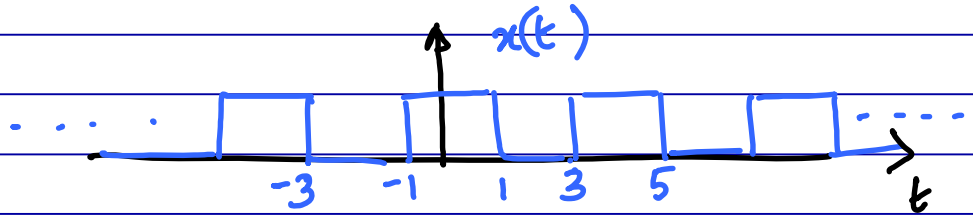
∴ F.T of periodic signal : from a_k's of the periodic signal

Eg:-



$\frac{2\sin\omega}{\omega}$





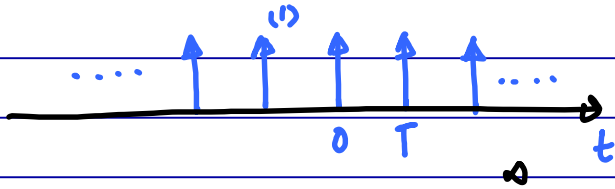
$T = 4$

$\omega_0 = \pi/2$

$$2\pi \sum_{k=-\infty}^{\infty} a_k\,\delta(\omega-k\omega_0)$$

Eg:- Find the F.T of an impulse train

$x(t)$



$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t-kT)$$

$$X(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} a_k\,\delta(\omega-k\omega_0)$$

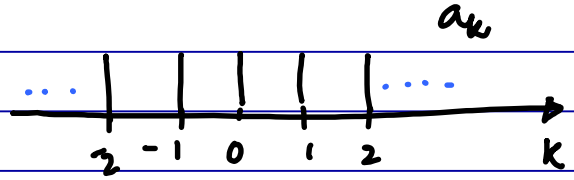
a_k's FS coeff

of the periodic fn

$$a_k = \frac{1}{T}\int_T \delta(t)\cdot e^{-jk\omega_0 t}\,dt$$

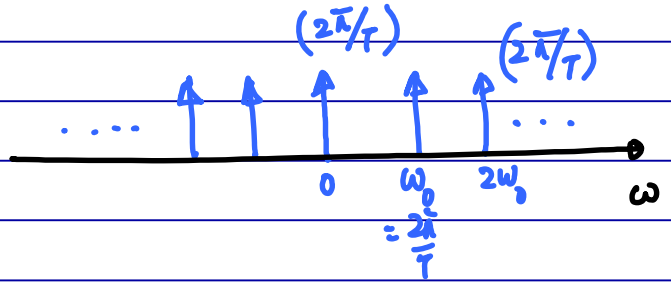
7.5

$$a_k = \frac{1}{T}$$

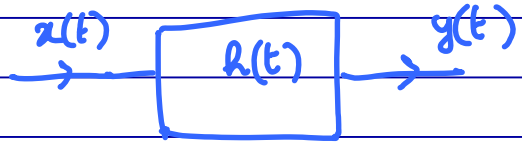


$$X(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} \frac{1}{T}\,\delta(\omega - k\omega_0)$$

$$= \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_0)$$



Eg:



$$Y(j\omega) = X(j\omega)\,H(j\omega)$$

$$\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 3y = 4\frac{dx}{dt} - x$$

Let $x(t) \overset{\text{7.T}}{\longleftrightarrow} X(j\omega)$

$y(t) \overset{\text{7.T}}{\longleftrightarrow} Y(j\omega)$

Qn: Find $H(j\omega)$

Rewriting in Fourier Domain / Freq Domain

$$(j\omega)^2 Y(j\omega) + 2j\omega Y(j\omega) + 3Y(j\omega) = 4j\omega X(j\omega) - X(j\omega)$$

$$H(j\omega) \quad \frac{Y(j\omega)}{X(j\omega)} = \frac{4j\omega - 1}{(j\omega)^2 + 2j\omega + 3}$$

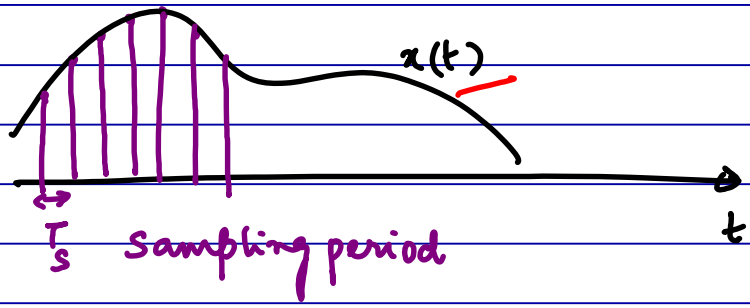
To find $h(t)$ express as partial fractions -

$$\frac{A}{a+j\omega} + \frac{B}{b+j\omega}$$

$$\downarrow$$

$$A e^{-at} u(t) \quad + \quad B e^{-bt} u(t)$$

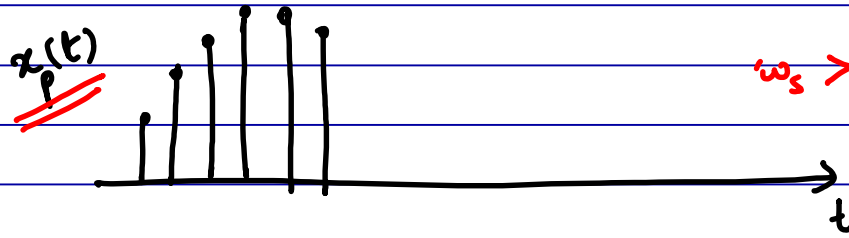
Sampling



$x(t)$

$x[n]$

$\omega_s = \frac{2\pi}{T_s}$: Sampling frequency



$\omega_s > 2\omega_M$

$$x_p(t) = x(t)\, p(t)$$

$$p(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT_s)$$

$$X_p(j\omega) = \frac{1}{2\pi}\left[X(j\omega) * P(j\omega)\right] \quad \text{[7.T properties]}$$

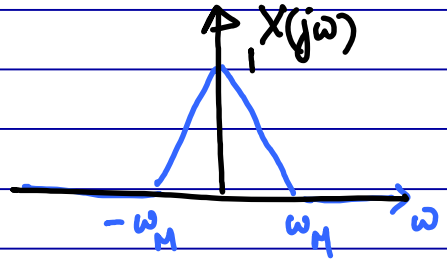
$$x(t) \longleftrightarrow X(j\omega)$$

$$p(t) \longleftrightarrow P(j\omega)$$

$$P(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s)$$

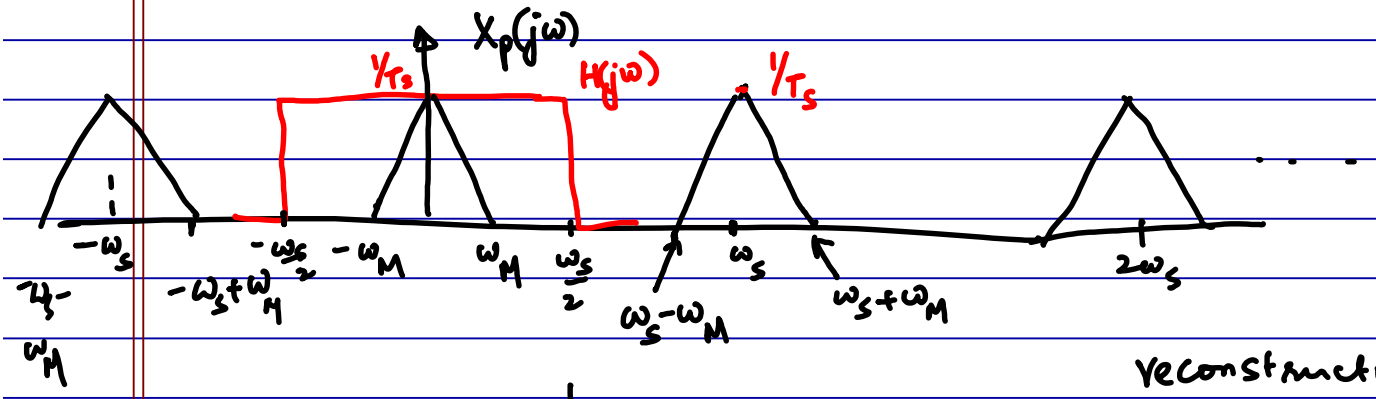
$$X_p(j\omega) = \frac{1}{2\pi} \cdot \frac{2\pi}{T} \underbrace{X(j\omega) * \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s)}$$

$$X_p(j\omega) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X\left[j(\omega - k\omega_s)\right]$$

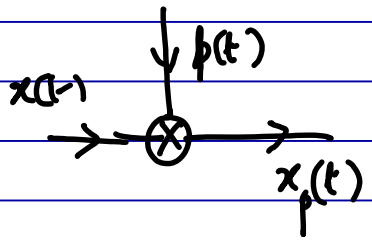


Assume $x(t)$ is band limited

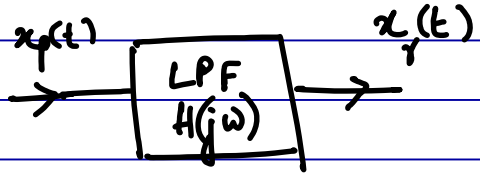
~~$\omega_s >> \omega_M$~~ $\omega_s > 2\omega_M$



reconstruction.

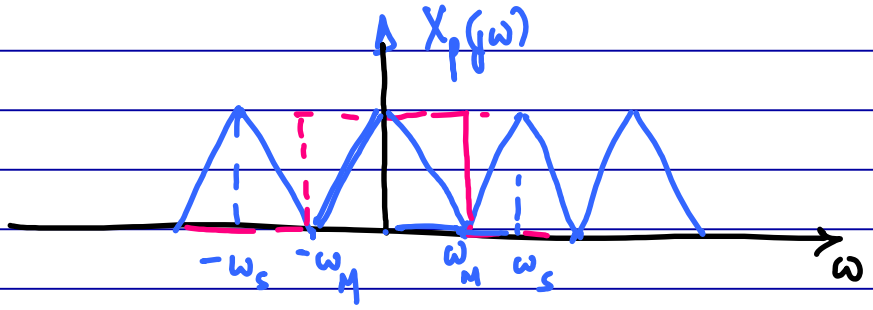


Sampled Signal

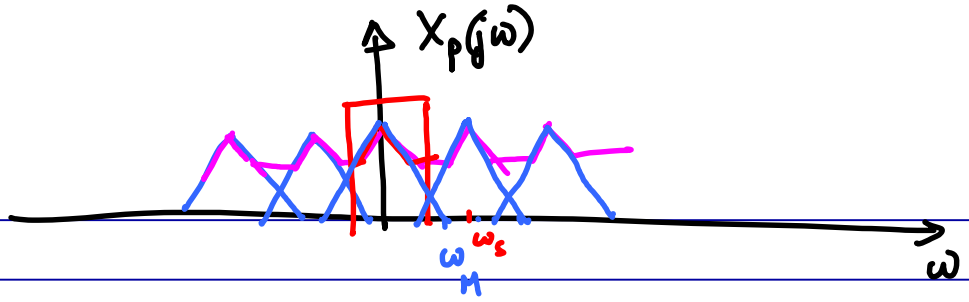


$$x_y(t) = x_p(t) * h(t)$$

$\omega_s = 2\omega_M$



$\omega_s < 2\omega_M$



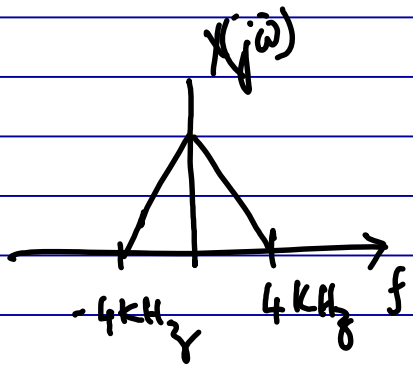
$\omega_s = 2\omega_M$: Nyquist rate

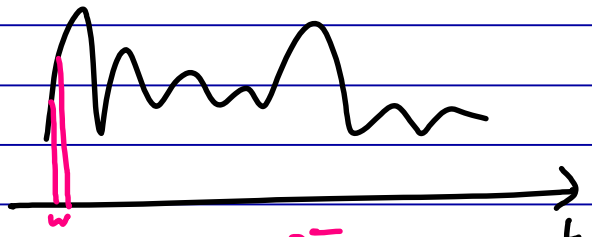
A band limited signal $x(t)$ can be uniquely represented by its samples,

$x(kT_s)$; $k = 0, \pm 1, \pm 2, \ldots$

provided the sampling freq, $\omega_s \left(= \frac{2\pi}{T_s}\right) \geq 2\omega_M$

where ω_M is the largest freq in the signal.

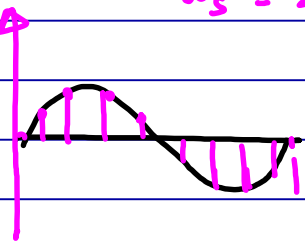




$T_s = \frac{2\pi}{\omega_s}$

$\omega_s > 2\omega_M$

$\omega_s = 2\omega_M$ $T = 1s$



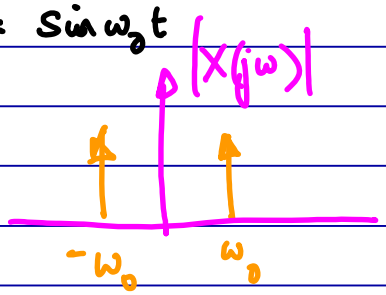
$T_s = \frac{1}{10}\ s$

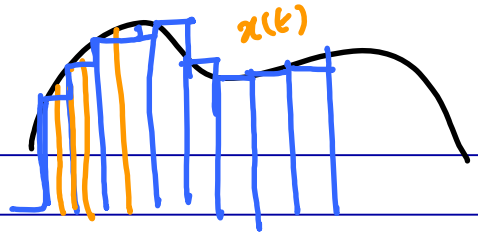
$\omega_s = 2\pi \cdot 10 \frac{\text{rad}}{s}$

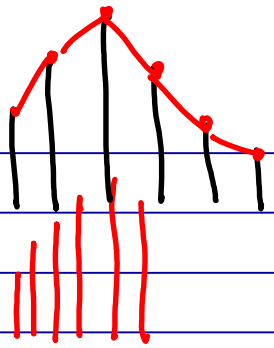
$\omega_M = 2\pi$

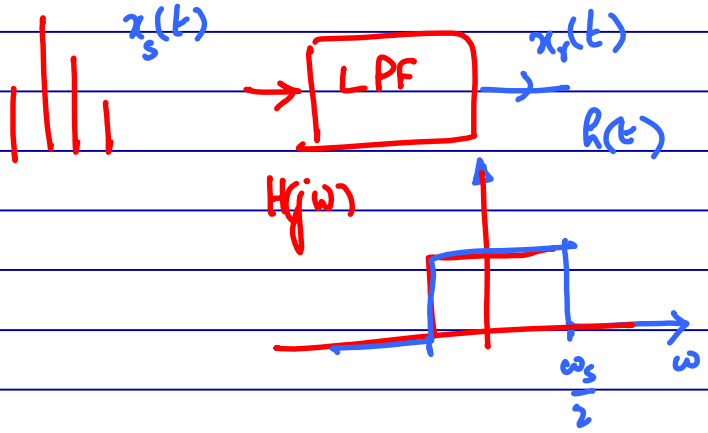
Aliasing

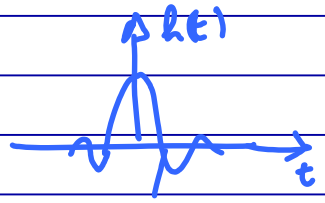
$x(t)$: $\sin \omega_0 t$



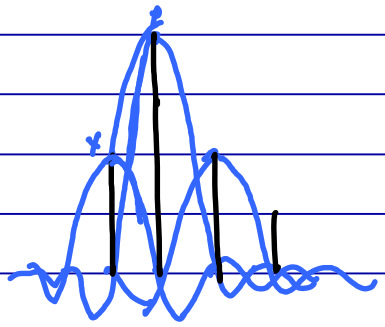


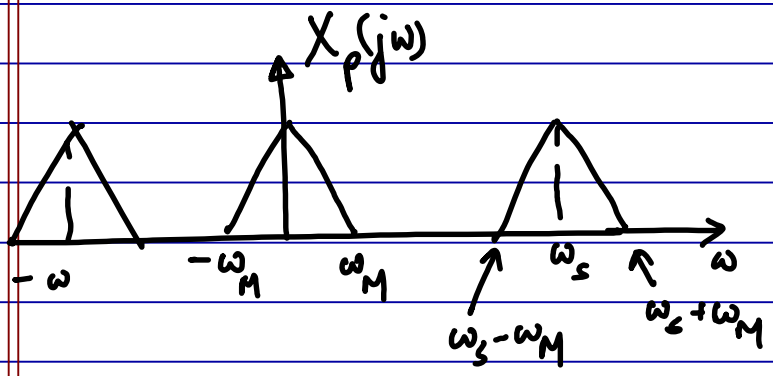






$$x_r(t) = x_s(t) * h(t)$$





ω_M : Nyquist freq

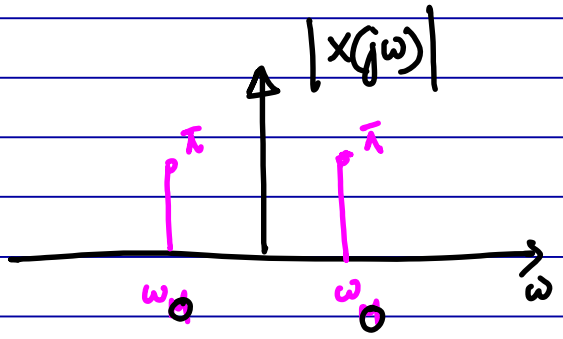
$\omega_s > 2\omega_M$ - Nyquist rate

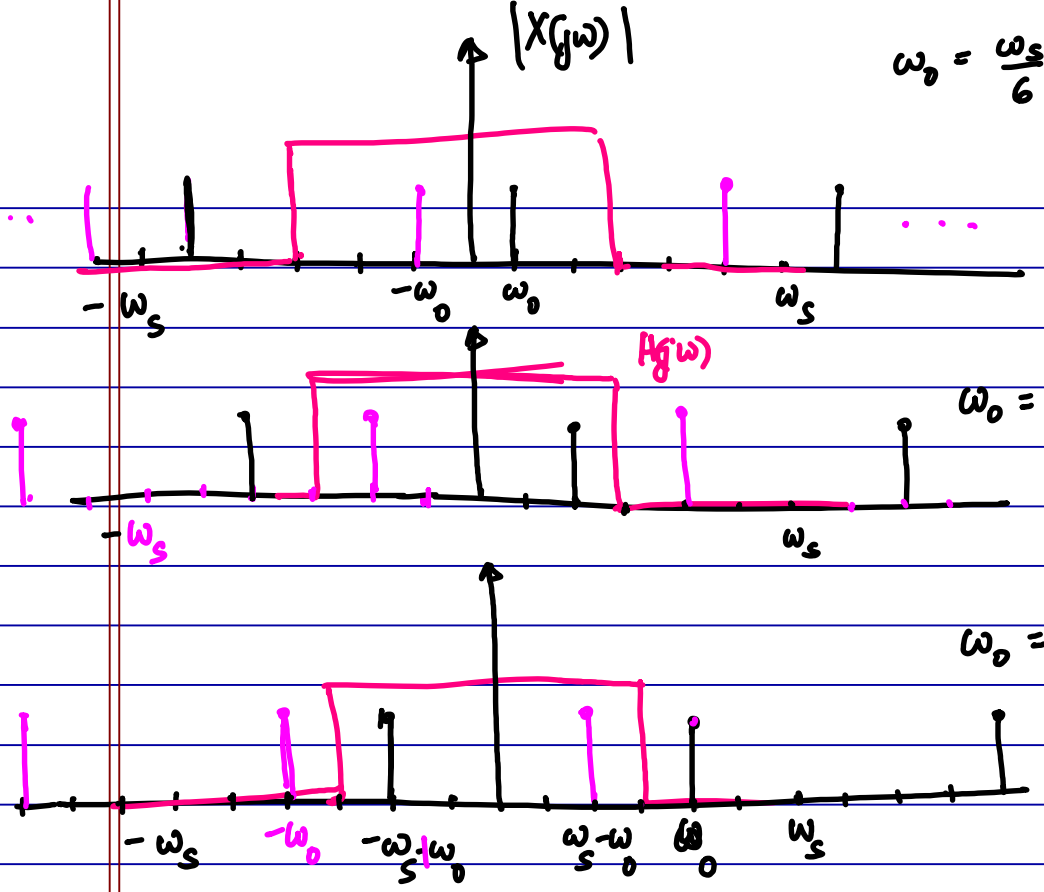
$\omega_s = 2\omega_M$

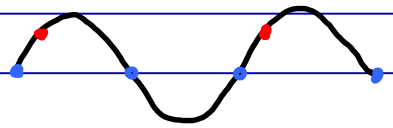
$\omega_s > 2\omega_M$

$x(t) = \sin(\omega_0 t)$

$=$







Laplace Transforms

- Helps to investigate system properties -

$x(t) = e^{st}$ → LTI, $h(t)$ → $y(t) = h(t) * x(t)$

$$= \int_{-\infty}^{\infty} h(\tau)\, e^{s(t-\tau)}\, d\tau$$

$$= e^{st} \int_{-\infty}^{\infty} h(\tau) e^{-s\tau}\, d\tau$$

$$y(t) = H(s)\, e^{st}$$

$$H(s) = \int_{-\infty}^{\infty} h(t) e^{-st}\, dt$$

Bilateral

In general, λ Laplace transform of $x(t)$ is,

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt \qquad s = \sigma + j\omega$$

$$x(t) \xleftrightarrow{LT} X(s)$$

$X(s)\Big|_{s=j\omega} = X(j\omega)$ is Fourier Transform of $x(t)$

$X(s)$; Laplace transform, is the Fourier Transform of $x(t).e^{-\sigma t}$

eg: $x(t) = e^{-at}\, u(t)$

$$X(s) = \int_{-\infty}^{\infty} e^{-at}\, u(t)\, e^{-st}\, dt = \int_{0}^{\infty} e^{-at}\, e^{-\sigma t}\, e^{-j\omega t}\, dt$$

$(\sigma + a) > 0$

$$= \int_{0}^{\infty} e^{-(\sigma+a)t}\, e^{-j\omega t}\, dt$$

$\sigma > -a$

$Re\{s\} > -a$

$$X(s) = \frac{1}{s+a} \quad ; \quad Re\{s\} > -a$$

Eg: $x(t) = -e^{-at}\, u(-t)$

$$X(s) = \int_{-\infty}^{0} -e^{-at}\, e^{-st}\, dt$$

$$= -\int_{-\infty}^{0} e^{-(s+a)t}\, dt = \frac{1}{s+a} \quad ; \quad Re\{s\} < -a$$

(1) $e^{-at}u(t) \xleftrightarrow{LT} \frac{1}{s+a}$; $Re\{s\} > -a$

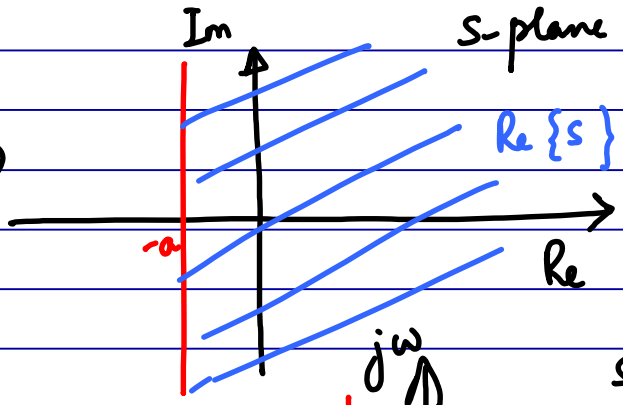
(2) $-e^{-at}u(-t) \xleftrightarrow{LT} \frac{1}{s+a}$; $Re\{s\} < -a$

$e^{bt}u(-t) \longleftrightarrow \frac{1}{s-b}$; $Re\{s\} < b$

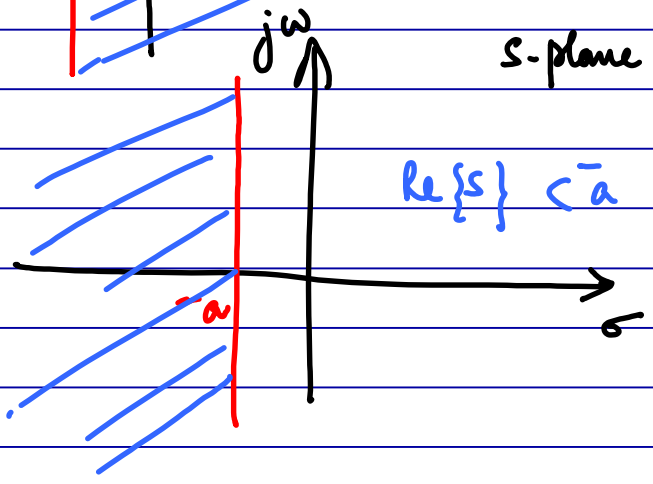
(3) $u(t) \xleftrightarrow{LT} \frac{1}{s}$; $Re\{s\} > 0$

ROC
Region of Convergence

(1)



(2)



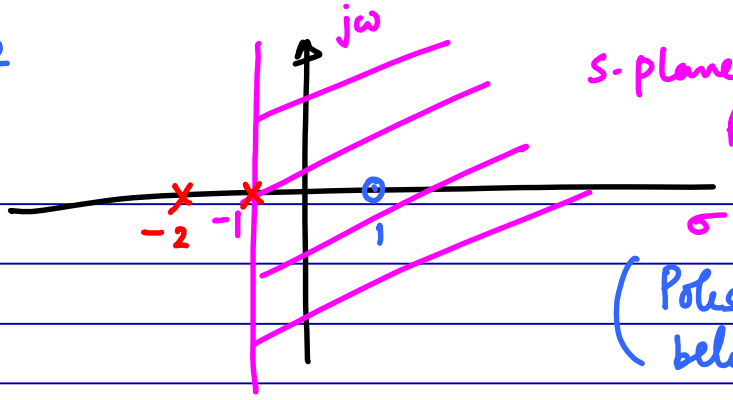
4: $x(t) = 3e^{-2t}u(t) - 2e^{-t}u(t)$ $\quad (3e^{-2t} - 2e^{-t})$

$X(s) = \frac{3}{s+2} - \frac{2}{s+1} = \frac{s-1}{s^2+3s+2}$

$Re\{s\} > -2$ $\quad$ $Re\{s\} > -1$

Poles : $s = -1, -2$

Zeros : $s = 1$



(Poles do not belong in ROC)

In general, $X(s) = \frac{N(s)}{D(s)}$

Values of s for which $N(s) = 0$ are 'zero's o

Values of s for which $D(s) = 0$ → poles x

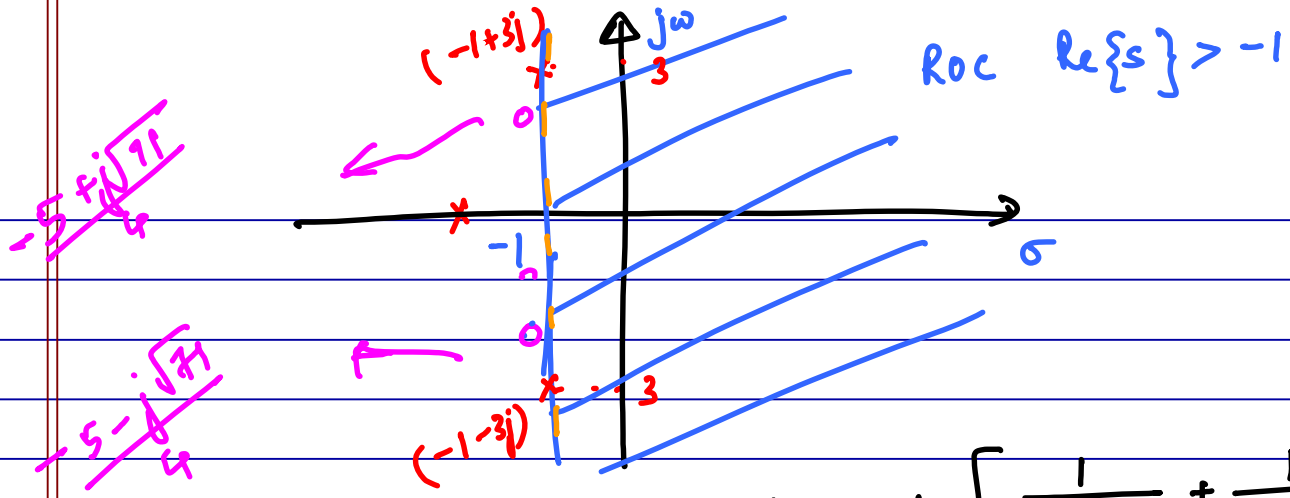
eg: $x(t) = e^{-2t} u(t) + e^{-t} \cos 3t \, u(t)$

$$x(t) = \left[e^{-2t} + \frac{e^{-t} e^{j3t}}{2} + \frac{e^{-t} e^{-3jt}}{2} \right] u(t)$$

$$= \left[e^{-2t} + \frac{1}{2} e^{-(1-3j)t} + \frac{1}{2} e^{-(1+3j)t} \right] u(t)$$

$$X(s) = \frac{1}{s+2} + \frac{1}{2} \cdot \frac{1}{s+1-3j} + \frac{1}{2} \frac{1}{s+1+3j}$$

$\text{Re}\{s\} > -2$ $\quad$ $\text{Re}\{s\} > -1$ $\quad$ $\text{Re}\{s\} > -1$



$$X(s) = \frac{1}{s+2} + \frac{1}{2}\left[\frac{1}{s+1+3j} + \frac{1}{s+1-3j}\right]$$

$$= \frac{2s^2+5s+12}{(s^2+2s+10)(s+2)}$$

Zero's : $2s^2 + 5s + 12 = 0$

$$s = \frac{-5 \pm j\sqrt{71}}{4}$$

Poles : $s = -2, \; -1+3j, \; -1-3j$

Convergence of FT $\int_{-\infty}^{\infty} |x(t)|\, e^{-j\omega t}\, dt < \infty$

What could be $x(t)$ such that LT converges?

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

Properties of ROC ?

e^{5t} e^{-1t}

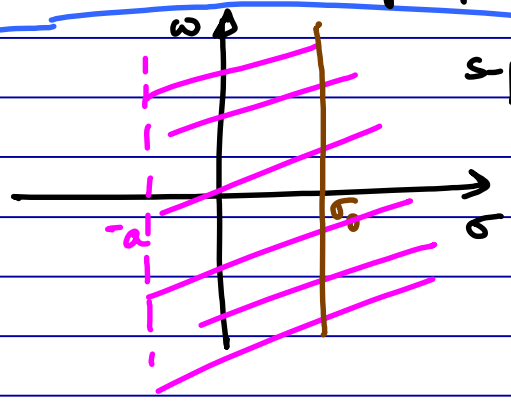
$$X(s) = \mathcal{FT}\left\{ x(t)\, e^{-\sigma t} \right\}$$

For convergence, $\int_{-\infty}^{\infty} |x(t)|\, e^{-\sigma t}\, dt < \infty$

Depends on σ

(I)

$\Rightarrow$ ROC is always plane/strip parallel to jω axis



s-plane

$\mathrm{Re}\{s\} > -a$

Eg: $x(t) = \delta(t) - \frac{4}{3} e^{-t} u(t) + \frac{1}{3} e^{2t} u(t)$

$$X(s) = 1 - \frac{4}{3}\cdot\frac{1}{s+1} + \frac{1}{3}\frac{1}{s-2}$$

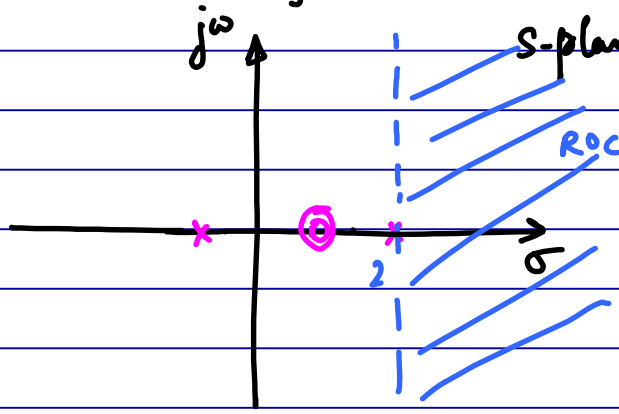
$$= \frac{(s-1)^2}{(s+1)(s-2)}$$

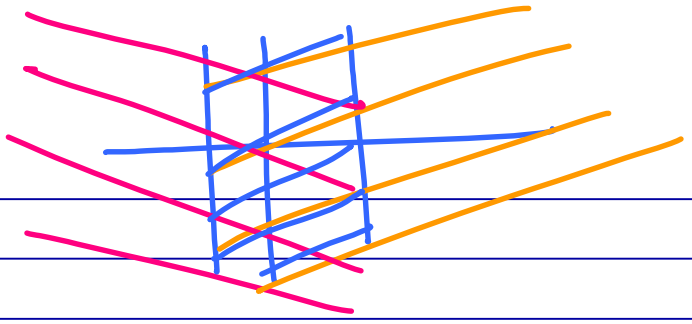
Pole-zero plot

Intersection $\begin{cases} \mathrm{Re}\{s\} > -1 \\ \mathrm{Re}\{s\} > 2 \end{cases}$

Poles : $s = -1, 2$

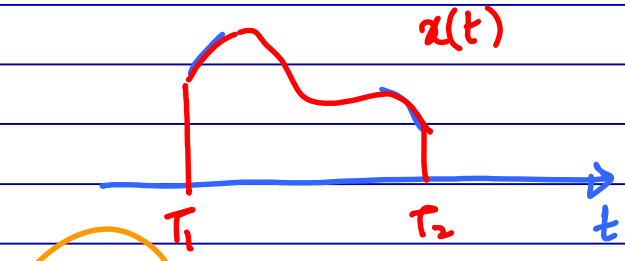
Zeros : $1, 1$



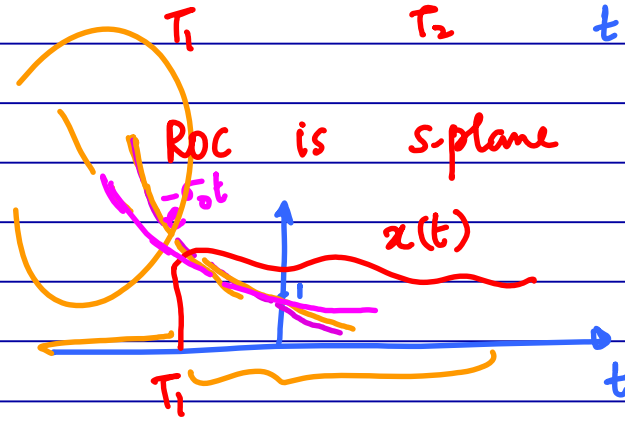


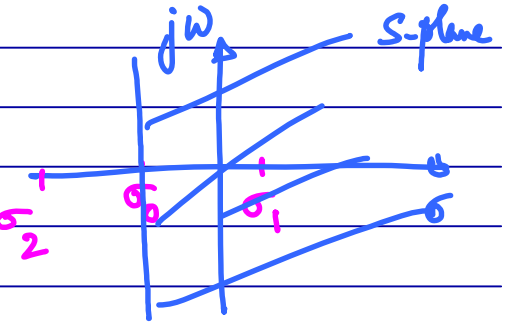
II. For rational LTs, ROC does not contain a pole

III.



III.





If x(t) is right sided and if the line

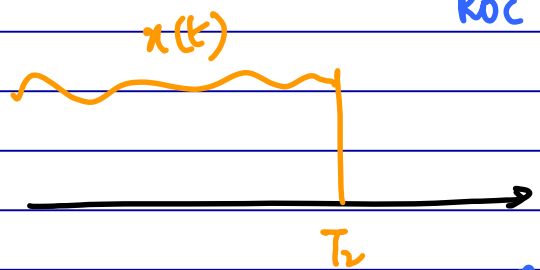
$\mathrm{Re}\{s\} = \sigma_0$ is in the ROC, then,

all value of $\sigma > \sigma_0$ will be in the ROC

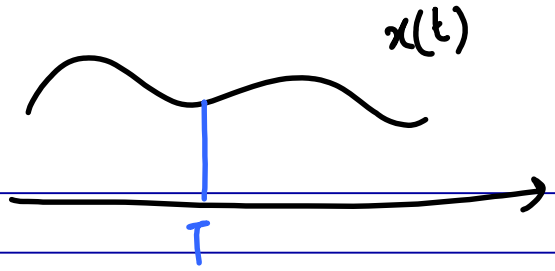
s for which $\mathrm{Re}\{s\} > \sigma_0$ " "

ROC is a right hand plane half

IV.

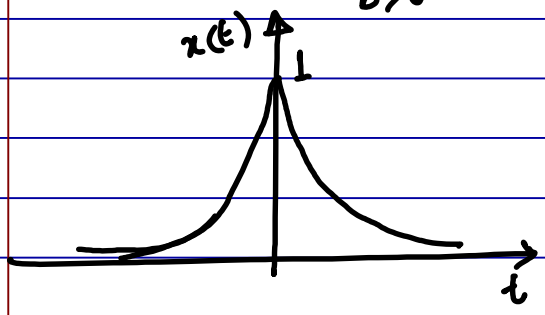


ROC is a left hand plane half

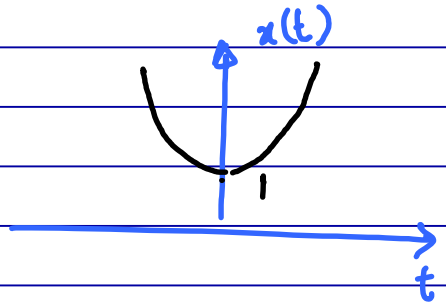


eg:- $x(t) = e^{-b|t|}$

$b>0$



$b<0$



No RoC

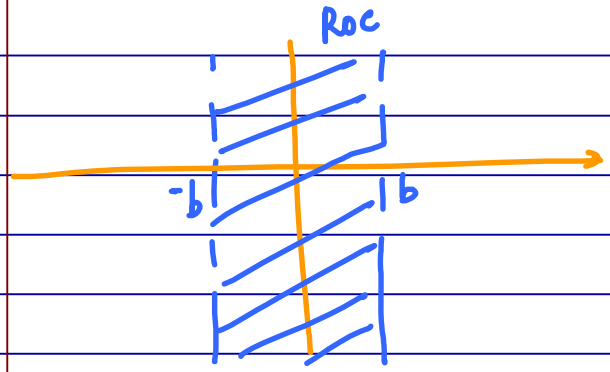
$x(t) = e^{-bt}u(t) + e^{+bt}u(-t)$

RoS $\{s > -b$

RoS $\{s < b$

$= \frac{1}{s+b} - \frac{1}{s-b}$

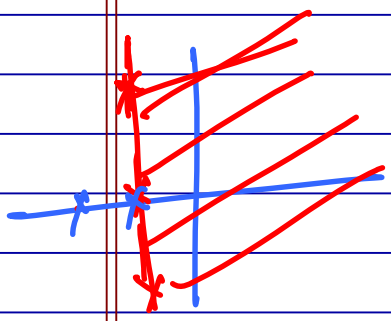
$= \frac{2b}{s^2 - b^2}$



	RoC
Finite length	s-plane
Right handed	Right half plane
Left handed	Left half plane
Neither R, or L Handed	ROC can be a strip or no ROC

VI Rational LT

ROC bounded by poles (either strip or extends to ∞)



ROC : to the right of the rightmost pole for a right handed signal

left of the leftmost pole for left handed signal

HW :- (1) $X(s) = \dfrac{1}{(s+1)(s+2)}$; $Re\{s\} > -1$

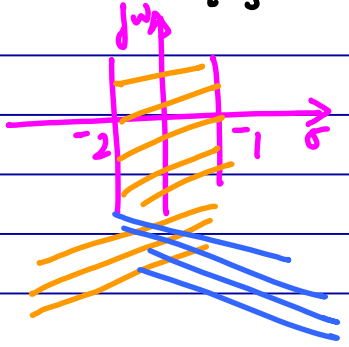
Find $x(t)$

$X(s) = \dfrac{1}{s+1} - \dfrac{1}{s+2}$

$x(t) = e^{-t}u(t) - e^{-2t}u(t)$

(ROC is right half plane; signal is right sided)

(2) $X(s) = \dfrac{1}{(s+1)(s+2)}$ $-2 < Re\{s\} < -1$



$X(s) = \underbrace{\dfrac{1}{s+1}}_{-e^{-t}u(-t)} - \dfrac{1}{s+2}$

$= -e^{-t}u(-t) - e^{-2t}u(t)$

(3) Find LT of $\cos \omega_0 t\, u(t)$, $\sin \omega_0 t\, u(t)$

(4) Given $x(t) = e^{-at}$, $0 < t < T$; $= 0$, elsewhere } Find $X(s)$ and ROC

$$LT\{\cos\omega_0 t\, u(t)\} = LT\left\{\left(\frac{e^{j\omega_0 t} + e^{-j\omega_0 t}}{2}\right)u(t)\right\}$$

$$= \frac{1}{2} LT\{e^{j\omega_0 t}u(t)\} + \frac{1}{2} LT\{e^{-j\omega_0 t}u(t)\}$$

$$= \frac{1}{2}\int_0^\infty e^{j\omega_0 t} e^{-st}\,dt + \frac{1}{2}\int_0^\infty e^{-j\omega_0 t} e^{-st}\,dt$$

$$= \frac{1}{2}\left[\frac{1}{s-j\omega_0} + \frac{1}{s+j\omega_0}\right] \qquad \text{Re}\{s\} > 0$$

$$\mathcal{LT}\{(\cos\omega_0 t)\,u(t)\} = \frac{s}{s^2+\omega_0^2} \qquad \text{Re}\{s\} > 0$$

$$\mathcal{LT}\{(\sin\omega_0 t)\,u(t)\} = \frac{\omega_0}{s^2+\omega_0^2} \qquad \text{Re}\{s\} > 0$$

(4) $x(t) = \begin{cases} e^{-at}; & 0 < t < T \\ 0 & \text{elsewhere} \end{cases}$

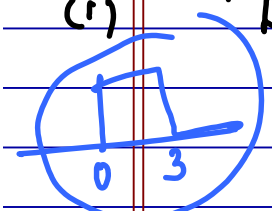
$$X(s) = \frac{1 - e^{-(s+a)T}}{s+a}$$

( ) + ( ) + ( ) + ( ) + ( )

- LT defn
- ROC, pole-zero plot
- Properties
- Sum of DE
- Appln of LT

Properties

(1) Linearity



$-u(t-3) + u(t)$

$$x_1(t) \overset{\mathcal{L}}{\longleftrightarrow} X_1(s); \quad \text{ROC: } R_1$$

$$x_2(t) \overset{\mathcal{L}}{\longleftrightarrow} X_2(s); \quad \text{ROC} = R_2$$

$$a x_1(t) + b x_2(t) \overset{\mathcal{L}}{\longleftrightarrow} a X_1(s) + b X_2(s)$$

ROC: $R_1 \cap R_2$

(2) Time shifting

$$x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s); \quad \text{ROC: } R$$

$$x(t-t_0) \longleftrightarrow e^{-st_0} X(s) \qquad \text{ROC: } R$$

$$\mathcal{L}\{x(t-t_0)\} = \int_{-\infty}^{\infty} x(t-t_0)\, e^{-st}\,dt$$

$$= \int_{-\infty}^{\infty} x(u)\, e^{-s(u+t_0)}\, du \qquad t - t_0 = u$$

$$= e^{-st_0} \underbrace{\int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt}_{} = e^{-st_0} X(s)$$

3. $x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s)$ ROC : R

$$x(t)\, e^{s_0 t} \overset{\mathcal{L}}{\longleftrightarrow} X(s - s_0) \qquad \text{ROC : } R + \text{Re}\{s_0\}$$

$$X(s - s_0) = \int_{-\infty}^{\infty} x(t)\, e^{-(s-s_0)t}\, dt$$

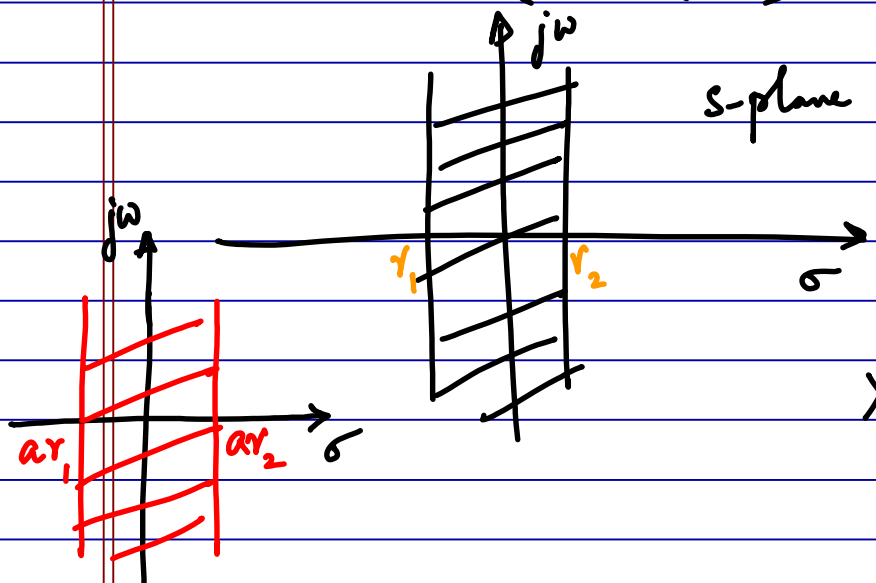
$$= \int_{-\infty}^{\infty} x(t) e^{s_0 t} e^{-st}\, dt$$

$$e^{j\omega_0 t}\, x(t) \longleftrightarrow X(s - j\omega_0) \; ; \quad \text{ROC : R}$$

4. Time Scaling

$$x(t) \overset{\mathcal{L}}{\longleftrightarrow} X(s) \; ; \quad \text{ROC : R}$$

$$x(at) \overset{\mathcal{L}}{\longleftrightarrow} X_1(s) = \frac{1}{|a|} X\left(\frac{s}{a}\right);$$



$u = at$

$du = a\, dt$

$$X_1(s) = \int_{-\infty}^{\infty} x(at)\, e^{-st}\, dt$$

$$X_1(s) = \int_{-\infty}^{\infty} x(u)\, e^{-(s/a)u}\, \frac{du}{a}$$

If a is -ve

$$\int_{\infty}^{-\infty} x(u)\, e^{-(s/a)u}\, \frac{du}{a}$$

$$\frac{1}{|a|} \int_{-\infty}^{\infty} x(u)\, e^{-(s/a)u}\, du$$

$$\frac{1}{|a|} X\left(\frac{s}{a}\right)$$

5 Differentiation in t

$$x(t) \xleftrightarrow{\mathcal{L}} X(s) \quad ; \text{ Roc : R}$$

$$\frac{d}{dt}\left(x(t)\right) \longleftrightarrow s\,X(s) \; ; \text{ Roc : R}$$

(Revisit after defn of inverse LT)

6 Differentiation in s

$$x(t) \xleftrightarrow{\mathcal{L}} X(s) \qquad \text{Roc R}$$

$$-t\,x(t) \longleftrightarrow \frac{dX(s)}{ds}$$

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

$$\frac{dX(s)}{ds} = \int_{-\infty}^{\infty} -t\,x(t)\, e^{-st}\, dt$$

L.T. $\{t \sin t\ u(t)\} = ?$

$$\sin \omega_0 t\ u(t) \longleftrightarrow \frac{\omega_0}{s^2+\omega_0^2} \qquad \omega_0 =$$

$$\sin t\ u(t) \longleftrightarrow \frac{1}{s^2+1}$$

$$-t \sin t\ u(t) \longleftrightarrow \frac{d}{ds}\left(\frac{1}{s^2+1}\right)$$

$$\frac{-2s}{(s^2+1)^2}$$

$$\therefore \quad t \sin t\ u(t) \longleftrightarrow \frac{2s}{(s^2+1)^2} \qquad \text{ROC}$$

eg:-

$$x(t) = t\, e^{-at} u(t)$$

$$e^{-at} u(t) \longleftrightarrow \frac{1}{s+a} \qquad \text{ROC. } \text{Re}\{s\} > -a$$

$$t\, e^{-at} u(t) \longleftrightarrow \frac{1}{(s+a)^2}$$

$$t^2 e^{-at} u(t) \longleftrightarrow \frac{+2}{(s+a)^3}$$

$$t^3 e^{-at} u(t) \longleftrightarrow \frac{2 \cdot 3}{(s+a)^4}$$

$$\vdots$$

$$t^n e^{-at} \longleftrightarrow \frac{n!}{(s+a)^{n+1}} \qquad \text{ROC } \text{Re}\{s\} > -a$$

Inverse Laplace Transforms

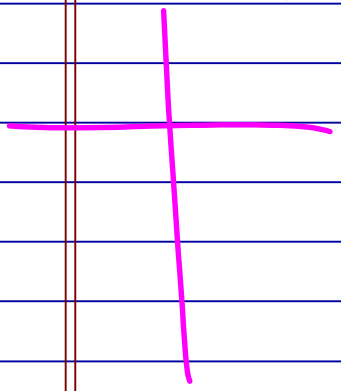
$$X(s) = \mathcal{FT}\left\{x(t)\, e^{-\sigma t}\right\}$$

$$x(t)\, e^{-\sigma t} = \text{Inverse F.T}\left\{X(s)\right\}$$

$$x(t)\, e^{-\sigma t} = \frac{1}{2\pi}\int_{\omega=-\infty}^{\omega=\infty} X(s)\, e^{j\omega t}\, d\omega$$

$$x(t) = \frac{1}{2\pi}\int_{\omega=-\infty}^{\omega=\infty} X(s)\, e^{j\omega t}\, e^{\sigma t}\, d\omega$$

S-plane



$$x(t) = \frac{1}{2\pi}\int_{\omega=-\infty}^{\infty} X(s)\, e^{st}\, d\omega$$

$$x(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)\, e^{st}\, ds \qquad \left| \begin{array}{l} s = \sigma + j\omega \\ d\omega = \frac{1}{j} ds \end{array} \right.$$

$$x(t) = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} X(s)\, e^{st}\, ds$$

$$\frac{dx(t)}{dt} = \frac{1}{2\pi j}\int_{\sigma-j\infty}^{\sigma+j\infty} sX(s)\, e^{st}\, ds$$

7 Convolution

$$x_1(t) \longleftrightarrow X_1(s) \qquad \text{ROC}: R_1$$

$$x_2(t) \longleftrightarrow X_2(s) \qquad \text{ROC}: R_2$$

$$x_1(t) * x_2(t) \longleftrightarrow X_1(s)\, X_2(s)$$

ROC: $R_1 \cap R_2$

$$X_1(s) = \frac{s+1}{s+2}, \quad X_2(s) = \frac{s+2}{s+1}$$

$$x_1(t) * x_2(t) \longleftrightarrow 1$$

8. Integration

$$x(t) \longleftrightarrow X(s) \quad ; \quad \text{ROC: } R$$

$$\int_{-\infty}^{t} x(\tau)\, d\tau \longleftrightarrow$$

$$x(t) * u(t) \longleftrightarrow X(s) \cdot \frac{1}{s}$$

ROC: $R \cap \mathrm{Re}\{s\} > 0$

9. Conjugation

$$x(t) \longleftrightarrow X(s)$$

$$x^*(t) \longleftrightarrow X^*(s^*)$$

Prove?

$s = \sigma + j\omega$

$$X(s) = \int_{-\infty}^{\infty} x(t)\, e^{-\sigma t}\, e^{-j\omega t}\, dt$$

$$X^*(s) = \int_{-\infty}^{\infty} x^*(t)\, e^{-\sigma t}\, e^{j\omega t}\, dt$$

$$X(s^*) = \int_{-\infty}^{\infty} x(t)\, e^{-\sigma t}\, e^{j\omega t}\, dt$$

$s^* = \sigma - j\omega$

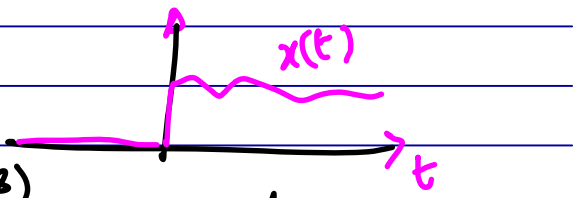
$$X^*(s^*) = \int_{-\infty}^{\infty} x^*(t)\, e^{-\sigma t}\, e^{-j\omega t}\, dt$$

$$X^*(s^*) = \int_{-\infty}^{\infty} x^*(t)\, e^{-st}\, dt$$

$$LT\{x^*(t)\} = X^*(s^*)$$

10/ Initial Value Theorem — } Valid for $x(t) = 0 \;\forall\; t<0$

Final Value Theorem — }



$$x(0^+) = \lim_{s\to\infty} sX(s)$$

$$\lim_{t\to\infty} x(t) = \lim_{s\to 0} sX(s)$$

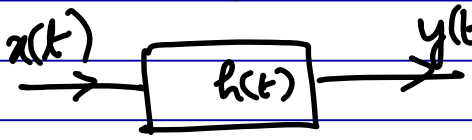
} Evaluate boundary conditions

Eg:

$$\frac{2s^2+5s+5}{(s+1)^2(s+2)} = \frac{A}{(s+1)} + \frac{B}{(s+1)^2} + \frac{C}{(s+2)}$$

Characterisation of LTI systems

— Stability — Causality



$$y(t) = x(t) * h(t)$$

H(s): System function

$$Y(s) = X(s)\,H(s)$$

— Stability

For stable systems, ROC contains jω axis,

Causality

For causal systems, h(t) is right-sided



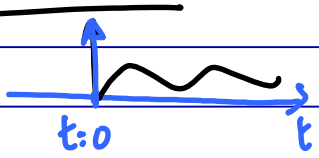
Is the converse true?

RoC of $H(s)$ is right half plane

If this system is causal, the RoC of LT of $h(t)$ is right half plane (ii)

Is converse true?

Unilateral LT



$$X(s) = \int_{0^-}^{\infty} x(t)\, e^{-st}\, dt$$

to include any singularity of $x(t)$ at $t:0$

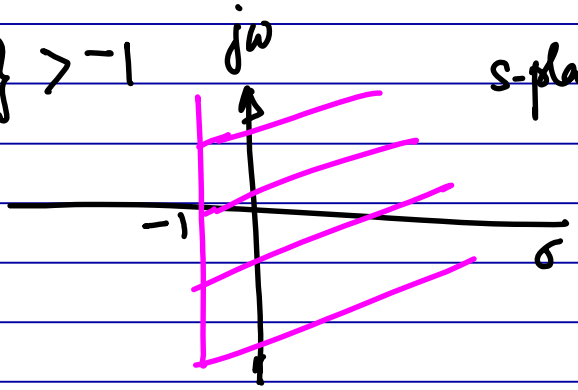
Bilateral → causal, anti-causal, non causal systems

Unilateral → only for causal, can take care of boundary condition

Eg:- (1) $H(s) = \dfrac{1}{s+1}$ $\quad \mathrm{Re}\{s\} > -1$

$h(t) = e^{-t} u(t)$

Causal



(2) $H(s) = \dfrac{e^{s}}{s+1}$ $\quad \mathrm{Re}\{s\} > -1$

$$e^{-t}\,u(t) \longleftrightarrow \frac{1}{s+1}$$

$$e^{-(t+1)}\,u(t+1) \longleftrightarrow \frac{e^{s}}{s+1}$$

$$h(t) = e^{-(t+1)}\,u(t+1)$$

If LT is rational, ROC is in right half plane, converse is true
q (ii)

Stability

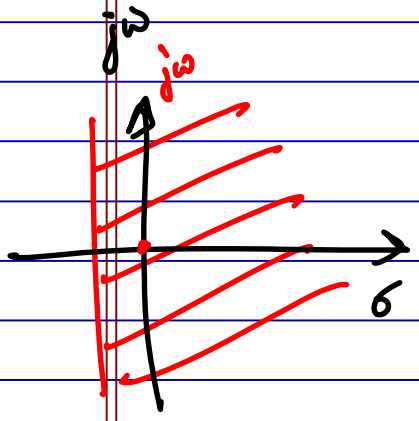
$h(t)$ is stable

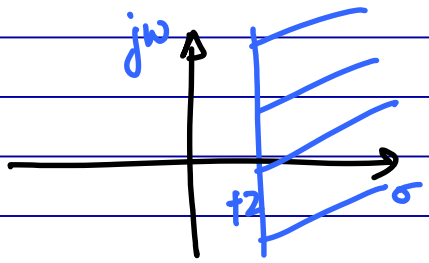
$\Rightarrow$ $h(t)$ is absolutely integrable

$\Rightarrow$ F.T of $h(t)$ converges

$\Rightarrow$ LT of $h(t)$ evaluated $\sigma = 0$ converges

$\Rightarrow$ ROC should contain $\sigma = 0$ ($j\omega$ axis)



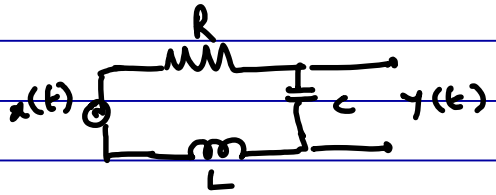


$\frac{1}{s-2}$

$$H(s) = \frac{s-1}{(s+1)(s-2)}$$

Find $h(t)$ such that the system is stable

LCR



Current through the circuit, $i(t) = C\frac{dy}{dt}$

$$x(t) = R.C\frac{dy}{dt} + LC\frac{d^2y}{dt^2} + y(t)$$

In Laplace domain

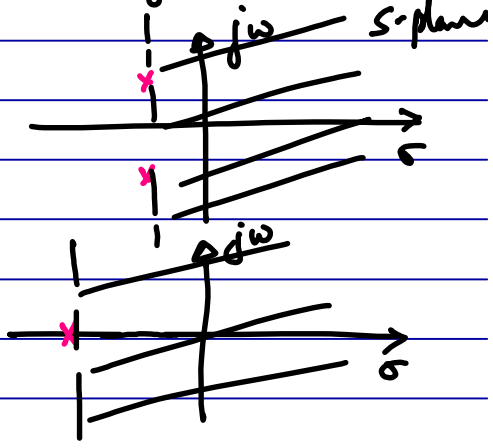
$$RC\,s\,Y(s) + LC\,s^2\,Y(s) + Y(s) = X(s)$$

$$\therefore \quad H(s) = \frac{1}{LC\,s^2 + RC\,s + 1}$$

Poles: $s = \dfrac{-RC \pm \sqrt{R^2C^2 - 4LC}}{2LC}$ $\quad H(s) = \dfrac{A}{(s-a)} + \dfrac{B}{(s-b)}$

System is causal, Real part of the poles are negative

$\therefore$ System is stable.



$$h(t) = \left(Ae^{at} + Be^{bt}\right)u(t)$$

Eg to illustrate convolution property of ULT

Convolution property holds good only when $x_1(t) = x_2(t) = 0 \;\forall\; t<0$

Given $x_1(t) = e^{-2t}u(t)$, $x_2(t) = e^{-3(t+1)}u(t+1)$

(1) Find $X_1(s)$ and $\chi_1(s)$

$$X_1(s) = \frac{1}{s+2} \quad \text{ROC: } Re\{s\} > -2 \;;\; \chi_1(s) = \frac{1}{s+2} \quad Re\{s\} > -2$$

(2) Find $X_2(s)$ and $\chi_2(s)$

$$X_2(s) = \frac{e^s}{s+3} \quad Re\{s\} > -3 \;;\; \chi_2(s) = \frac{e^{-3}}{s+3} \quad Re\{s\} > 3$$

(3) Find $x_1(t) * x_2(t)$ from the product $X_1(s)\,X_2(s)$;

$$X_3(s) = \frac{e^s}{(s+3)} \cdot \frac{1}{(s+2)} \qquad x_3(t) =$$

(4) Find inverse ULT of $X_1(s)\,X_2(s)$

Show that ULT of $X_1(s)\,X_2(s)$ is not the same as $x_1(t) * x_2(t)$ for $t > 0^-$

$$x_1(t) * x_2(t) \xleftrightarrow{\text{BLT}} X_1(s)\,X_2(s)$$

$$x_1(t) * x_2(t) \xleftrightarrow{\text{ULT}} X_1(s)\,X_2(s)$$

Convolution property holds good only when $x_1(t) = x_2(t) = 0 \quad \forall\ t < 0$

Differentiation

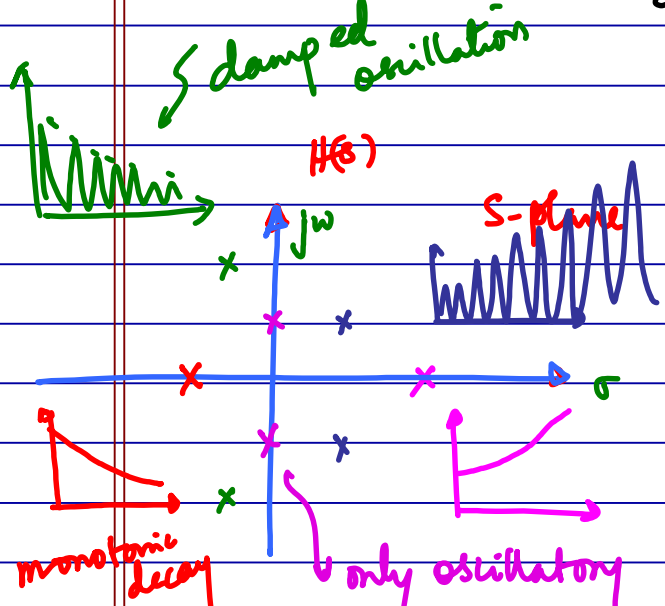
$$x(t) \xleftrightarrow{\text{ULT}} X(s)$$

$$\frac{dx}{dt} \xleftrightarrow{} s\,X(s) - x(0^-)$$

$$\frac{d^2x}{dt^2} = \frac{d}{dt}\left(\frac{dx}{dt}\right) \xleftrightarrow{} s\left(s\,X(s) - x(0^-)\right) - x'(0^-)$$

$$= s^2 X(s) - s\,x(0^-) - x'(0^-)$$

$$x'(0^-) = \left.\frac{dx}{dt}\right|_{t=0^-}$$





Eg:- $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = x$

(a) Find $H(s) = \frac{Y(s)}{X(s)} = \frac{1}{s^2+3s+2}$ **Causal**

(b) Input is $x(t) = \alpha u(t)$; α is a constant

Find $y(t)$? $\quad Y(s) = H(s)X(s) = \frac{\alpha}{s(s^2+3s+2)}$

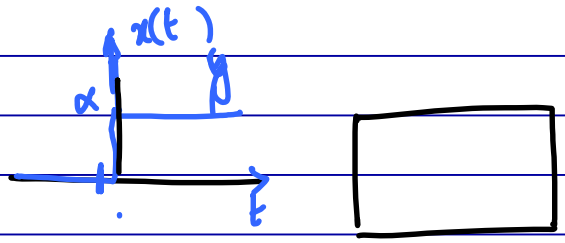
$$Y(s) = \frac{(\alpha/2)}{s} - \frac{\alpha}{(s+1)} + \frac{\alpha/2}{(s+2)}$$

$$y(t) = \left[\left(\frac{\alpha}{2}\right) - \alpha e^{-t} + \frac{\alpha}{2} e^{-2t}\right] u(t)$$

(c) Given initial conditions $\quad x(t) = \alpha u(t)$

$y(0^-) = \beta$, $y'(0^-) = \gamma$

Find $y(t)$



ULT since initial conditions to be taken care of

$$s^2Y(s) - sy(0^-) - y'(0^-) + 3sY(s) - 3y(0^-) + 2Y(s) = X(s)$$

$$s^2Y(s) - \beta s - \gamma + 3sY(s) - 3\beta + 2Y(s) = \frac{\alpha}{s}$$

$$Y(s)\left[s^2+3s+2\right] - \beta s - (\gamma + 3\beta) = \frac{\alpha}{s}$$

$$Y(s) = \frac{\alpha}{s(s^2+3s+2)} + \frac{\beta s}{(s^2+3s+2)} + \frac{\gamma+3\beta}{s^2+3s+2}$$

$$Y(s) = \frac{\alpha}{s(s^2+3s+2)} + \frac{\beta(s+3)}{(s^2+3s+2)} + \frac{\gamma}{(s^2+3s+2)}$$

Zero state response: when the response is at rest / system is at rest / $\beta = 0$, $\gamma = 0$

If input is zero, LT of o/p is

Zero input response

$y(t) =$

Qn 4. Find the natural and forced response :-

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 6y = \frac{dx}{dt} + 6x$$

$x(t) = u(t)$

Given $y(0^-) = 1$, $y'(0^-) = 2$

$$s^2Y(s) - s\,y(0^-) - y'(0^-) + 5sY(s) - 5\,y(0^-) + 6Y(s) = sX(s) - x(0^-) + 6X(s)$$

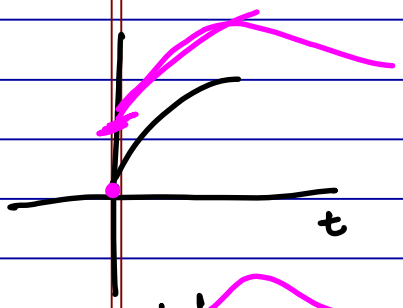
$$Y(s) = \underbrace{\frac{X(s)(s+6)}{s^2+5s+6}}_{\text{Zero state response / forced response}} + \underbrace{\frac{y(0^-)(s+5) + y'(0^-)}{s^2+5s+6}}_{\text{zero input response / natural response}}$$

$$= \underbrace{\frac{(s+6)}{s(s+3)(s+2)}} + \frac{s+7}{(s+3)(s+2)}$$

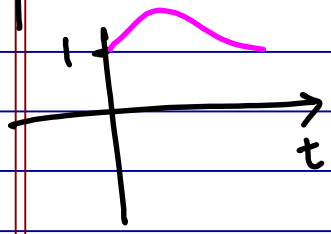
$$Y^f(s) = \frac{s+6}{s(s+3)(s+2)} = \frac{1}{s} - \frac{2}{s+2} + \frac{1}{s+3}$$

$$y^f(t) = 1u(t) - 2e^{-2t}u(t) + e^{-3t}u(t)$$

$\forall\ t > 0$



$$Y^n(s) = \frac{s+7}{(s+3)(s+2)} = \frac{5}{s+2} - \frac{4}{s+3}$$



$$y^n(t) = 5e^{-2t}u(t) - 4e^{-3t}u(t)$$

$$y(t) = y^f(t) + y^n(t)$$

Qn

1. System is causal :- ROC is right half plane

System fn is rational $\Rightarrow$ $H(s) = \frac{P(s)}{Q(s)} = \frac{s\,R(s)}{Q(s)}$

Two poles $s=-2$, $s=4$ $\Rightarrow$ $Q(s) = K(s+2)(s-4)$

$$\boxed{y(t) = H(s)\,x(t)}$$

$x(t) : e^{st}$

$y(t) = H(0)\,1$ $\qquad x(t) = e^{st}$

0 $\qquad H(0) = 0$ $\qquad s=0$

$$h(t)\Big|_{t=0^+} = \lim_{s\to\infty} s\,H(s)$$

$$= \lim_{s\to\infty} \frac{s^2 R(s)}{K(s+2)(s-4)}$$

$$= \lim_{s\to\infty} \frac{R(s)\, s^2}{K(s^2 - 2s - 8)}$$

$$4 = \lim_{s\to\infty} \frac{R(s)}{K\left(1 - \frac{2}{s} - \frac{8}{s^2}\right)}$$

$\Rightarrow R(s)$ is const $= K_2$

$$\therefore H(s) = \frac{4s}{(s^2 - 2s - 8)}$$

Block Diagrams – Introduction

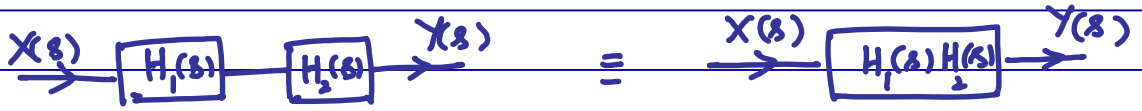
Large systems may have enormous number of components / elements

- It is convenient to represent the large system as subsystems – each subsystem has its own characteristic input and output.

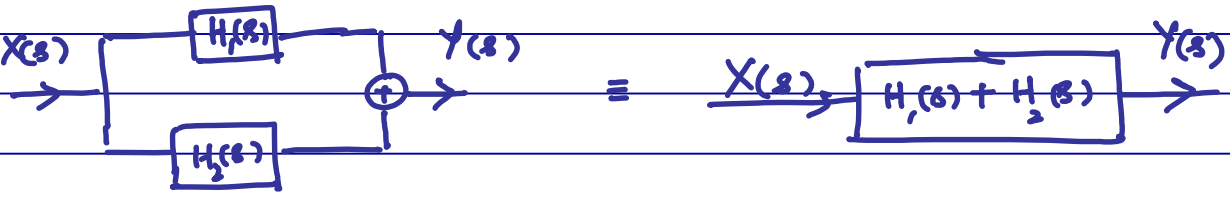


$Y(s) = H(s)\,X(s)$

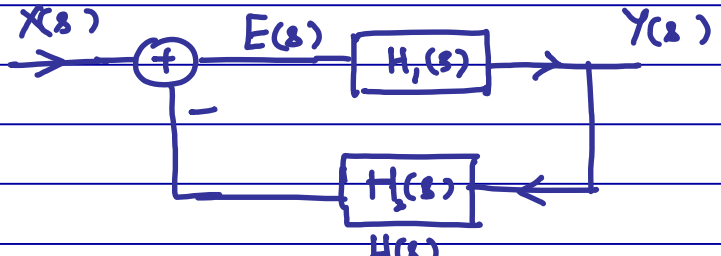
Series



Parallel



Feedback System



Equivalent System function $\hat{H(s)}$ can be arrived at, as follows:

$$E(s) = X(s) - H_2(s)\,Y(s)$$

$$Y(s) = H_1(s)\,E(s)$$

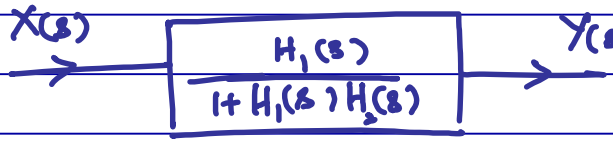
$$= H_1(s)\left(X(s) - H_2(s)\,Y(s)\right)$$

$$Y(s) = H_1(s)\,X(s) - H_1(s)\,H_2(s)\,Y(s)$$

$$\frac{Y(s)}{X(s)} = H_1(s) - H_1(s)\,H_2(s)\,\frac{Y(s)}{X(s)}$$

$$\therefore \quad H(s) = \frac{Y(s)}{X(s)} = \frac{H_1(s)}{1 + H_1(s)\,H_2(s)} = \frac{1}{\left(\frac{1}{H_1(s)}\right) + H_2(s)}$$

Equivalent system:



Example:

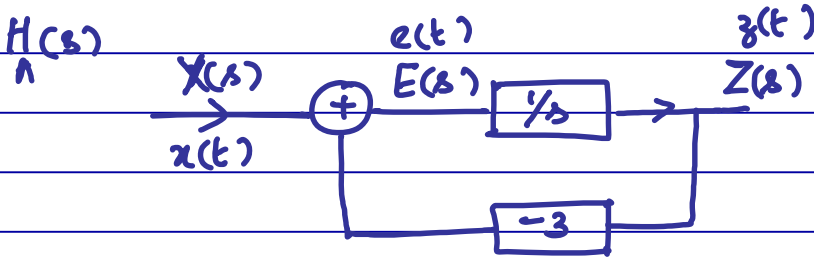
Find the block diagram representation of subsystem representing $H(s) = \frac{s+2}{s+3}$

Soln $H(s) = \frac{1}{(s+3)} \cdot (s+2) = \underbrace{\left(\frac{1/s}{1+(3/s)}\right)}_{H_A(s)} \underbrace{(s+2)}_{H_B(s)}$

Cascade of two systems

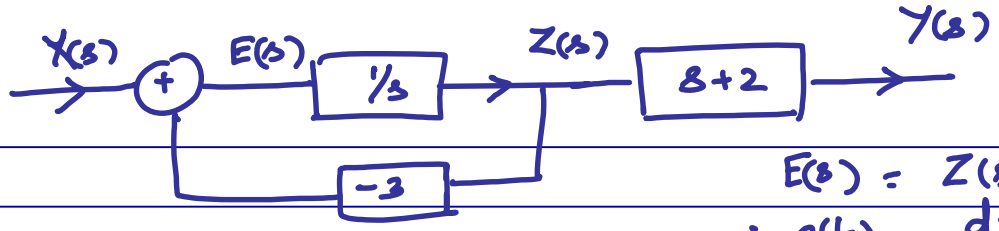
$H_A(s)$ is in the form of feedback system;

with $H_1(s) = (1/s)$, $H_2(s) = -3$; [$H_1(s)$ $H_2(s)$ – notations from feed back system representation]



Note: $H_1(s) = (1/s)$ represents integration in time

H(s):



$E(s) = Z(s) \cdot s$

$\therefore e(t) = \frac{dz}{dt}$

Alternate realisation:-

$Y(s) = Z(s) \cdot (s+2)$

$Y(s) = s\,Z(s) + 2\,Z(s)$

$y(t) = \frac{dz}{dt} + 2\,z(t)$

$y(t) = e(t) + 2\,z(t)$

